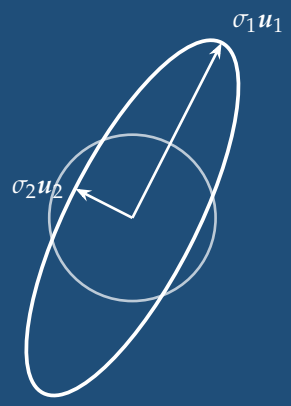
The SVD and the Pseudo-Inverse



A hands-on textbook, from vectors to minimum-norm least squares

Part 1 of a three-part course

1. **Textbook** (this volume) — theory, worked hand calculations, exercises with full solutions.
2. **Python Lab Manual** — every algorithm implemented from scratch in Python, with real output.
3. **Coding Assignments** — six graded assignments with an automatic `pytest` test suite.

Contents at a glance: vectors & norms · the four fundamental subspaces · orthogonality & QR · eigenvalues of symmetric matrices · positive semidefinite matrices · the singular value decomposition · low-rank approximation · least squares · the Moore–Penrose pseudo-inverse · regularisation and PCA.

Edition of October 2026

How to Use This Course

This book takes you from the very first ideas of linear algebra (vectors, dot products, matrices) to two of its most useful tools: the **singular value decomposition** (SVD) and the **Moore–Penrose pseudo-inverse**. Every concept is built only from the ones before it, so if you work through the chapters in order you will never meet an idea that has not been prepared.

The three components

Textbook (this volume). Definitions, theorems with proofs, and above all *worked examples done by hand*. Every example uses small matrices with clean numbers so you can reproduce each line with pen and paper. Each chapter ends with exercises graded by difficulty; complete solutions are in Appendix B.

Python Lab Manual. A separate PDF in which every algorithm in this book is implemented *from scratch* — no `numpy.linalg` — and checked against the hand calculations you did here. You see the code, the output it really produces, and plots of how the algorithms behave.

Coding Assignments. Six assignments, each a Python module of function stubs for you to fill in, with an automatic test suite (`pytest`) and a grading script that prints your score per assignment. The test suite blocks `numpy.linalg` inside your code, so you really do implement everything yourself.

Reading the boxes

Definition: Definitions. Precise statements of new terms. Learn these word for word.

Theorem: Theorems and propositions. Facts that are proved. The proofs are short and worth reading: they show *why* the recipes work.

Worked Example: Worked examples.
Complete hand calculations. Cover the solution, try it yourself, then compare line by line.

Key idea. The one thing to remember from a section.

Common pitfall. Mistakes that almost everybody makes once.

Exercises are marked by difficulty: $\star$ routine practice, $\star\star$ needs several steps or a short argument, $\star\star\star$ a proof or a challenge.

Tips for hand calculations

- **Keep exact values.** Write $\sqrt{45} = 3\sqrt{5}$, not 6.708. Rounding early destroys the checks below.
- **Normalise last.** Find an eigenvector with integer entries first, e.g. $(1, 1)$, then divide by its length.
- **Check orthogonality as you go.** If two vectors that should be orthogonal have a non-zero dot product, stop and find the slip now.
- **Multiply back.** After computing an SVD, multiply $U\Sigma V^{\mathsf{T}}$ out. After a pseudo-inverse, check $AA^{+}A = A$. These checks take one minute and catch almost every error.
- **Use trace and determinant.** For a 2×2 matrix the eigenvalues satisfy $\lambda_1 + \lambda_2 = \operatorname{tr}$ and $\lambda_1\lambda_2 = \det$ — a free check.

Study Plan

The course is designed as **six weeks at roughly 6–8 hours per week**. If you have more time, compress it to three weeks by doing two rows of the table per week; if you are new to linear algebra or studying alongside a job, stretch it to ten or twelve weeks and spend extra time on Weeks 1–3. Do not skip the foundations: almost every difficulty people have with the SVD is really a difficulty with orthogonality or with eigenvectors of symmetric matrices.

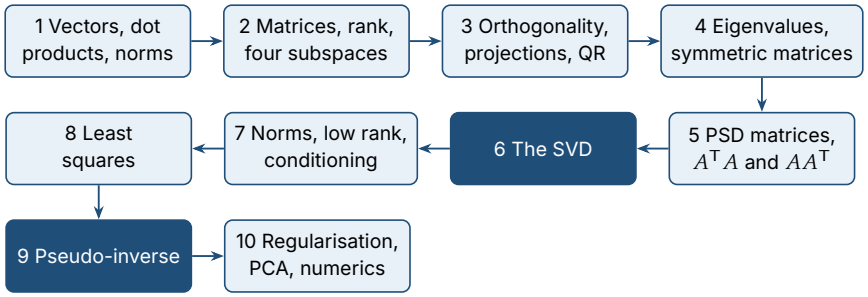


Figure 1: The dependency chain of the course. The two dark boxes are the destinations.

Week-by-week plan

Week	Read (Textbook)	Hand practice	Python Lab	Assignment
1	Ch. 1 Vectors, inner products, norms; Ch. 2 Matrices, rank, four subspaces	Ex. 1.1–1.7, 2.1–2.6; redo Worked Ex. 2.3 without looking	Part 0 (setup, floating point), Part 1 (vectors, matrices, elimination)	**A1** Vectors & matrices
2	Ch. 3 Orthogonal matrices, projections, Gram–Schmidt, QR	Ex. 3.1–3.6; compute one QR fully by hand	Part 2 (Gram–Schmidt, Householder QR, projections)	**A2** Orthogonality & QR
3	Ch. 4 Eigenvalues, spectral theorem; Ch. 5 Positive semidefinite matrices, $A^\mathsf{T}A$	Ex. 4.1–4.6, 5.1–5.6; three power-iteration steps by hand	Part 3 (power iteration, Jacobi eigenvalue method)	**A3** Eigenvalues
4	Ch. 6 The SVD; Ch. 7 Norms, low-rank approximation, conditioning	Ex. 6.1–6.8, 7.1–7.7; four full SVDs by hand	Part 4 (SVD via $A^\mathsf{T}A$, one-sided Jacobi SVD)	**A4** The SVD
5	Ch. 8 Least squares; Ch. 9 The pseudo-inverse	Ex. 8.1–8.7, 9.1–9.9; verify Penrose conditions by hand	Part 5 (pseudo-inverse), Part 6 (least squares)	**A5** Pseudo-inverse & least squares

Week	Read (Textbook)	Hand practice	Python Lab	Assignment
6	Ch. 10 Regularisation, PCA, numerical practice; review	Ex. 10.1–10.6; final self-test below	Part 7 (image compression, PCA)	**A6** Applications

A suggested weekly rhythm

Day 1. Read the chapter once quickly, then again slowly with pen and paper. Redo every worked example with the solution covered.

Day 2. Do the ⋆ and ⋆⋆ exercises. Check against Appendix B only after a genuine attempt.

Day 3. Read the matching part of the Python Lab Manual. Type the code in yourself rather than copying; run it; confirm it reproduces your hand calculations.

Day 4. Start the assignment. Run the tests often — they are designed to tell you which case you got wrong.

Day 5. Finish the assignment, attempt the ⋆ ⋆ ⋆ exercises, and work through the checkpoint below.

Weekly checkpoints

Move on when you can do everything in the checkpoint *without notes*.

Week 1 — Foundations

- Compute dot products, lengths, angles, and the 1-, 2- and ∞-norms of a vector.
- Project a vector onto a line and check that the error is perpendicular to the line.
- Multiply matrices in all four ways, in particular as a sum of outer products.
- Find the rank and a basis for each of the four fundamental subspaces of a 3×4 matrix.

Week 2 — Orthogonality

- Explain why $Q^{\mathsf{T}}Q = I$ means "orthonormal columns" and why $\|Q\boldsymbol{x}\| = \|\boldsymbol{x}\|$.
- Build the projection matrix onto a column space and verify $P^2 = P = P^{\mathsf{T}}$.
- Run Gram–Schmidt by hand on three vectors in $\mathbb{R}^3$ and write the resulting $A = QR$.

Week 3 — Eigenvalues

- Find eigenvalues and eigenvectors of 2×2 and simple 3×3 matrices.
- Write a symmetric matrix as $\sum \lambda_i \boldsymbol{q}_i \boldsymbol{q}_i^{\mathsf{T}}$ and check it.
- Decide whether a symmetric matrix is positive definite, semidefinite or indefinite.
- Prove that $A^{\mathsf{T}}A$ is positive semidefinite and that $A^{\mathsf{T}}A$ and AA^{T} share their non-zero eigenvalues.

Week 4 — The SVD

- Compute the full SVD of a 2×2, a 2×3 and a 3×2 matrix by hand, including the null-space columns.
- Explain why you must compute $\boldsymbol{u}_i = A\boldsymbol{v}_i/\sigma_i$ rather than finding U and V separately.
- State the Eckart–Young theorem and compute the best rank-one approximation of a matrix and its error.
- Read off $\|A\|_2$, $\|A\|_F$, the rank and the condition number from the singular values.

Week 5 — Least squares and A^+

- Fit a line or parabola with the normal equations, with QR, and with the SVD.
- Compute A^+ from the SVD and from the full-rank formulas; verify the four Penrose conditions.
- Explain why $A^+\boldsymbol{b}$ is the *minimum-norm* least-squares solution.

Week 6 — Applications

- Explain why tiny singular values amplify noise, and how truncation and Tikhonov regularisation filter them.
- Perform PCA on a small data set by hand and state the fraction of variance explained.
- Explain why production software never forms $A^{\mathsf{T}}A$ to compute an SVD.

Prerequisite self-check

You need only school algebra: solving two linear equations in two unknowns, working with square roots and fractions, and the formula for the roots of a quadratic. Try these five questions; if any feels hard, review it before Week 1.

1. Solve $2x + y = 5,\ x - y = 1$.
2. Simplify $\sqrt{45}$ and $\dfrac{6}{\sqrt{3}}$.
3. Solve $\lambda^2 - 50\lambda + 225 = 0$.
4. Expand $(a-b)^2$ and use it to show $a^2 + b^2 \geq 2ab$.
5. What is $\cos 60^\circ$? What angle has cosine 0?

Answers: (1) $x = 2,\ y = 1$. (2) $3\sqrt{5}$ and $2\sqrt{3}$. (3) $\lambda = 45$ or $\lambda = 5$ — you will meet this exact equation in Chapter 6. (4) $(a-b)^2 = a^2 - 2ab + b^2 \geq 0$. (5) $\frac{1}{2}$; 90°.

Notation

$\boldsymbol{x}$, $\boldsymbol{b}$, $\boldsymbol{u}$	column vectors (bold lower case); $\boldsymbol{x} \in \mathbb{R}^n$ has entries $x_1, \ldots, x_n$
A, Q, Σ	matrices (upper case); A is $m \times n$: m rows, n columns
A^{T}	transpose of A
$\boldsymbol{x}^{\mathsf{T}}\boldsymbol{y}$, $\boldsymbol{x} \cdot \boldsymbol{y}$	dot (inner) product of $\boldsymbol{x}$ and $\boldsymbol{y}$
$\|\boldsymbol{x}\|$	Euclidean length $\|\boldsymbol{x}\|_2$ unless another norm is indicated
$\mathcal{C}(A)$, $\mathcal{N}(A)$	column space and null space of A
rank A	rank of A, usually written r
I, I_n	identity matrix (of size n)
$\boldsymbol{e}_i$	i-th standard basis vector
$\operatorname{diag}(d_1, \ldots, d_k)$	diagonal matrix with the listed diagonal entries
$\sigma_1 \geq \sigma_2 \geq \ldots$	singular values, always in decreasing order
$A = U\Sigma V^{\mathsf{T}}$	singular value decomposition; $\boldsymbol{u}_i$, $\boldsymbol{v}_i$ are columns of U, V
A^{+}	Moore–Penrose pseudo-inverse of A
$\kappa(A)$	condition number σ_1 / σ_r

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CHAPTER 1

Vectors, Inner Products and Norms

Everything in this course — the SVD, the pseudo-inverse, least squares — is about *lengths* and *angles*. The SVD finds the directions a matrix stretches most; least squares finds the closest point; the pseudo-inverse returns the shortest answer. "Closest", "shortest" and "direction" are all defined through one operation, the dot product. This chapter builds it carefully.

1.1 Vectors and linear combinations

A **vector** in $\mathbb{R}^n$ is an ordered list of n real numbers, written as a column:

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \qquad \text{or, to save space, } \boldsymbol{x} = (x_1, x_2, \ldots, x_n).$$

Vectors are added entry by entry and multiplied by scalars entry by entry. The two operations combine into the most important construction in linear algebra:

Definition 1.1: Linear combination. Given vectors $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_k \in \mathbb{R}^n$ and scalars $c_1, \ldots, c_k$, the vector

$$c_1\boldsymbol{a}_1 + c_2\boldsymbol{a}_2 + \cdots + c_k\boldsymbol{a}_k$$

is a **linear combination** of $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_k$. The set of all linear combinations is the **span** of the vectors.

For example $2\,(1,0,1) - 3\,(0,1,1) = (2,-3,-1)$. The span of one non-zero vector is a line through the origin; the span of two vectors that do not lie on a common line is a plane through the origin.

1.2 The dot product, length and angle

Definition 1.2: Dot product. The **dot product** (or inner product) of $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ is the number

$$\boldsymbol{x} \cdot \boldsymbol{y} = \boldsymbol{x}^{\mathsf{T}} \boldsymbol{y} = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n = \sum_{i=1}^{n} x_i y_i.$$

The notation $\boldsymbol{x}^{\mathsf{T}} \boldsymbol{y}$ treats $\boldsymbol{x}^{\mathsf{T}}$ as a $1 \times n$ row and $\boldsymbol{y}$ as an $n \times 1$ column; the product is 1×1, a number. We use both notations. The dot product is

- **symmetric**: $\boldsymbol{x} \cdot \boldsymbol{y} = \boldsymbol{y} \cdot \boldsymbol{x}$;
- **linear** in each argument: $(\alpha \boldsymbol{x} + \beta \boldsymbol{0}) \cdot \boldsymbol{y} = \alpha\, \boldsymbol{x} \cdot \boldsymbol{y} + \beta\, \boldsymbol{0} \cdot \boldsymbol{y}$;
- **positive**: $\boldsymbol{x} \cdot \boldsymbol{x} = \sum x_i^2 \geq 0$, with equality only for $\boldsymbol{x} = \boldsymbol{0}$.

Definition 1.3: Length and unit vectors. The **length** (Euclidean norm) of $\boldsymbol{x}$ is $\|\boldsymbol{x}\| = \sqrt{\boldsymbol{x} \cdot \boldsymbol{x}} = \sqrt{x_1^2 + \cdots + x_n^2}$. A **unit vector** has length 1. For $\boldsymbol{x} \neq \boldsymbol{0}$ the vector $\boldsymbol{x} / \|\boldsymbol{x}\|$ is the unit vector in the direction of $\boldsymbol{x}$; computing it is called **normalising**.

The dot product measures how much two vectors point in the same direction:

Theorem 1.4: Angle formula. For non-zero $\boldsymbol{u}, \boldsymbol{v}$ the angle $\theta \in [0, \pi]$ between them satisfies

$$\cos\theta = \frac{\boldsymbol{u} \cdot \boldsymbol{v}}{\|\boldsymbol{u}\|\, \|\boldsymbol{v}\|}.$$

For the plane this is the law of cosines: $\|\boldsymbol{u} - \boldsymbol{v}\|^2 = \|\boldsymbol{u}\|^2 + \|\boldsymbol{v}\|^2 - 2\boldsymbol{u} \cdot \boldsymbol{v}$ and also $= \|\boldsymbol{u}\|^2 + \|\boldsymbol{v}\|^2 - 2\,\|\boldsymbol{u}\|\,\|\boldsymbol{v}\| \cos\theta$. In $\mathbb{R}^n$ we *define* the angle by this formula — which only makes sense if the right-hand side lies in $[-1, 1]$. That is the content of the most useful inequality in the subject.

Theorem 1.5: Cauchy–Schwarz inequality. For all $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$: $|\boldsymbol{u} \cdot \boldsymbol{v}| \leq \|\boldsymbol{u}\|\, \|\boldsymbol{v}\|$, with equality exactly when one vector is a multiple of the other.

Proof. If $\boldsymbol{v} = \boldsymbol{0}$ both sides are 0. Otherwise, for every real t,

$$0 \leq \|\boldsymbol{u} - t\boldsymbol{v}\|^2 = \|\boldsymbol{u}\|^2 - 2t\, \boldsymbol{u} \cdot \boldsymbol{v} + t^2 \|\boldsymbol{v}\|^2.$$

Choose $t = \boldsymbol{u} \cdot \boldsymbol{v} / \|\boldsymbol{v}\|^2$ (the value minimising the right side). Then $0 \leq \|\boldsymbol{u}\|^2 - (\boldsymbol{u} \cdot \boldsymbol{v})^2 / \|\boldsymbol{v}\|^2$, which rearranges to the inequality. Equality forces $\boldsymbol{u} - t\boldsymbol{v} = \boldsymbol{0}$. □

A consequence is the **triangle inequality** $\|\boldsymbol{u} + \boldsymbol{v}\| \leq \|\boldsymbol{u}\| + \|\boldsymbol{v}\|$: expand $\|\boldsymbol{u} + \boldsymbol{v}\|^2 = \|\boldsymbol{u}\|^2 + 2\boldsymbol{u} \cdot \boldsymbol{v} + \|\boldsymbol{v}\|^2$ and bound the middle term by Cauchy–Schwarz.

Worked Example 1.1: Dot product, lengths and angle.

Let $\boldsymbol{u} = (1, 2, 2)$ and $\boldsymbol{v} = (2, 0, 1)$.

Dot product. $\boldsymbol{u} \cdot \boldsymbol{v} = 1 \cdot 2 + 2 \cdot 0 + 2 \cdot 1 = 4$.

Lengths. $\|\boldsymbol{u}\| = \sqrt{1+4+4} = 3, \quad \|\boldsymbol{v}\| = \sqrt{4+0+1} = \sqrt{5}$.

Angle. $\cos\theta = \dfrac{4}{3\sqrt{5}} = \dfrac{4\sqrt{5}}{15} \approx 0.5963$, so $\theta \approx 53.40°$.

Check. Cauchy–Schwarz says $4 \le 3\sqrt{5} \approx 6.71$. ✓

Unit vector. $\boldsymbol{u}/\|\boldsymbol{u}\| = (\frac{1}{3}, \frac{2}{3}, \frac{2}{3})$; its length is $\sqrt{1/9 + 4/9 + 4/9} = 1$. ✓

1.3 Norms

Length is one way to measure size; there are others, and they appear in numerical work.

Definition 1.6: Norm. A **norm** on $\mathbb{R}^n$ is a function $\|\cdot\|$ with, for all $\boldsymbol{x}, \boldsymbol{y}$ and scalars α: (i) $\|\boldsymbol{x}\| \ge 0$, with equality only for $\boldsymbol{x} = \boldsymbol{0}$; (ii) $\|\alpha\boldsymbol{x}\| = |\alpha|\,\|\boldsymbol{x}\|$; (iii) $\|\boldsymbol{x} + \boldsymbol{y}\| \le \|\boldsymbol{x}\| + \|\boldsymbol{y}\|$.

The three you will meet are

$$\|\boldsymbol{x}\|_1 = \sum_i |x_i|, \qquad \|\boldsymbol{x}\|_2 = \Big(\sum_i x_i^2\Big)^{1/2}, \qquad \|\boldsymbol{x}\|_\infty = \max_i |x_i|.$$

Their "unit balls" $\{\boldsymbol{x} : \|\boldsymbol{x}\| \le 1\}$ in the plane are a diamond, a disc and a square (Figure 1.1). They always satisfy $\|\boldsymbol{x}\|_\infty \le \|\boldsymbol{x}\|_2 \le \|\boldsymbol{x}\|_1$.

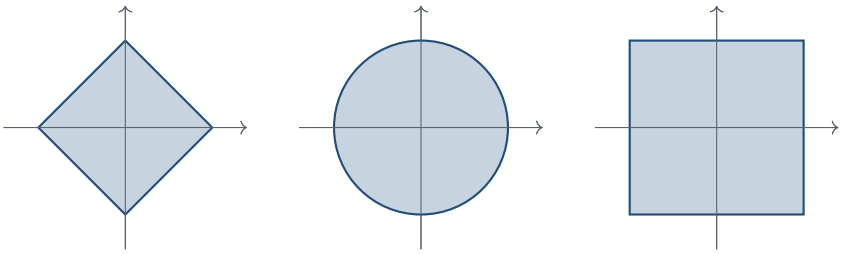


Figure 1.1: Unit balls of the 1-, 2- and ∞-norms in $\mathbb{R}^2$.

Worked Example 1.2: Three norms of one vector.

For $\boldsymbol{x} = (3, -4, 0)$: $\|\boldsymbol{x}\|_1 = 3 + 4 + 0 = 7$, $\|\boldsymbol{x}\|_2 = \sqrt{9 + 16 + 0} = 5$, $\|\boldsymbol{x}\|_\infty = \max(3, 4, 0) = 4$. Indeed $4 \le 5 \le 7$.

Unless stated otherwise, $\|\cdot\|$ means $\|\cdot\|_2$ in this book: it is the only one of the three that comes from a dot product, and therefore the only one with angles and orthogonality — which is what the SVD is built on.

1.4 Orthogonality and orthonormal bases

Definition 1.7: Orthogonal and orthonormal. Vectors u, v are **orthogonal**, written $u \perp v$, if $u \cdot v = 0$. A set $\{q_1, \dots, q_k\}$ is **orthonormal** if $q_i \cdot q_j = 0$ for $i \neq j$ and $q_i \cdot q_i = 1$. Compactly: $q_i^{\mathsf{T}} q_j = \delta_{ij}$, where δ_{ij} is 1 if $i = j$ and 0 otherwise.

For orthogonal vectors, $\|u + v\|^2 = \|u\|^2 + 2u \cdot v + \|v\|^2 = \|u\|^2 + \|v\|^2$: **Pythagoras' theorem** in any dimension.

Orthonormal vectors are automatically linearly independent, and they make coordinates trivial to compute:

Theorem 1.8: Coordinates in an orthonormal basis. If $q_1, \dots, q_n$ is an orthonormal basis of $\mathbb{R}^n$, then every $x \in \mathbb{R}^n$ satisfies

$$x = (q_1 \cdot x)\, q_1 + (q_2 \cdot x)\, q_2 + \cdots + (q_n \cdot x)\, q_n, \qquad \|x\|^2 = \sum_i (q_i \cdot x)^2.$$

Proof. Write $x = \sum_j c_j q_j$ and dot both sides with q_i: every term with $j \neq i$ vanishes, leaving $q_i \cdot x = c_i$. The length formula is Pythagoras applied repeatedly. □

Key idea. In an orthonormal basis, the i-th coordinate of x is just $q_i \cdot x$. No system of equations to solve. The SVD gives orthonormal bases adapted to a matrix, and this is why every formula involving the SVD is so simple.

Worked Example 1.3: Coordinates in a rotated basis.

Take $q_1 = \frac{1}{\sqrt{2}}(1, 1)$, $q_2 = \frac{1}{\sqrt{2}}(1, -1)$ and $x = (3, 1)$.

Check orthonormal. $q_1 \cdot q_2 = \frac{1}{2}(1 - 1) = 0$, $q_1 \cdot q_1 = \frac{1}{2}(1 + 1) = 1$, and likewise $q_2 \cdot q_2 = 1$.

Coordinates. $c_1 = q_1 \cdot x = \frac{4}{\sqrt{2}} = 2\sqrt{2}$, $\quad c_2 = q_2 \cdot x = \frac{2}{\sqrt{2}} = \sqrt{2}$.

Reassemble. $2\sqrt{2} \cdot \frac{1}{\sqrt{2}}(1, 1) + \sqrt{2} \cdot \frac{1}{\sqrt{2}}(1, -1) = (2, 2) + (1, -1) = (3, 1)$. ✓

Length. $c_1^2 + c_2^2 = 8 + 2 = 10 = 3^2 + 1^2$. ✓

1.5 Projection onto a line

Given a non-zero vector $\boldsymbol{a}$ and any $\boldsymbol{b}$, which multiple $\hat{t}\,\boldsymbol{a}$ is closest to $\boldsymbol{b}$? Geometrically the error $\boldsymbol{e} = \boldsymbol{b} - \hat{t}\boldsymbol{a}$ must be *perpendicular* to $\boldsymbol{a}$ (Figure 1.2). So

$$\boldsymbol{a} \cdot (\boldsymbol{b} - \hat{t}\boldsymbol{a}) = 0 \quad \Longrightarrow \quad \hat{t} = \frac{\boldsymbol{a} \cdot \boldsymbol{b}}{\boldsymbol{a} \cdot \boldsymbol{a}}.$$

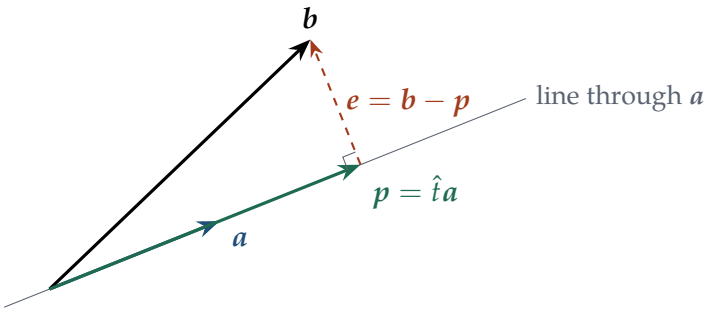


Figure 1.2: Projecting $\boldsymbol{b}$ onto the line through $\boldsymbol{a}$. The error $\boldsymbol{e}$ is perpendicular to $\boldsymbol{a}$.

Theorem 1.9: Projection onto a line. The point on the line through $\boldsymbol{a} \neq \boldsymbol{0}$ closest to $\boldsymbol{b}$ is

$$\boldsymbol{p} = \frac{\boldsymbol{a} \cdot \boldsymbol{b}}{\boldsymbol{a} \cdot \boldsymbol{a}}\,\boldsymbol{a} = \frac{\boldsymbol{a}\boldsymbol{a}^{\mathsf{T}}}{\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a}}\,\boldsymbol{b}.$$

The error $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p}$ is orthogonal to $\boldsymbol{a}$, and $\|\boldsymbol{b}\|^2 = \|\boldsymbol{p}\|^2 + \|\boldsymbol{e}\|^2$.

The second form is the first hint of what is to come: the matrix $P = \boldsymbol{a}\boldsymbol{a}^{\mathsf{T}}/(\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a})$ — a column times a row — *does* the projection. Matrices of the form "column times row" are called *outer products*, and the SVD writes every matrix as a sum of them.

Worked Example 1.4: Projecting onto a line.

Project $\boldsymbol{b} = (1,1,1)$ onto the line through $\boldsymbol{a} = (1,2,2)$.

Coefficient. $\boldsymbol{a} \cdot \boldsymbol{b} = 5$, $\boldsymbol{a} \cdot \boldsymbol{a} = 9$, so $\hat{t} = \frac{5}{9}$.

Projection. $\boldsymbol{p} = \frac{5}{9}(1,2,2) = \left(\frac{5}{9}, \frac{10}{9}, \frac{10}{9}\right)$.

Error. $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p} = \left(\frac{4}{9}, -\frac{1}{9}, -\frac{1}{9}\right)$.

Check $\boldsymbol{e} \perp \boldsymbol{a}$. $\frac{1}{9}(4 - 2 - 2) = 0$. ✓

Check Pythagoras. $\|\boldsymbol{p}\|^2 = \frac{25}{81} \cdot 9 = \frac{25}{9}$, $\|\boldsymbol{e}\|^2 = \frac{16+1+1}{81} = \frac{2}{9}$; sum $= \frac{27}{9} = 3 = \|\boldsymbol{b}\|^2$. ✓

Projection matrix. $P = \dfrac{1}{9}\begin{bmatrix}1\\2\\2\end{bmatrix}\begin{bmatrix}1 & 2 & 2\end{bmatrix} = \dfrac{1}{9}\begin{bmatrix}1 & 2 & 2\\2 & 4 & 4\\2 & 4 & 4\end{bmatrix}$, and $P\boldsymbol{b} = \frac{1}{9}(5,10,10) = \boldsymbol{p}$. ✓

Common pitfall. $\boldsymbol{a}^{\mathsf{T}}\boldsymbol{b}$ is a number (an inner product), while $\boldsymbol{a}\boldsymbol{b}^{\mathsf{T}}$ is an $n \times n$ matrix (an outer product). Check the shapes: $(1 \times n)(n \times 1) = 1 \times 1$ but $(n \times 1)(1 \times n) = n \times n$.

Exercises

Exercise 1.1 $\star$ Let $\boldsymbol{u} = (1, -2, 2)$ and $\boldsymbol{v} = (2, 1, 0)$. Compute $\boldsymbol{u} \cdot \boldsymbol{v}$, $\|\boldsymbol{u}\|$, $\|\boldsymbol{v}\|$ and the angle between $\boldsymbol{u}$ and $\boldsymbol{v}$. Normalise both vectors.

Exercise 1.2 $\star$ Compute $\|\boldsymbol{x}\|_1$, $\|\boldsymbol{x}\|_2$ and $\|\boldsymbol{x}\|_\infty$ for $\boldsymbol{x} = (3, -4, 12)$ and check that $\|\boldsymbol{x}\|_\infty \le \|\boldsymbol{x}\|_2 \le \|\boldsymbol{x}\|_1$.

Exercise 1.3 $\star$ Project $\boldsymbol{b} = (2, 3, 4)$ onto the line through $\boldsymbol{a} = (1, 0, 1)$. Give $\boldsymbol{p}$ and $\boldsymbol{e}$, and verify $\boldsymbol{e} \perp \boldsymbol{a}$ and $\|\boldsymbol{b}\|^2 = \|\boldsymbol{p}\|^2 + \|\boldsymbol{e}\|^2$.

Exercise 1.4 $\star\star$ Find a unit vector orthogonal to both $(1, 1, 0)$ and $(0, 1, 1)$. How many such unit vectors are there?

Exercise 1.5 $\star\star$ Verify the Cauchy–Schwarz inequality for $\boldsymbol{u} = (1, 2, 3)$, $\boldsymbol{v} = (4, 5, 6)$ and compute the angle between them.

Exercise 1.6 $\star\star$ Let $\boldsymbol{q}_1 = \frac{1}{3}(1, 2, 2)$, $\boldsymbol{q}_2 = \frac{1}{3}(2, 1, -2)$, $\boldsymbol{q}_3 = \frac{1}{3}(2, -2, 1)$. Show these form an orthonormal basis of $\mathbb{R}^3$, and write $\boldsymbol{x} = (3, 3, 3)$ in this basis using Theorem 1.8.

Exercise 1.7 $\star\star\star$ (a) Prove that if $\boldsymbol{u} \perp \boldsymbol{v}$ then $\|\boldsymbol{u} + \boldsymbol{v}\|^2 = \|\boldsymbol{u}\|^2 + \|\boldsymbol{v}\|^2$. (b) For unit vectors $\boldsymbol{u}, \boldsymbol{v}$ with angle θ show $\|\boldsymbol{u} - \boldsymbol{v}\|^2 = 2 - 2\cos\theta$. (c) Prove the triangle inequality from Cauchy–Schwarz.

CHAPTER 2

Matrices, Rank and the Four Fundamental Subspaces

A matrix is a machine that turns vectors into vectors. This chapter studies what that machine does: which outputs it can produce, which inputs it destroys, and how its rows and columns are related. The answer — four subspaces with a beautiful orthogonality between them — is exactly the structure the SVD will reveal with orthonormal bases in Chapter 6.

2.1 Matrix–vector multiplication: two pictures

An $m \times n$ matrix A has m rows and n columns. It multiplies $\boldsymbol{x} \in \mathbb{R}^n$ and produces $A\boldsymbol{x} \in \mathbb{R}^m$. There are two ways to compute $A\boldsymbol{x}$, and you need both.

Column picture. $A\boldsymbol{x}$ is a linear combination of the columns of A, with weights from $\boldsymbol{x}$:

$$A\boldsymbol{x} = \begin{bmatrix} | & & | \\ \boldsymbol{a}_1 & \cdots & \boldsymbol{a}_n \\ | & & | \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = x_1\boldsymbol{a}_1 + \cdots + x_n\boldsymbol{a}_n.$$

Row picture. The i-th entry of $A\boldsymbol{x}$ is the dot product of row i of A with $\boldsymbol{x}$.

Worked Example 2.1: Both pictures.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}, \boldsymbol{x} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

Column picture. $A\boldsymbol{x} = 2\begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} - 1\begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 4 \end{bmatrix}.$

Row picture. $(1,2)\cdot(2,-1) = 0$, $(3,4)\cdot(2,-1) = 2$, $(5,6)\cdot(2,-1) = 4$. Same answer. ✓

The column picture tells us immediately which vectors $\boldsymbol{b}$ can be written as $A\boldsymbol{x}$: exactly the linear combinations of the columns. The row picture tells us when $A\boldsymbol{x} = \boldsymbol{0}$: exactly when $\boldsymbol{x}$ is orthogonal to every row. Keep these two sentences in mind; they are the whole of Section 2.5.

Multiplication by A is **linear**: $A(\alpha\boldsymbol{x} + \beta\boldsymbol{y}) = \alpha A\boldsymbol{x} + \beta A\boldsymbol{y}$. Conversely, every linear map $\mathbb{R}^n \to \mathbb{R}^m$ is multiplication by some matrix (column j is the image of $\boldsymbol{e}_j$). Some 2×2 examples:

Matrix	Effect on the plane	Preserves lengths?
$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$	rotation by θ	yes
$\begin{bmatrix} 3 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$	stretch x by 3, squash y by $\frac{1}{2}$	no
$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	reflection in the x-axis	yes
$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	projection onto the x-axis	no
$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$	shear	no

One of the remarkable consequences of the SVD is that *every* matrix is a rotation (or reflection), followed by a stretch along the axes, followed by another rotation.

2.2 Matrix multiplication: four views

If A is $m \times n$ and B is $n \times p$, then AB is $m \times p$. There are four equivalent ways to compute it:

1. **Entry by entry:** $(AB)_{ij} = (\text{row } i \text{ of } A) \cdot (\text{column } j \text{ of } B) = \sum_k a_{ik} b_{kj}$.
2. **Column by column:** column j of AB is A times column j of B.
3. **Row by row:** row i of AB is row i of A times B.
4. **Sum of outer products:** $AB = \sum_{k=1}^{n} (\text{column } k \text{ of } A)(\text{row } k \text{ of } B)$.

View 4 deserves attention. An **outer product** $\boldsymbol{u}\boldsymbol{v}^{\mathsf{T}}$ of $\boldsymbol{u} \in \mathbb{R}^m$ and $\boldsymbol{v} \in \mathbb{R}^p$ is the $m \times p$ matrix with entries $u_i v_j$. Every column of $\boldsymbol{u}\boldsymbol{v}^{\mathsf{T}}$ is a multiple of $\boldsymbol{u}$, and every row is a multiple of $\boldsymbol{v}^{\mathsf{T}}$ — it is the simplest possible non-zero matrix, a **rank-one** matrix. View 4 says a product of matrices is a sum of rank-one matrices. The SVD will say the same of every single matrix: $A = \sigma_1 \boldsymbol{u}_1 \boldsymbol{v}_1^{\mathsf{T}} + \cdots + \sigma_r \boldsymbol{u}_r \boldsymbol{v}_r^{\mathsf{T}}$.

Worked Example 2.2: Multiplication as a sum of outer products.

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 3 & 1 \\ 1 & 0 \end{bmatrix}.$$

Entry by entry. $AB = \begin{bmatrix} 1\cdot 3 + 2\cdot 1 & 1\cdot 1 + 2\cdot 0 \\ 0\cdot 3 + 1\cdot 1 & 0\cdot 1 + 1\cdot 0 \end{bmatrix} = \begin{bmatrix} 5 & 1 \\ 1 & 0 \end{bmatrix}$.

Outer products. Column 1 of A times row 1 of B plus column 2 times row 2:

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 3 & 1 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 5 & 1 \\ 1 & 0 \end{bmatrix}. \checkmark$$

2.3 The transpose and symmetric matrices

The **transpose** A^{T} turns rows into columns: $(A^{\mathsf{T}})_{ij} = a_{ji}$. Its key rules are

$$(A^{\mathsf{T}})^{\mathsf{T}} = A, \qquad (A+B)^{\mathsf{T}} = A^{\mathsf{T}} + B^{\mathsf{T}}, \qquad (AB)^{\mathsf{T}} = B^{\mathsf{T}} A^{\mathsf{T}} \text{ (order reverses)}.$$

The transpose is tied to the dot product by the identity

$$(A\boldsymbol{x}) \cdot \boldsymbol{y} = (A\boldsymbol{x})^{\mathsf{T}} \boldsymbol{y} = \boldsymbol{x}^{\mathsf{T}} A^{\mathsf{T}} \boldsymbol{y} = \boldsymbol{x} \cdot (A^{\mathsf{T}} \boldsymbol{y}), \tag{2.1}$$

which we will use constantly. A square matrix is **symmetric** if $S^{\mathsf{T}} = S$. For any matrix A, both $A^{\mathsf{T}}A$ ($n \times n$) and AA^{T} ($m \times m$) are symmetric, since $(A^{\mathsf{T}}A)^{\mathsf{T}} = A^{\mathsf{T}}(A^{\mathsf{T}})^{\mathsf{T}} = A^{\mathsf{T}}A$. These two matrices are the doorway to the SVD.

2.4 Independence, basis and dimension

Definition 2.1: Independence, basis, dimension. Vectors $\boldsymbol{a}_1, \dots, \boldsymbol{a}_k$ are **linearly independent** if the only combination $c_1\boldsymbol{a}_1 + \cdots + c_k\boldsymbol{a}_k$ equal to $\mathbf{0}$ is the one with all $c_i = 0$. A **subspace** is a set of vectors closed under addition and scalar multiplication (lines and planes through the origin, for example). A **basis** of a subspace is an independent set that spans it, and every basis has the same number of vectors: the **dimension**.

For example, $(1,0,1)$, $(0,1,1)$ and $(1,1,2)$ are *dependent*, since the third is the sum of the first two; they span only a plane, of dimension 2.

2.5 The four fundamental subspaces

Definition 2.2: The four fundamental subspaces. For an $m \times n$ matrix A:

- the **column space** $\mathcal{C}(A) = \{A\boldsymbol{x} : \boldsymbol{x} \in \mathbb{R}^n\} \subseteq \mathbb{R}^m$ — all combinations of the columns;
- the **null space** $\mathcal{N}(A) = \{\boldsymbol{x} : A\boldsymbol{x} = \mathbf{0}\} \subseteq \mathbb{R}^n$;
- the **row space** $\mathcal{C}(A^\mathsf{T}) \subseteq \mathbb{R}^n$ — all combinations of the rows;
- the **left null space** $\mathcal{N}(A^\mathsf{T}) = \{\boldsymbol{y} : A^\mathsf{T}\boldsymbol{y} = \mathbf{0}\} \subseteq \mathbb{R}^m$.

The **rank** $r = \operatorname{rank} A$ is the dimension of the column space.

Theorem 2.3: The fundamental theorem of linear algebra. Let A be $m \times n$ with rank r. Then

1. $\dim \mathcal{C}(A) = \dim \mathcal{C}(A^\mathsf{T}) = r$ (row rank equals column rank);
2. $\dim \mathcal{N}(A) = n - r$ and $\dim \mathcal{N}(A^\mathsf{T}) = m - r$;
3. $\mathcal{N}(A) \perp \mathcal{C}(A^\mathsf{T})$ in $\mathbb{R}^n$, and $\mathcal{N}(A^\mathsf{T}) \perp \mathcal{C}(A)$ in $\mathbb{R}^m$.

Proof of part 3. If $A\boldsymbol{x} = \mathbf{0}$ then, by the row picture, $\boldsymbol{x}$ is orthogonal to every row of A, hence to every combination of rows — the whole row space. Apply the same argument to A^T for the second statement. □

Parts 1 and 2 follow from elimination (below). Part 2 for $\mathcal{N}(A)$ is often called the **rank–nullity theorem**: $r + \dim \mathcal{N}(A) = n$. Figure 2.1 shows the picture to carry in your head. The input space $\mathbb{R}^n$ splits into two perpendicular pieces, the row space and the null space; A sends the null space to $\mathbf{0}$ and sends the row space onto the column space.

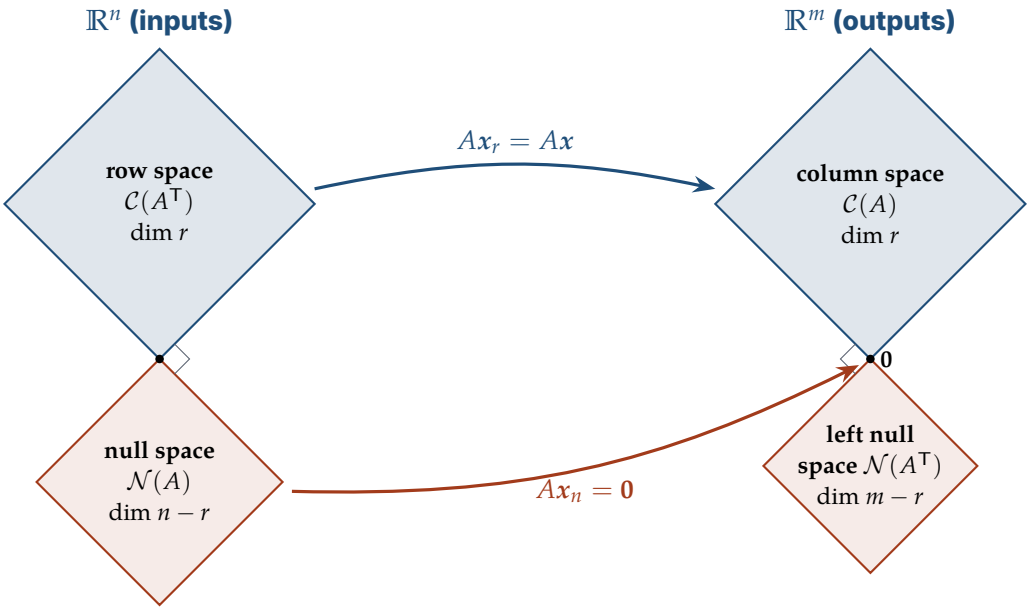


Figure 2.1: The four fundamental subspaces. In $\mathbb{R}^n$ the row space and null space are perpendicular; in $\mathbb{R}^m$ the column space and left null space are perpendicular. A maps the row space onto the column space and the null space to $\mathbf{0}$.

Finding the subspaces by elimination

Row-reduce A to its **reduced row echelon form** R (Gauss–Jordan elimination). Row operations do not change the row space or the null space. Then:

- the number of pivots is the rank r;
- the non-zero rows of R are a basis of the row space;
- the *pivot columns of the original* A are a basis of the column space;
- setting one free variable to 1 and the others to 0 and solving $R\boldsymbol{x} = \mathbf{0}$ gives one **special solution**; the $n - r$ special solutions are a basis of $\mathcal{N}(A)$;
- $\mathcal{N}(A^\mathsf{T})$ is found the same way from A^T (or by spotting which combination of rows of A vanishes).

Worked Example 2.3: All four subspaces of a 3×3 matrix.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{bmatrix}.$$

Eliminate. $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - 3R_1$ give $\left[\begin{smallmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{smallmatrix}\right]$; then $R_3 \leftarrow R_3 - R_2$ and $R_1 \leftarrow R_1 - 3R_2$ give

$$R = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Rank. Two pivots (columns 1 and 3), so $r = 2$.

Row space. Basis $(1, 2, 0)$ and $(0, 0, 1)$.

Column space. Pivot columns of A: $(1,2,3)$ and $(3,7,10)$.

Null space. Free variable x_2; with $x_2 = 1$: $x_3 = 0$ and $x_1 = -2$. Basis $(-2,1,0)$. Dimension $n - r = 1$. ✓

Left null space. Row 3 of A equals row 1 plus row 2, so $1 \cdot \text{row}_1 + 1 \cdot \text{row}_2 - 1 \cdot \text{row}_3 = \mathbf{0}$: $\boldsymbol{y} = (1,1,-1)$ satisfies $A^{\mathsf{T}}\boldsymbol{y} = \mathbf{0}$. Dimension $m - r = 1$. ✓

Check orthogonality. $(-2,1,0) \cdot (1,2,0) = 0$ and $(-2,1,0) \cdot (0,0,1) = 0$; $(1,1,-1) \cdot (1,2,3) = 0$ and $(1,1,-1) \cdot (3,7,10) = 0$. ✓

2.6 Inverses — and when they fail

A square matrix A is **invertible** if there is a matrix A^{-1} with $A^{-1}A = AA^{-1} = I$. For an $n \times n$ matrix the following are equivalent: A is invertible; $\operatorname{rank} A = n$; $\mathcal{N}(A) = \{\mathbf{0}\}$; the columns are independent; $\det A \neq 0$. For 2×2 matrices,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Products invert in reverse order: $(AB)^{-1} = B^{-1}A^{-1}$, and $(A^{\mathsf{T}})^{-1} = (A^{-1})^{\mathsf{T}}$.

But most matrices in applications are *not* invertible: they are rectangular, or square but singular. Then $A\boldsymbol{x} = \boldsymbol{b}$ may have no solution (if $\boldsymbol{b} \notin \mathcal{C}(A)$) or infinitely many (if $\mathcal{N}(A) \neq \{\mathbf{0}\}$), or both. The pseudo-inverse of Chapter 9 handles both problems at once: it picks the *best* approximate solution and, among those, the *shortest*. The four subspaces tell us exactly what is going wrong: components of $\boldsymbol{b}$ in $\mathcal{N}(A^{\mathsf{T}})$ can never be reached, and components of $\boldsymbol{x}$ in $\mathcal{N}(A)$ are invisible to A.

Exercises

Exercise 2.1 ⋆ Compute $A\boldsymbol{x}$ by the column picture and by the row picture, where $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$ and $\boldsymbol{x} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

Exercise 2.2 ⋆ Compute AB for $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$ as a sum of two outer products, and check the result entry by entry.

Exercise 2.3 ⋆⋆ Find the rank of $A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 4 \\ 3 & 6 & 1 & 5 \end{bmatrix}$ and a basis for each of its four fundamental subspaces. Check the dimensions against Theorem 2.3 and verify the two orthogonality relations.

Exercise 2.4 ⋆⋆ (a) Show that $A^{\mathsf{T}}A$ is symmetric for every matrix A. (b) Show that $\boldsymbol{x}^{\mathsf{T}}A^{\mathsf{T}}A\boldsymbol{x} = \|A\boldsymbol{x}\|^2$. (c) Compute $A^{\mathsf{T}}A$ and AA^{T} for $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}$.

Exercise 2.5 ⋆⋆ Show that $M = \begin{bmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \end{bmatrix}$ has rank one by writing it as an outer product $\boldsymbol{u}\boldsymbol{v}^{\mathsf{T}}$. What are its four fundamental subspaces?

Exercise 2.6 ⋆⋆⋆ Prove that $\operatorname{rank}(AB) \le \min(\operatorname{rank} A, \operatorname{rank} B)$. *Hint:* relate $\mathcal{C}(AB)$ to $\mathcal{C}(A)$ and the row space of AB to the row space of B.

CHAPTER 3

Orthogonality: Orthogonal Matrices, Projections and QR

The U and V in $A = U\Sigma V^\mathsf{T}$ are orthogonal matrices, and the pseudo-inverse is defined through projections. This chapter develops both. It ends with Gram–Schmidt and the QR factorisation, the workhorse for turning any basis into an orthonormal one.

3.1 Orthonormal columns and orthogonal matrices

Put orthonormal vectors $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n \in \mathbb{R}^m$ into the columns of an $m \times n$ matrix Q. Entry (i, j) of $Q^\mathsf{T}Q$ is $\boldsymbol{q}_i^\mathsf{T}\boldsymbol{q}_j = \delta_{ij}$, so

$$\boxed{Q^\mathsf{T}Q = I_n} \qquad \text{(orthonormal columns)}.$$

Definition 3.1: Orthogonal matrix. A *square* matrix Q with $Q^\mathsf{T}Q = I$ is called an **orthogonal matrix**. Then $Q^{-1} = Q^\mathsf{T}$, and also $QQ^\mathsf{T} = I$, so the rows are orthonormal too.

Theorem 3.2: Orthogonal matrices preserve geometry. If Q has orthonormal columns then for all $\boldsymbol{x}, \boldsymbol{y}$:

$$(Q\boldsymbol{x}) \cdot (Q\boldsymbol{y}) = \boldsymbol{x} \cdot \boldsymbol{y}, \qquad \|Q\boldsymbol{x}\| = \|\boldsymbol{x}\|.$$

Proof. $(Q\boldsymbol{x})^\mathsf{T}(Q\boldsymbol{y}) = \boldsymbol{x}^\mathsf{T}Q^\mathsf{T}Q\boldsymbol{y} = \boldsymbol{x}^\mathsf{T}\boldsymbol{y}$. Take $\boldsymbol{y} = \boldsymbol{x}$ for the length. □

So orthogonal matrices are *rigid motions*: rotations and reflections. Standard examples:

- **Rotation** $Q = \left[\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right]$; indeed $Q^\mathsf{T}Q = \left[\begin{smallmatrix} \cos^2\theta+\sin^2\theta & 0 \\ 0 & \sin^2\theta+\cos^2\theta \end{smallmatrix}\right] = I$.
- **Permutation** matrices, which reorder the entries of a vector.
- **Householder reflections** $H = I - 2\boldsymbol{u}\boldsymbol{u}^\mathsf{T}$ with $\|\boldsymbol{u}\| = 1$. H is symmetric and $H^\mathsf{T}H = H^2 = I - 4\boldsymbol{u}\boldsymbol{u}^\mathsf{T} + 4\boldsymbol{u}(\boldsymbol{u}^\mathsf{T}\boldsymbol{u})\boldsymbol{u}^\mathsf{T} = I$. It reflects across the plane perpendicular to $\boldsymbol{u}$. With $\boldsymbol{u} = \frac{1}{\sqrt{2}}(1, 1)$: $H = \left[\begin{smallmatrix} 0 & -1 \\ -1 & 0 \end{smallmatrix}\right]$, the reflection across the line $y = -x$.

Common pitfall. If Q is *rectangular* ($m > n$) with orthonormal columns, then $Q^\mathsf{T}Q = I_n$ but $QQ^\mathsf{T} \neq I_m$. For example $Q = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ has $Q^\mathsf{T}Q = [1]$ but $QQ^\mathsf{T} = \left[\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right]$. The matrix QQ^T is not the identity — it is a *projection*, as we now show.

3.2 Projection onto a subspace

Let A be $m \times n$ with *independent* columns and let $\boldsymbol{b} \in \mathbb{R}^m$. The point $\boldsymbol{p} = A\hat{\boldsymbol{x}}$ in $\mathcal{C}(A)$ closest to $\boldsymbol{b}$ is characterised, as for a line, by the error $\boldsymbol{e} = \boldsymbol{b} - A\hat{\boldsymbol{x}}$ being perpendicular to the whole column space — that is, to every column of A:

$$A^{\mathsf{T}}(\boldsymbol{b} - A\hat{\boldsymbol{x}}) = \boldsymbol{0} \quad \Longleftrightarrow \quad \boxed{A^{\mathsf{T}}A\,\hat{\boldsymbol{x}} = A^{\mathsf{T}}\boldsymbol{b}}$$

These are the **normal equations**. Since the columns are independent, $A^{\mathsf{T}}A$ is invertible (Chapter 5 proves this), and

$$\hat{\boldsymbol{x}} = (A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}\boldsymbol{b}, \qquad \boldsymbol{p} = A\hat{\boldsymbol{x}} = P\boldsymbol{b}, \qquad P = A(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}.$$

Theorem 3.3: Projection matrices. $P = A(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}$ is the orthogonal projection onto $\mathcal{C}(A)$. It satisfies $P^2 = P$ and $P^{\mathsf{T}} = P$. The matrix $I - P$ is the projection onto the orthogonal complement $\mathcal{N}(A^{\mathsf{T}})$. If the columns of Q are an orthonormal basis of $\mathcal{C}(A)$, then $P = QQ^{\mathsf{T}}$.

Proof. $P^2 = A(A^{\mathsf{T}}A)^{-1}(A^{\mathsf{T}}A)(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}} = P$, and $P^{\mathsf{T}} = A\big((A^{\mathsf{T}}A)^{-1}\big)^{\mathsf{T}}A^{\mathsf{T}} = P$ because $A^{\mathsf{T}}A$ is symmetric. If A has orthonormal columns Q, then $Q^{\mathsf{T}}Q = I$ and the formula collapses to QQ^{T}. Finally $(I - P)\boldsymbol{b} = \boldsymbol{e}$ is the error, which lies in $\mathcal{N}(A^{\mathsf{T}})$. $\square$

Key idea. Projecting onto a subspace with an *orthonormal* basis $\boldsymbol{q}_1, \dots, \boldsymbol{q}_k$ is easy: $P\boldsymbol{b} = QQ^{\mathsf{T}}\boldsymbol{b} = \sum_{i=1}^{k} (\boldsymbol{q}_i \cdot \boldsymbol{b})\,\boldsymbol{q}_i$. Just add up the projections onto each basis vector. In Chapter 9 we will see that $AA^{+} = U_r U_r^{\mathsf{T}}$ is precisely this, with the first r left singular vectors.

Worked Example 3.1: Projecting onto a plane.

Project $\boldsymbol{b} = (6, 0, 0)$ onto the column space of $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$.

Normal equations. $A^{\mathsf{T}}A = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}$, $A^{\mathsf{T}}\boldsymbol{b} = \begin{bmatrix} 6 \\ 0 \end{bmatrix}$.

Solve. $\det(A^{\mathsf{T}}A) = 15 - 9 = 6$, so $(A^{\mathsf{T}}A)^{-1} = \frac{1}{6}\begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix}$ and $\hat{\boldsymbol{x}} = \frac{1}{6}\begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix}\begin{bmatrix} 6 \\ 0 \end{bmatrix} = \begin{bmatrix} 5 \\ -3 \end{bmatrix}$.

Projection and error. $\boldsymbol{p} = A\hat{\boldsymbol{x}} = (5, 2, -1)$, $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p} = (1, -2, 1)$.

Check $e \perp \mathcal{C}(A)$. $A^{\mathsf{T}}\boldsymbol{e} = (1 - 2 + 1,\ 0 - 2 + 2) = (0, 0)$. ✓

Projection matrix. $P = A(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}} = \dfrac{1}{6}\begin{bmatrix} 5 & 2 & -1 \\ 2 & 2 & 2 \\ -1 & 2 & 5 \end{bmatrix}$; then $P\boldsymbol{b} = \frac{1}{6}(30, 12, -6) = (5, 2, -1)$. ✓ P is symmetric and you can check that $P^2 = P$.

We will meet this exact computation again in Chapter 8: it fits the best straight line through three data points.

3.3 Gram–Schmidt orthogonalisation

Given independent vectors $\boldsymbol{a}_1, \dots, \boldsymbol{a}_n$, Gram–Schmidt produces orthonormal $\boldsymbol{q}_1, \dots, \boldsymbol{q}_n$ spanning the same spaces step by step: $\operatorname{span}(\boldsymbol{q}_1, \dots, \boldsymbol{q}_k) = \operatorname{span}(\boldsymbol{a}_1, \dots, \boldsymbol{a}_k)$ for each k. The idea is: take the next vector, subtract its projections onto the $\boldsymbol{q}$'s found so far, and normalise what is left.

Gram–Schmidt.

For $j = 1, 2, \dots, n$:

1. $\boldsymbol{w}_j = \boldsymbol{a}_j - (\boldsymbol{q}_1 \cdot \boldsymbol{a}_j)\boldsymbol{q}_1 - \cdots - (\boldsymbol{q}_{j-1} \cdot \boldsymbol{a}_j)\boldsymbol{q}_{j-1}$ (remove components along earlier $\boldsymbol{q}$'s)
2. $\boldsymbol{q}_j = \boldsymbol{w}_j / \|\boldsymbol{w}_j\|$ (normalise; $\boldsymbol{w}_j \neq \boldsymbol{0}$ by independence)

Worked Example 3.2: Gram–Schmidt in $\mathbb{R}^3$.

Orthonormalise $\boldsymbol{a}_1 = (1, 1, 0)$, $\boldsymbol{a}_2 = (1, 0, 1)$, $\boldsymbol{a}_3 = (0, 1, 1)$.

$j = 1$. $\|\boldsymbol{a}_1\| = \sqrt{2}$, so $\boldsymbol{q}_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$.

$j = 2$. $\boldsymbol{q}_1 \cdot \boldsymbol{a}_2 = \frac{1}{\sqrt{2}}$, so $\boldsymbol{w}_2 = (1, 0, 1) - \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}(1, 1, 0) = (1, 0, 1) - (\frac{1}{2}, \frac{1}{2}, 0) = (\frac{1}{2}, -\frac{1}{2}, 1)$.
$\|\boldsymbol{w}_2\| = \sqrt{\frac{1}{4} + \frac{1}{4} + 1} = \sqrt{\frac{3}{2}}$, so $\boldsymbol{q}_2 = \frac{1}{\sqrt{6}}(1, -1, 2)$.

$j = 3$. $\boldsymbol{q}_1 \cdot \boldsymbol{a}_3 = \frac{1}{\sqrt{2}}$ and $\boldsymbol{q}_2 \cdot \boldsymbol{a}_3 = \frac{1}{\sqrt{6}}(0 - 1 + 2) = \frac{1}{\sqrt{6}}$. So

$$\boldsymbol{w}_3 = (0, 1, 1) - \tfrac{1}{2}(1, 1, 0) - \tfrac{1}{6}(1, -1, 2) = \left(-\tfrac{2}{3}, \tfrac{2}{3}, \tfrac{2}{3}\right),$$

so $\|\boldsymbol{w}_3\| = \frac{2}{\sqrt{3}}$ and $\boldsymbol{q}_3 = \frac{1}{\sqrt{3}}(-1, 1, 1)$.

Check. $\boldsymbol{q}_1 \cdot \boldsymbol{q}_2 \propto 1 - 1 + 0 = 0$, $\boldsymbol{q}_1 \cdot \boldsymbol{q}_3 \propto -1 + 1 + 0 = 0$, $\boldsymbol{q}_2 \cdot \boldsymbol{q}_3 \propto -1 - 1 + 2 = 0$. ✓

Common pitfall. In exact arithmetic it does not matter whether you compute $\boldsymbol{q}_i \cdot \boldsymbol{a}_j$ (the *original* vector, "classical" Gram–Schmidt) or $\boldsymbol{q}_i \cdot \boldsymbol{w}$ (the *partially reduced* vector, "modified" Gram–Schmidt). In floating point it matters a lot: classical Gram–Schmidt can lose orthogonality badly. The Python Lab Manual (Part 2) shows this experimentally.

3.4 The QR factorisation

Gram–Schmidt can be recorded as a matrix factorisation. Each $\boldsymbol{a}_j$ is a combination of $\boldsymbol{q}_1, \dots, \boldsymbol{q}_j$ only, with coefficients $r_{ij} = \boldsymbol{q}_i \cdot \boldsymbol{a}_j$ (for $i < j$) and $r_{jj} = \|\boldsymbol{w}_j\|$:

$$\boldsymbol{a}_j = r_{1j}\boldsymbol{q}_1 + \cdots + r_{jj}\boldsymbol{q}_j.$$

In matrix form this is $A = QR$ with R *upper triangular*.

Theorem 3.4: QR factorisation. Every $m \times n$ matrix A with independent columns can be written $A = QR$, where Q $(m \times n)$ has orthonormal columns and R $(n \times n)$ is upper triangular with positive diagonal. Extending Q to a square orthogonal matrix and padding R with zero rows gives the **full** QR factorisation, Q: $m \times m$, R: $m \times n$.

Worked Example 3.3: The QR factorisation from Worked Example 3.2.
Reading off the coefficients:

$$\underbrace{\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix}}_{Q} \underbrace{\begin{bmatrix} \sqrt{2} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & \sqrt{\frac{3}{2}} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{2}{\sqrt{3}} \end{bmatrix}}_{R}.$$

Check column 3. $\frac{1}{\sqrt{2}}\boldsymbol{q}_1 + \frac{1}{\sqrt{6}}\boldsymbol{q}_2 + \frac{2}{\sqrt{3}}\boldsymbol{q}_3 = (\frac{1}{2}, \frac{1}{2}, 0) + \frac{1}{6}(1, -1, 2) + \frac{2}{3}(-1, 1, 1) = (0, 1, 1) = \boldsymbol{a}_3$. ✓

Why QR matters. With $A = QR$ the normal equations become $R^{\mathsf{T}}Q^{\mathsf{T}}QR\hat{\boldsymbol{x}} = R^{\mathsf{T}}Q^{\mathsf{T}}\boldsymbol{b}$, i.e.

$$R\hat{\boldsymbol{x}} = Q^{\mathsf{T}}\boldsymbol{b},$$

a triangular system solved by back substitution. This avoids forming $A^{\mathsf{T}}A$, which (Chapter 8) squares the sensitivity to rounding errors. In practice QR is computed not by Gram–Schmidt but by a sequence of Householder reflections, which keeps Q orthogonal to machine precision; you will implement both in the Lab.

3.5 Orthogonal complements

The **orthogonal complement** of a subspace $S \subseteq \mathbb{R}^n$ is $S^{\perp} = \{\boldsymbol{x} : \boldsymbol{x} \cdot \boldsymbol{s} = 0 \text{ for all } \boldsymbol{s} \in S\}$. Theorem 2.3 can now be sharpened: the row space and null space are not just perpendicular, they are each other's complements,

$$\mathcal{N}(A) = \mathcal{C}(A^{\mathsf{T}})^{\perp}, \qquad \mathcal{N}(A^{\mathsf{T}}) = \mathcal{C}(A)^{\perp}.$$

Consequently every $\boldsymbol{x} \in \mathbb{R}^n$ splits uniquely as

$$\boldsymbol{x} = \boldsymbol{x}_r + \boldsymbol{x}_n, \qquad \boldsymbol{x}_r \in \mathcal{C}(A^{\mathsf{T}}),\ \boldsymbol{x}_n \in \mathcal{N}(A), \qquad A\boldsymbol{x} = A\boldsymbol{x}_r.$$

This splitting is the reason a *minimum-norm* solution exists: adding any $\boldsymbol{x}_n$ to a solution changes nothing in $A\boldsymbol{x}$ but increases the length (by Pythagoras), so the shortest solution is the one with $\boldsymbol{x}_n = \boldsymbol{0}$, lying entirely in the row space.

Exercises

Exercise 3.1 $\star$ Show that $Q = \dfrac{1}{3}\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$ is orthogonal. What is Q^{-1}?

Exercise 3.2 $\star$ Write down the projection matrix P onto the line through $\boldsymbol{a} = (1,2,2)$. Verify $P^2 = P$, find $\operatorname{tr} P$, and compute $P\boldsymbol{b}$ for $\boldsymbol{b} = (1,1,1)$.

Exercise 3.3 $\star\star$ Apply Gram–Schmidt to $\boldsymbol{a}_1 = (1,1,1,1)$, $\boldsymbol{a}_2 = (1,1,0,0)$, $\boldsymbol{a}_3 = (1,0,1,0)$ and write the result as $A = QR$.

Exercise 3.4 $\star\star$ Project $\boldsymbol{b} = (1,2,3)$ onto the column space of $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$. Give $\hat{\boldsymbol{x}}$, $\boldsymbol{p}$ and $\boldsymbol{e}$, and check $A^{\mathsf{T}}\boldsymbol{e} = \boldsymbol{0}$.

Exercise 3.5 $\star\star$ Let P satisfy $P^2 = P$ and $P^{\mathsf{T}} = P$. (a) Show $I - P$ has the same two properties. (b) Show every eigenvalue of P is 0 or 1. (c) Show $P\boldsymbol{b} \perp (I - P)\boldsymbol{b}$ for every $\boldsymbol{b}$.

Exercise 3.6 $\star\star\star$ Let Q be an $n \times n$ orthogonal matrix. Prove that (a) $\det Q = \pm 1$; (b) every real eigenvalue of Q is ± 1; (c) the product of two orthogonal matrices is orthogonal.

CHAPTER 4

Eigenvalues, Eigenvectors and Symmetric Matrices

The SVD is computed from the eigenvalues and eigenvectors of the symmetric matrix $A^{\mathsf{T}}A$. Symmetric matrices are the best-behaved matrices there are: real eigenvalues, orthonormal eigenvectors, and a decomposition $S = Q\Lambda Q^{\mathsf{T}}$ that is the SVD's direct ancestor. This chapter develops what we need — no more.

4.1 Eigenvalues and eigenvectors

Definition 4.1: Eigenvalue and eigenvector. Let A be $n \times n$. A non-zero vector $\boldsymbol{x}$ with $A\boldsymbol{x} = \lambda\boldsymbol{x}$ is an **eigenvector** of A, and the scalar λ is its **eigenvalue**.

An eigenvector is a direction that A does not turn; it only stretches it by λ (or flips it if $\lambda < 0$). Rewriting $A\boldsymbol{x} = \lambda\boldsymbol{x}$ as $(A - \lambda I)\boldsymbol{x} = \boldsymbol{0}$ shows that λ is an eigenvalue exactly when $A - \lambda I$ is singular:

$$\det(A - \lambda I) = 0 \qquad \text{(the } \textbf{characteristic equation}\text{)}.$$

For each eigenvalue, the eigenvectors (with $\boldsymbol{0}$) form the null space $\mathcal{N}(A - \lambda I)$, the **eigenspace**.

Two facts give free checks: the eigenvalues sum to the trace and multiply to the determinant,

$$\lambda_1 + \cdots + \lambda_n = \operatorname{tr} A = \sum_i a_{ii}, \qquad \lambda_1\lambda_2\cdots\lambda_n = \det A.$$

For a 2×2 matrix the characteristic equation is simply $\lambda^2 - (\operatorname{tr} A)\lambda + \det A = 0$.

Worked Example 4.1: A non-symmetric matrix.

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}.$$

Eigenvalues. $\operatorname{tr} A = 4$, $\det A = 3$: $\lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3) = 0$, so $\lambda_1 = 3$, $\lambda_2 = 1$. (Triangular: the eigenvalues are the diagonal entries.)

$\lambda = 3$. $A - 3I = \left[\begin{smallmatrix} -2 & 2 \\ 0 & 0 \end{smallmatrix}\right]$, null space spanned by $\boldsymbol{x}_1 = (1, 1)$.

$\lambda = 1$. $A - I = \left[\begin{smallmatrix} 0 & 2 \\ 0 & 2 \end{smallmatrix}\right]$, null space spanned by $\boldsymbol{x}_2 = (1, 0)$.

Observe. $\boldsymbol{x}_1 \cdot \boldsymbol{x}_2 = 1 \neq 0$: the eigenvectors are *not* orthogonal. This is typical of non-symmetric matrices.

4.2 Diagonalisation and its limits

If A has n independent eigenvectors, put them in the columns of X and the eigenvalues in $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$. Then $AX = X\Lambda$, so

$$A = X\Lambda X^{-1}, \qquad A^k = X\Lambda^k X^{-1}.$$

Two things can go wrong:

- There may not be enough eigenvectors. $A = \left[\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]$ has $\lambda = 0$ twice but only the eigenvector direction $(1, 0)$; it cannot be diagonalised.
- Eigenvalues can be complex: the rotation $\left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$ has $\lambda = \pm i$.

And of course rectangular matrices have no eigenvalues at all. The SVD avoids every one of these problems; the price is using *two* bases, U and V, instead of one. Its construction rests on matrices where none of the problems arise: symmetric matrices.

4.3 Symmetric matrices and the spectral theorem

Theorem 4.2: Spectral theorem. Let S be a real symmetric $n \times n$ matrix. Then

1. all eigenvalues of S are real;
2. eigenvectors belonging to different eigenvalues are orthogonal;
3. S has an orthonormal basis of eigenvectors $\boldsymbol{q}_1, \dots, \boldsymbol{q}_n$, so

$$\boxed{S = Q\Lambda Q^{\mathsf{T}} = \lambda_1 \boldsymbol{q}_1 \boldsymbol{q}_1^{\mathsf{T}} + \lambda_2 \boldsymbol{q}_2 \boldsymbol{q}_2^{\mathsf{T}} + \cdots + \lambda_n \boldsymbol{q}_n \boldsymbol{q}_n^{\mathsf{T}}}$$

with Q orthogonal and Λ diagonal and real.

Proof of part 2. Let $S\boldsymbol{x} = \lambda \boldsymbol{x}$ and $S\boldsymbol{y} = \mu \boldsymbol{y}$ with $\lambda \neq \mu$. Using $S^{\mathsf{T}} = S$ and (2.1),

$$\lambda\, \boldsymbol{x} \cdot \boldsymbol{y} = (S\boldsymbol{x}) \cdot \boldsymbol{y} = \boldsymbol{x} \cdot (S\boldsymbol{y}) = \mu\, \boldsymbol{x} \cdot \boldsymbol{y}.$$

So $(\lambda - \mu)\, \boldsymbol{x} \cdot \boldsymbol{y} = 0$, and since $\lambda \neq \mu$, $\boldsymbol{x} \cdot \boldsymbol{y} = 0$. □

Part 1 is Exercise 4.6. Part 3 needs an induction argument we omit; for a repeated eigenvalue, it says the eigenspace has the full dimension, and we may choose an orthonormal basis inside it (by Gram–Schmidt).

The sum form says: a symmetric matrix is a weighted sum of projections $\boldsymbol{q}_i \boldsymbol{q}_i^{\mathsf{T}}$ onto perpendicular lines. Acting on $\boldsymbol{x}$, it splits $\boldsymbol{x}$ into its components along the $\boldsymbol{q}_i$, scales the i-th component by λ_i, and adds them back.

Worked Example 4.2: Spectral decomposition of a 2×2 matrix.

$$S = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Eigenvalues. $\operatorname{tr} = 4$, $\det = 3$: $\lambda^2 - 4\lambda + 3 = 0$, so $\lambda_1 = 3$, $\lambda_2 = 1$.

Eigenvectors. $S - 3I = \left[\begin{smallmatrix} -1 & 1 \\ 1 & -1 \end{smallmatrix}\right]$ gives $(1,1)$; $S - I = \left[\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right]$ gives $(1,-1)$. Orthogonal, as promised. Normalise: $\boldsymbol{q}_1 = \frac{1}{\sqrt{2}}(1,1)$, $\boldsymbol{q}_2 = \frac{1}{\sqrt{2}}(1,-1)$.

Decomposition.

$$3\boldsymbol{q}_1\boldsymbol{q}_1^{\mathsf{T}} + 1\boldsymbol{q}_2\boldsymbol{q}_2^{\mathsf{T}} = \frac{3}{2}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + \frac{1}{2}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}. \checkmark$$

Worked Example 4.3: A repeated eigenvalue.

$$S = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}.$$

Eigenvalues. $S = I + J$ where J is the all-ones matrix. J has eigenvalue 3 on $(1,1,1)$ and 0 on the plane $x + y + z = 0$, so S has $\lambda = 4$ (once) and $\lambda = 1$ (twice). Check: $\operatorname{tr} S = 6 = 4 + 1 + 1$. ✓

$\lambda = 4$. $\boldsymbol{q}_1 = \frac{1}{\sqrt{3}}(1,1,1)$.

$\lambda = 1$. The eigenspace is the whole plane $x + y + z = 0$ — any vector in it works. Pick $(1,-1,0)$, then a vector in the plane orthogonal to it, $(1,1,-2)$. Normalise: $\boldsymbol{q}_2 = \frac{1}{\sqrt{2}}(1,-1,0)$, $\boldsymbol{q}_3 = \frac{1}{\sqrt{6}}(1,1,-2)$.

Check. $S\boldsymbol{q}_3 \propto (2+1-2,\ 1+2-2,\ 1+1-4) = (1,1,-2)$, eigenvalue 1. ✓

Non-uniqueness. Any other orthonormal pair in the plane would do equally well. With a repeated eigenvalue the individual eigenvectors are not unique — only the eigenspace is. The same will be true of singular vectors with repeated singular values.

4.4 The Rayleigh quotient

Definition 4.3: Rayleigh quotient. For symmetric S and $\boldsymbol{x} \neq \mathbf{0}$, the **Rayleigh quotient** is $R(\boldsymbol{x}) = \dfrac{\boldsymbol{x}^{\mathsf{T}} S \boldsymbol{x}}{\boldsymbol{x}^{\mathsf{T}} \boldsymbol{x}}$.

If $\boldsymbol{x}$ is an eigenvector, $R(\boldsymbol{x}) = \lambda$. In general:

Theorem 4.4: Extremal eigenvalues. Let S be symmetric with eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$. Then for every $\boldsymbol{x} \neq \mathbf{0}$

$$\lambda_n \leq R(\boldsymbol{x}) \leq \lambda_1,$$

with $R(\boldsymbol{x}) = \lambda_1$ when $\boldsymbol{x} = \boldsymbol{q}_1$ and $R(\boldsymbol{x}) = \lambda_n$ when $\boldsymbol{x} = \boldsymbol{q}_n$. Thus $\lambda_1 = \max_{\boldsymbol{x} \neq \mathbf{0}} R(\boldsymbol{x})$.

Proof. Expand $\boldsymbol{x} = \sum c_i \boldsymbol{q}_i$ in the orthonormal eigenbasis. Then $S\boldsymbol{x} = \sum \lambda_i c_i \boldsymbol{q}_i$, so

$$R(\boldsymbol{x}) = \frac{\sum_i \lambda_i c_i^2}{\sum_i c_i^2},$$

a weighted average of the eigenvalues with weights $c_i^2 \geq 0$. A weighted average lies between the smallest and largest values. □

Key idea. "The largest eigenvalue of a symmetric matrix is the maximum of $\boldsymbol{x}^{\mathsf{T}} S \boldsymbol{x}$ over unit vectors." Apply it to $S = A^{\mathsf{T}} A$: since $\boldsymbol{x}^{\mathsf{T}} A^{\mathsf{T}} A \boldsymbol{x} = \|A\boldsymbol{x}\|^2$, the largest eigenvalue of $A^{\mathsf{T}} A$ is the maximum of $\|A\boldsymbol{x}\|^2$ over unit $\boldsymbol{x}$ — the square of the largest stretching factor of A. That stretching factor will be called σ_1.

4.5 Power iteration

How does a computer find λ_1 and $\boldsymbol{q}_1$? The simplest method: multiply by S repeatedly. Writing $\boldsymbol{x}_0 = \sum c_i \boldsymbol{q}_i$,

$$S^k \boldsymbol{x}_0 = \sum_i c_i \lambda_i^k \boldsymbol{q}_i = \lambda_1^k \Big(c_1 \boldsymbol{q}_1 + \sum_{i \geq 2} c_i \big(\tfrac{\lambda_i}{\lambda_1}\big)^k \boldsymbol{q}_i \Big).$$

If $|\lambda_1| > |\lambda_2|$ and $c_1 \neq 0$, the other terms die out like $|\lambda_2/\lambda_1|^k$, and the direction of $S^k \boldsymbol{x}_0$ approaches $\pm \boldsymbol{q}_1$. Normalising at each step prevents overflow:

Power iteration.
Start with a random unit $\boldsymbol{x}_0$. Repeat: $\boldsymbol{y} = S\boldsymbol{x}_k$, $\boldsymbol{x}_{k+1} = \boldsymbol{y} / \|\boldsymbol{y}\|$, $\lambda^{(k+1)} = \boldsymbol{x}_{k+1}^{\mathsf{T}} S \boldsymbol{x}_{k+1}$. Stop when $\lambda^{(k)}$ stops changing.

Worked Example 4.4: Power iteration by hand.
$S = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ (eigenvalues 3, 1), $\boldsymbol{x}_0 = (1, 0)$. Without normalising (the direction is all that matters):

k	$\boldsymbol{x}_k$	$R(\boldsymbol{x}_k)$	error $3 - R(\boldsymbol{x}_k)$
0	$(1,0)$	2	1
1	$(2,1)$	$\frac{14}{5} = 2.8$	0.2
2	$(5,4)$	$\frac{122}{41} \approx 2.9756$	0.0244
3	$(14,13)$	$\frac{1094}{365} \approx 2.9973$	0.0027

For instance, $R(5,4) = \dfrac{2\cdot 25 + 2\cdot 5\cdot 4 + 2\cdot 16}{25+16} = \dfrac{122}{41}$. The direction approaches $(1,1)$ and the error shrinks by a factor of about $9 = (\lambda_1/\lambda_2)^2$ per step: for symmetric matrices the Rayleigh quotient converges twice as fast as the vector.

Once $\boldsymbol{q}_1$ is known, **deflation** $S \leftarrow S - \lambda_1 \boldsymbol{q}_1 \boldsymbol{q}_1^{\mathsf{T}}$ removes it, and power iteration on the result finds λ_2. The Lab implements this, and also the **Jacobi eigenvalue method**, which finds all eigenvalues at once by applying plane rotations that zero the off-diagonal entries one by one.

Exercises

Exercise 4.1 $\star$ Find the eigenvalues and eigenvectors of $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$. Are the eigenvectors orthogonal? Why is that not surprising?

Exercise 4.2 $\star$ Find the spectral decomposition $S = \lambda_1 \boldsymbol{q}_1 \boldsymbol{q}_1^{\mathsf{T}} + \lambda_2 \boldsymbol{q}_2 \boldsymbol{q}_2^{\mathsf{T}}$ of $S = \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}$ and verify it.

Exercise 4.3 $\star\star$ Find the eigenvalues and an orthonormal basis of eigenvectors of $S = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$. *Hint:* compare with Worked Example 4.3.

Exercise 4.4 $\star\star$ Run three steps of power iteration on $S = \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}$ starting from $\boldsymbol{x}_0 = (1,0)$, without normalising. Give the Rayleigh quotient at each step and compare with the true eigenvalue from Exercise 4.2. What convergence factor do you expect?

Exercise 4.5 $\star\star$ Let $A\boldsymbol{x} = \lambda \boldsymbol{x}$. Show that (a) $A^k \boldsymbol{x} = \lambda^k \boldsymbol{x}$; (b) if A is invertible, $\lambda \neq 0$ and $A^{-1}\boldsymbol{x} = \lambda^{-1}\boldsymbol{x}$; (c) $(A + cI)\boldsymbol{x} = (\lambda + c)\boldsymbol{x}$. Use (c) to explain the trick in Exercise 4.3.

Exercise 4.6 ⋆⋆⋆ Prove part 1 of the spectral theorem: every eigenvalue of a real symmetric matrix is real. *Hint:* let $S\boldsymbol{x} = \lambda \boldsymbol{x}$ with possibly complex $\boldsymbol{x}, \lambda$, and compute $\bar{\boldsymbol{x}}^{\mathsf{T}} S \boldsymbol{x}$ in two ways, where the bar denotes complex conjugation.

CHAPTER 5

Positive Semidefinite Matrices and $A^\mathsf{T}A$

We now meet the two matrices from which the SVD is built, $A^\mathsf{T}A$ and AA^T. They are symmetric, so the spectral theorem applies, and they are *positive semidefinite*, so their eigenvalues are non-negative and have square roots. Those square roots are the singular values.

5.1 Positive definite and semidefinite matrices

Definition 5.1: Definiteness. A symmetric matrix S is

- **positive definite** (PD) if $\boldsymbol{x}^\mathsf{T}S\boldsymbol{x} > 0$ for every $\boldsymbol{x} \neq \mathbf{0}$;
- **positive semidefinite** (PSD) if $\boldsymbol{x}^\mathsf{T}S\boldsymbol{x} \geq 0$ for every $\boldsymbol{x}$;
- **negative (semi)definite** if $-S$ is positive (semi)definite;
- **indefinite** if $\boldsymbol{x}^\mathsf{T}S\boldsymbol{x}$ takes both signs.

The number $\boldsymbol{x}^\mathsf{T}S\boldsymbol{x} = \sum_{i,j} s_{ij}x_ix_j$ is called the **quadratic form** or *energy* of S.

Theorem 5.2: Tests for positive definiteness. For a symmetric matrix S the following are equivalent:

1. S is positive definite;
2. all eigenvalues of S are positive;
3. all leading principal minors (determinants of the upper-left $k \times k$ blocks) are positive;
4. $S = B^\mathsf{T}B$ for some matrix B with independent columns.

The PSD version: eigenvalues ≥ 0; $S = B^\mathsf{T}B$ for some B (any columns).

Proof of $1 \Leftrightarrow 2$. Write $\boldsymbol{x} = \sum c_i\boldsymbol{q}_i$ in an orthonormal eigenbasis. Then $\boldsymbol{x}^\mathsf{T}S\boldsymbol{x} = \sum_i \lambda_i c_i^2$. If every $\lambda_i > 0$ this is positive whenever some $c_i \neq 0$, i.e. whenever $\boldsymbol{x} \neq \mathbf{0}$. Conversely, taking $\boldsymbol{x} = \boldsymbol{q}_i$ gives $\boldsymbol{q}_i^\mathsf{T}S\boldsymbol{q}_i = \lambda_i$, which must be positive. □

For a 2×2 matrix $\left[\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right]$, test 3 reads: $a > 0$ and $ac - b^2 > 0$.

Worked Example 5.1: Three tests on one matrix.

$S = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$.

Eigenvalues. $\operatorname{tr} = 4$, $\det = 3$: $\lambda = 3, 1 > 0$. PD.

Minors. $2 > 0$ and $\det S = 3 > 0$. PD.

Energy. $\boldsymbol{x}^\mathsf{T} S \boldsymbol{x} = 2x^2 - 2xy + 2y^2 = x^2 + y^2 + (x-y)^2$, a sum of squares, zero only when $x = y = 0$. PD.

By contrast, $\left[\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right]$ has eigenvalues 3 and -1 and is indefinite: $\boldsymbol{x} = (1, -1)$ gives energy $1 - 4 + 1 = -2 < 0$. And $\left[\begin{smallmatrix} 1 & 2 \\ 2 & 4 \end{smallmatrix}\right]$ has eigenvalues 5 and 0: PSD but not PD, since $\boldsymbol{x} = (2, -1)$ gives energy 0.

Geometry. For PD S the level set $\{\boldsymbol{x} : \boldsymbol{x}^\mathsf{T} S \boldsymbol{x} = 1\}$ is an ellipse (ellipsoid in higher dimensions) whose axes point along the eigenvectors, with semi-axis lengths $1/\sqrt{\lambda_i}$: a large eigenvalue means steep energy and a short axis. Figure 5.1 shows the case $S = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$.

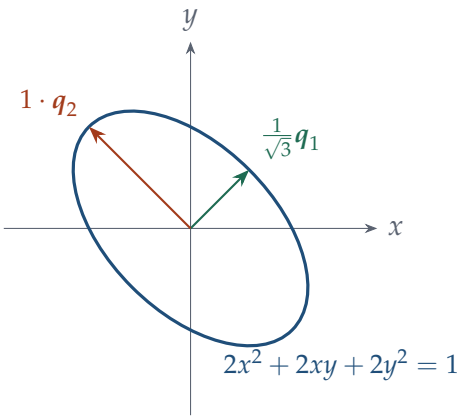


Figure 5.1: The ellipse $\boldsymbol{x}^\mathsf{T} S \boldsymbol{x} = 1$ for $S = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$: axes along $\boldsymbol{q}_1 = \frac{1}{\sqrt{2}}(1, 1)$ ($\lambda_1 = 3$, semi-axis $1/\sqrt{3}$) and $\boldsymbol{q}_2 = \frac{1}{\sqrt{2}}(1, -1)$ ($\lambda_2 = 1$, semi-axis 1; drawn as $-\boldsymbol{q}_2$).

5.2 The matrix $A^\mathsf{T}A$

Theorem 5.3: $A^\mathsf{T}A$ is positive semidefinite. For any $m \times n$ matrix A:

1. $A^\mathsf{T}A$ is symmetric and positive semidefinite;
2. $\mathcal{N}(A^\mathsf{T}A) = \mathcal{N}(A)$, hence $\operatorname{rank}(A^\mathsf{T}A) = \operatorname{rank} A$;
3. $A^\mathsf{T}A$ is positive definite (and so invertible) if and only if the columns of A are independent.

Proof. 1. Symmetry was shown in Chapter 2. For the energy: $\boldsymbol{x}^\mathsf{T} A^\mathsf{T} A \boldsymbol{x} = (A\boldsymbol{x})^\mathsf{T}(A\boldsymbol{x}) = \|A\boldsymbol{x}\|^2 \geq 0$.

2. If $A\boldsymbol{x} = \mathbf{0}$ then $A^\mathsf{T}A\boldsymbol{x} = \mathbf{0}$. Conversely, if $A^\mathsf{T}A\boldsymbol{x} = \mathbf{0}$ then $0 = \boldsymbol{x}^\mathsf{T}A^\mathsf{T}A\boldsymbol{x} = \|A\boldsymbol{x}\|^2$, so $A\boldsymbol{x} = \mathbf{0}$. Equal null spaces mean equal nullity $n - r$, and so equal rank.

3. $\boldsymbol{x}^\mathsf{T}A^\mathsf{T}A\boldsymbol{x} = \|A\boldsymbol{x}\|^2$ is zero only for $\boldsymbol{x} \in \mathcal{N}(A)$, which is $\{\mathbf{0}\}$ exactly when the columns are independent. □

Key idea. The energy of $A^\mathsf{T}A$ is $\|A\boldsymbol{x}\|^2$. So the eigenvalues of $A^\mathsf{T}A$ measure *how much* A *stretches* in each eigenvector direction: if $A^\mathsf{T}A\boldsymbol{v} = \lambda\boldsymbol{v}$ with $\|\boldsymbol{v}\| = 1$, then $\|A\boldsymbol{v}\|^2 = \boldsymbol{v}^\mathsf{T}A^\mathsf{T}A\boldsymbol{v} = \lambda$. The stretch factor is $\sqrt{\lambda}$.

5.3 $A^\mathsf{T}A$ and AA^T share their non-zero eigenvalues

$A^\mathsf{T}A$ is $n \times n$ and AA^T is $m \times m$ — different sizes, yet intimately related.

Theorem 5.4: Shared eigenvalues. $A^\mathsf{T}A$ and AA^T have the same non-zero eigenvalues, with the same multiplicities. More precisely, if $A^\mathsf{T}A\boldsymbol{v} = \lambda\boldsymbol{v}$ with $\lambda > 0$ and $\|\boldsymbol{v}\| = 1$, then

$$\boldsymbol{u} = \frac{A\boldsymbol{v}}{\sqrt{\lambda}}$$

is a unit vector with $AA^\mathsf{T}\boldsymbol{u} = \lambda\boldsymbol{u}$; and if $\boldsymbol{v}_1 \perp \boldsymbol{v}_2$ then $\boldsymbol{u}_1 \perp \boldsymbol{u}_2$.

Proof. $AA^\mathsf{T}(A\boldsymbol{v}) = A(A^\mathsf{T}A\boldsymbol{v}) = A(\lambda\boldsymbol{v}) = \lambda(A\boldsymbol{v})$, so $A\boldsymbol{v}$ is an eigenvector of AA^T provided $A\boldsymbol{v} \neq \mathbf{0}$; and $\|A\boldsymbol{v}\|^2 = \lambda > 0$. Dividing by $\|A\boldsymbol{v}\| = \sqrt{\lambda}$ makes it a unit vector. For orthogonality: $\boldsymbol{u}_1^\mathsf{T}\boldsymbol{u}_2 = \boldsymbol{v}_1^\mathsf{T}A^\mathsf{T}A\boldsymbol{v}_2/\sqrt{\lambda_1\lambda_2} = \lambda_2\boldsymbol{v}_1^\mathsf{T}\boldsymbol{v}_2/\sqrt{\lambda_1\lambda_2} = 0$. Exchanging the roles of A and A^T gives the reverse map, so the multiplicities agree. □

This theorem is the SVD in embryo: the eigenvectors $\boldsymbol{v}_i$ of $A^\mathsf{T}A$ and $\boldsymbol{u}_i$ of AA^T are paired by $A\boldsymbol{v}_i = \sqrt{\lambda_i}\,\boldsymbol{u}_i$. Chapter 6 simply writes this down for all i at once.

Worked Example 5.2: Shared eigenvalues.

$A = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$.

The two products. $A^\mathsf{T}A = \begin{bmatrix} 3 & 4 \\ 0 & 5 \end{bmatrix}\begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 25 & 20 \\ 20 & 25 \end{bmatrix}$, $AA^\mathsf{T} = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}\begin{bmatrix} 3 & 4 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 9 & 12 \\ 12 & 41 \end{bmatrix}$.

Eigenvalues. Both have trace 50 and determinant $625 - 400 = 225 = 369 - 144$. So both have $\lambda^2 - 50\lambda + 225 = 0$, i.e. $\lambda = 45$ and $\lambda = 5$.

Pairing. $A^{\mathsf{T}}A$ has eigenvector $\boldsymbol{v}_1 = \frac{1}{\sqrt{2}}(1,1)$ for $\lambda = 45$. Then $A\boldsymbol{v}_1 = \frac{1}{\sqrt{2}}(3,9)$ and

$$AA^{\mathsf{T}}\begin{bmatrix}3\\9\end{bmatrix} = \begin{bmatrix}27+108\\36+369\end{bmatrix} = \begin{bmatrix}135\\405\end{bmatrix} = 45\begin{bmatrix}3\\9\end{bmatrix}. \checkmark$$

Dividing by $\|A\boldsymbol{v}_1\| = \sqrt{45}$: $\boldsymbol{u}_1 = \frac{1}{\sqrt{2}\cdot 3\sqrt{5}}(3,9) = \frac{1}{\sqrt{10}}(1,3)$, a unit eigenvector of AA^{T}.

Exercises

Exercise 5.1 $\star$ Classify each symmetric matrix as positive definite, positive semidefinite, negative definite or indefinite: $S_1 = \left[\begin{smallmatrix}2&1\\1&2\end{smallmatrix}\right]$, $S_2 = \left[\begin{smallmatrix}1&2\\2&4\end{smallmatrix}\right]$, $S_3 = \left[\begin{smallmatrix}1&3\\3&1\end{smallmatrix}\right]$, $S_4 = \left[\begin{smallmatrix}-2&1\\1&-2\end{smallmatrix}\right]$.

Exercise 5.2 $\star$ For $A = \begin{bmatrix}1&0\\1&1\\0&1\end{bmatrix}$ compute $A^{\mathsf{T}}A$ and AA^{T} and their eigenvalues. Confirm Theorem 5.4.

Exercise 5.3 $\star\star$ Find all values of c for which $S = \begin{bmatrix}c&2\\2&c\end{bmatrix}$ is (a) positive definite, (b) positive semidefinite.

Exercise 5.4 $\star\star$ Show that if S is positive definite then so is S^{-1}, and that if S and T are positive definite then so is $S+T$. Is ST always positive definite?

Exercise 5.5 $\star\star$ Prove directly that $\mathcal{N}(A^{\mathsf{T}}A) = \mathcal{N}(A)$, and deduce that $A^{\mathsf{T}}A\boldsymbol{x} = A^{\mathsf{T}}\boldsymbol{b}$ always has at least one solution, for every $\boldsymbol{b}$. *Hint:* show $\mathcal{C}(A^{\mathsf{T}}A) = \mathcal{C}(A^{\mathsf{T}})$.

Exercise 5.6 $\star\star\star$ Let $\boldsymbol{v}_1,\dots,\boldsymbol{v}_r$ be orthonormal eigenvectors of $A^{\mathsf{T}}A$ with positive eigenvalues $\lambda_1,\dots,\lambda_r$. Show that $\boldsymbol{u}_i = A\boldsymbol{v}_i/\sqrt{\lambda_i}$ are orthonormal eigenvectors of AA^{T}, and that $A^{\mathsf{T}}\boldsymbol{u}_i = \sqrt{\lambda_i}\,\boldsymbol{v}_i$.

CHAPTER 6

The Singular Value Decomposition

Every matrix — square or rectangular, invertible or not — has a singular value decomposition. It says that for a well-chosen orthonormal basis of the input space and a well-chosen orthonormal basis of the output space, the matrix simply stretches: the i-th input basis vector goes to σ_i times the i-th output basis vector. This chapter states the theorem, proves it, and — most importantly — shows you how to compute it by hand.

6.1 The goal

For a symmetric matrix we found orthonormal $\boldsymbol{q}_i$ with $S\boldsymbol{q}_i = \lambda_i \boldsymbol{q}_i$. For a general $m \times n$ matrix A that is impossible ($A\boldsymbol{v}$ lives in $\mathbb{R}^m$, $\boldsymbol{v}$ in $\mathbb{R}^n$), but we can ask for the next best thing:

$$\boxed{A\boldsymbol{v}_i = \sigma_i \boldsymbol{u}_i} \qquad \boldsymbol{v}_1, \dots, \boldsymbol{v}_n \text{ orthonormal in } \mathbb{R}^n, \quad \boldsymbol{u}_1, \dots, \boldsymbol{u}_m \text{ orthonormal in } \mathbb{R}^m, \quad \sigma_i \geq 0.$$

Chapter 5 already told us where to find them: $\boldsymbol{v}_i$ are eigenvectors of $A^\mathsf{T}A$, $\boldsymbol{u}_i$ are eigenvectors of AA^T, and $\sigma_i = \sqrt{\lambda_i}$.

6.2 The theorem

Theorem 6.1: Singular value decomposition. Let A be a real $m \times n$ matrix of rank r. Then

$$\boxed{A = U\Sigma V^\mathsf{T}}$$

where

- $U = [\boldsymbol{u}_1 \cdots \boldsymbol{u}_m]$ is an $m \times m$ orthogonal matrix (the **left singular vectors**);
- $V = [\boldsymbol{v}_1 \cdots \boldsymbol{v}_n]$ is an $n \times n$ orthogonal matrix (the **right singular vectors**);
- Σ is $m \times n$, zero except for its diagonal entries $\Sigma_{ii} = \sigma_i$, the **singular values**, ordered $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0 = \sigma_{r+1} = \dots$.

Equivalently, $A\boldsymbol{v}_i = \sigma_i \boldsymbol{u}_i$ and $A^\mathsf{T}\boldsymbol{u}_i = \sigma_i \boldsymbol{v}_i$ for each i, and

$$A = \sigma_1 \boldsymbol{u}_1 \boldsymbol{v}_1^\mathsf{T} + \sigma_2 \boldsymbol{u}_2 \boldsymbol{v}_2^\mathsf{T} + \cdots + \sigma_r \boldsymbol{u}_r \boldsymbol{v}_r^\mathsf{T}.$$

The singular values are uniquely determined by A: σ_i^2 are the eigenvalues of $A^\mathsf{T}A$.

Proof. **Step 1 (V and σ_i).** $A^\mathsf{T}A$ is symmetric PSD (Theorem 5.3), so by the spectral theorem it has an orthonormal eigenbasis $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$ with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n \geq 0$. Exactly r of them are positive, since $\operatorname{rank} A^\mathsf{T}A = \operatorname{rank} A = r$. Put $\sigma_i = \sqrt{\lambda_i}$.

Step 2 (first r columns of U). For $i \leq r$ set $\boldsymbol{u}_i = A\boldsymbol{v}_i/\sigma_i$. By Theorem 5.4 (and Exercise 5.6) these are orthonormal.

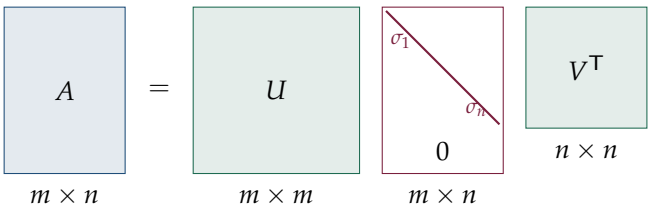


Figure 6.1: Shapes in the full SVD of a tall matrix.

Step 3 (complete U). Extend $\boldsymbol{u}_1, \dots, \boldsymbol{u}_r$ to an orthonormal basis $\boldsymbol{u}_1, \dots, \boldsymbol{u}_m$ of $\mathbb{R}^m$ (e.g. by Gram–Schmidt).

Step 4 (check $AV = U\Sigma$). For $i \le r$, $A\boldsymbol{v}_i = \sigma_i \boldsymbol{u}_i$ by construction. For $i > r$, $\|A\boldsymbol{v}_i\|^2 = \boldsymbol{v}_i^{\mathsf{T}} A^{\mathsf{T}} A \boldsymbol{v}_i = \lambda_i = 0$, so $A\boldsymbol{v}_i = \boldsymbol{0} = 0 \cdot \boldsymbol{u}_i$. Column by column, this is $AV = U\Sigma$, and multiplying on the right by $V^{\mathsf{T}} = V^{-1}$ gives $A = U\Sigma V^{\mathsf{T}}$. □

6.3 Computing the SVD by hand

The proof is an algorithm. Here it is as a checklist.

Computing the SVD of an $m \times n$ matrix by hand.

1. Form the *smaller* of $A^{\mathsf{T}}A$ ($n \times n$) and AA^{T} ($m \times m$). Below we assume $A^{\mathsf{T}}A$; for AA^{T} swap the roles of U and V.
2. Find its eigenvalues $\lambda_1 \ge \lambda_2 \ge \dots$. The singular values are $\sigma_i = \sqrt{\lambda_i}$.
3. Find orthonormal eigenvectors $\boldsymbol{v}_1, \boldsymbol{v}_2, \dots$ in the same order. These are the columns of V.
4. For every $\sigma_i > 0$ compute $\boldsymbol{u}_i = A\boldsymbol{v}_i / \sigma_i$. **Do not** compute the $\boldsymbol{u}_i$ independently from AA^{T}.
5. If U needs more columns ($r < m$), complete it with an orthonormal basis of $\mathcal{N}(A^{\mathsf{T}})$.
6. **Check**: multiply out $U\Sigma V^{\mathsf{T}}$, or at least verify $A\boldsymbol{v}_i = \sigma_i \boldsymbol{u}_i$ for each i.

Worked Example 6.1: A full 2×2 SVD.

Find the SVD of $A = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$.

Step 1. $A^{\mathsf{T}}A = \begin{bmatrix} 25 & 20 \\ 20 & 25 \end{bmatrix}$.

Step 2. $\operatorname{tr} = 50$, $\det = 625 - 400 = 225$: $\lambda^2 - 50\lambda + 225 = (\lambda - 45)(\lambda - 5) = 0$. So

$$\sigma_1 = \sqrt{45} = 3\sqrt{5} \approx 6.708, \qquad \sigma_2 = \sqrt{5} \approx 2.236.$$

Step 3. $A^{\mathsf{T}}A - 45I = \left[\begin{smallmatrix} -20 & 20 \\ 20 & -20 \end{smallmatrix}\right]$ gives $\boldsymbol{v}_1 = \frac{1}{\sqrt{2}}(1, 1)$. $A^{\mathsf{T}}A - 5I = \left[\begin{smallmatrix} 20 & 20 \\ 20 & 20 \end{smallmatrix}\right]$ gives $\boldsymbol{v}_2 = \frac{1}{\sqrt{2}}(1, -1)$.

Step 4.

$$\boldsymbol{u}_1 = \frac{A\boldsymbol{v}_1}{\sigma_1} = \frac{1}{3\sqrt{5}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 3 \\ 9 \end{bmatrix} = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \qquad \boldsymbol{u}_2 = \frac{A\boldsymbol{v}_2}{\sigma_2} = \frac{1}{\sqrt{5}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{10}} \begin{bmatrix} 3 \\ -1 \end{bmatrix}.$$

$\boldsymbol{u}_1 \cdot \boldsymbol{u}_2 = \frac{1}{10}(3-3) = 0$. ✓ Step 5 is not needed ($r = m = 2$).

Result.

$$A = \underbrace{\frac{1}{\sqrt{10}} \begin{bmatrix} 1 & 3 \\ 3 & -1 \end{bmatrix}}_{U} \underbrace{\begin{bmatrix} 3\sqrt{5} & 0 \\ 0 & \sqrt{5} \end{bmatrix}}_{\Sigma} \underbrace{\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}}_{V^{\mathsf{T}}}.$$

Step 6 (check). $U\Sigma = \dfrac{\sqrt{5}}{\sqrt{10}} \begin{bmatrix} 3 & 3 \\ 9 & -1 \end{bmatrix} = \dfrac{1}{\sqrt{2}} \begin{bmatrix} 3 & 3 \\ 9 & -1 \end{bmatrix}$, and then

$$U\Sigma V^{\mathsf{T}} = \frac{1}{2} \begin{bmatrix} 3 & 3 \\ 9 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 6 & 0 \\ 8 & 10 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}. \checkmark$$

Worked Example 6.2: A wide 2×3 matrix.

Find the full SVD of $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$.

Step 1 (use AA^{T}, it is smaller). $AA^{\mathsf{T}} = \begin{bmatrix} 17 & 8 \\ 8 & 17 \end{bmatrix}$.

Step 2. $\operatorname{tr} = 34$, $\det = 289 - 64 = 225$: $\lambda^2 - 34\lambda + 225 = (\lambda - 25)(\lambda - 9) = 0$. So $\sigma_1 = 5$, $\sigma_2 = 3$.

Step 3 (now these give U). $\lambda = 25$: $\boldsymbol{u}_1 = \frac{1}{\sqrt{2}}(1,1)$; $\lambda = 9$: $\boldsymbol{u}_2 = \frac{1}{\sqrt{2}}(1,-1)$.

Step 4 (roles swapped: $v_i = A^{\mathsf{T}}\boldsymbol{u}_i/\sigma_i$).

$$\boldsymbol{v}_1 = \frac{1}{5} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 5 \\ 5 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \qquad \boldsymbol{v}_2 = \frac{1}{3} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix} = \frac{1}{3\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}.$$

Step 5 (complete V). $\boldsymbol{v}_3$ must span $\mathcal{N}(A)$. The cross product of the two rows, $(3,2,2) \times (2,3,-2) = (-10,10,5)$, is orthogonal to both; normalising, $\boldsymbol{v}_3 = \frac{1}{3}(2,-2,-1)$ (sign is free). Check $\boldsymbol{v}_3 \cdot \boldsymbol{v}_1 \propto 2 - 2 = 0$ and $\boldsymbol{v}_3 \cdot \boldsymbol{v}_2 \propto 2 + 2 - 4 = 0$. ✓

Result.

$$A = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{3\sqrt{2}} & -\frac{1}{3\sqrt{2}} & \frac{4}{3\sqrt{2}} \\ \frac{2}{3} & -\frac{2}{3} & -\frac{1}{3} \end{bmatrix}.$$

Step 6 (check with outer products).

$$5\boldsymbol{u}_1\boldsymbol{v}_1^{\mathsf{T}} + 3\boldsymbol{u}_2\boldsymbol{v}_2^{\mathsf{T}} = \frac{5}{2} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 & -1 & 4 \\ -1 & 1 & -4 \end{bmatrix} = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}. \checkmark$$

The third right singular vector does not appear in the sum: it is multiplied by a zero column of Σ.

Worked Example 6.3: A rank-one matrix.

Find the full SVD of $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix}$.

Steps 1–2. $A^{\mathsf{T}}A = \begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix}$, eigenvalues 6 and 0. So $\sigma_1 = \sqrt{6}$ and $\sigma_2 = 0$; rank $r = 1$.

Step 3. $v_1 = \frac{1}{\sqrt{2}}(1,1)$, $v_2 = \frac{1}{\sqrt{2}}(1,-1)$.

Step 4. $u_1 = Av_1/\sqrt{6} = \frac{1}{\sqrt{2}\sqrt{6}}(2,2,2) = \frac{1}{\sqrt{3}}(1,1,1)$. There is no u_2 from this formula, because $\sigma_2 = 0$.

Step 5. U needs two more columns: an orthonormal basis of $\mathcal{N}(A^{\mathsf{T}}) = \{\boldsymbol{y} : y_1 + y_2 + y_3 = 0\}$. As in Worked Example 4.3: $u_2 = \frac{1}{\sqrt{2}}(1,-1,0)$, $u_3 = \frac{1}{\sqrt{6}}(1,1,-2)$.

Result.

$$A = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & -\frac{2}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} \sqrt{6} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \sqrt{6}\, u_1 v_1^{\mathsf{T}}.$$

Indeed $\sqrt{6} \cdot \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{2}} = 1$, and $u_1 v_1^{\mathsf{T}}$ has all entries equal. A rank-one matrix $\boldsymbol{x}\boldsymbol{y}^{\mathsf{T}}$ always has $\sigma_1 = \|\boldsymbol{x}\|\,\|\boldsymbol{y}\|$, $u_1 = \boldsymbol{x}/\|\boldsymbol{x}\|$, $v_1 = \boldsymbol{y}/\|\boldsymbol{y}\|$.

Worked Example 6.4: A tall 3×2 matrix of full column rank.

Find the full SVD of $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$.

Steps 1–3. $A^{\mathsf{T}}A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, eigenvalues $3, 1$ (Worked Example 4.2): $\sigma_1 = \sqrt{3}$, $\sigma_2 = 1$, $v_1 = \frac{1}{\sqrt{2}}(1,1)$, $v_2 = \frac{1}{\sqrt{2}}(1,-1)$.

Step 4. $u_1 = \dfrac{Av_1}{\sqrt{3}} = \dfrac{1}{\sqrt{3}\sqrt{2}}\begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} = \dfrac{1}{\sqrt{6}}\begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$, $\quad u_2 = \dfrac{Av_2}{1} = \dfrac{1}{\sqrt{2}}\begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$.

Step 5. u_3 spans $\mathcal{N}(A^{\mathsf{T}})$: $A^{\mathsf{T}}\boldsymbol{y} = \mathbf{0}$ means $y_1 + y_3 = 0$ and $y_1 + y_2 = 0$, giving $u_3 = \frac{1}{\sqrt{3}}(1,-1,-1)$. Check $u_3 \cdot u_1 \propto 2 - 1 - 1 = 0$, $u_3 \cdot u_2 \propto 0 + 1 - 1 = 0$. ✓

Result. $U = \begin{bmatrix} \frac{2}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} \end{bmatrix}$, $\Sigma = \begin{bmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$, $V = \dfrac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$.

6.4 Signs, uniqueness and a classic mistake

The singular values are unique. The singular vectors are not quite:

- **Signs.** Replacing (u_i, v_i) by $(-u_i, -v_i)$ leaves $\sigma_i u_i v_i^{\mathsf{T}}$ unchanged. So each pair is determined only up to a *simultaneous* sign flip.
- **Repeated singular values.** If $\sigma_i = \sigma_{i+1}$, any rotation of v_i, v_{i+1} (with the matching rotation of u_i, u_{i+1}) works too. For an orthogonal matrix all $\sigma_i = 1$, and *every* orthonormal basis can serve as V.
- **Zero singular values.** The columns of U beyond r (and of V beyond r) can be any orthonormal basis of $\mathcal{N}(A^{\mathsf{T}})$ (respectively $\mathcal{N}(A)$).

The independent-eigenvector mistake. It is tempting to find V from the eigenvectors of $A^{\mathsf{T}}A$ and, *separately*, U from the eigenvectors of AA^{T}. Each eigenvector is only determined up to sign, and the two sign choices are not coordinated. For $A = \left[\begin{smallmatrix} 3 & 0 \\ 4 & 5 \end{smallmatrix}\right]$, the vector $\tilde{u}_2 = \frac{1}{\sqrt{10}}(-3,1)$ is a perfectly good unit eigenvector of AA^{T} for $\lambda = 5$. But with $v_2 = \frac{1}{\sqrt{2}}(1,-1)$,

$$\sigma_1 u_1 v_1^{\mathsf{T}} + \sigma_2 \tilde{u}_2 v_2^{\mathsf{T}} = \frac{3}{2}\begin{bmatrix} 1 & 1 \\ 3 & 3 \end{bmatrix} + \frac{1}{2}\begin{bmatrix} -3 & 3 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 3 \\ 5 & 4 \end{bmatrix} \neq A.$$

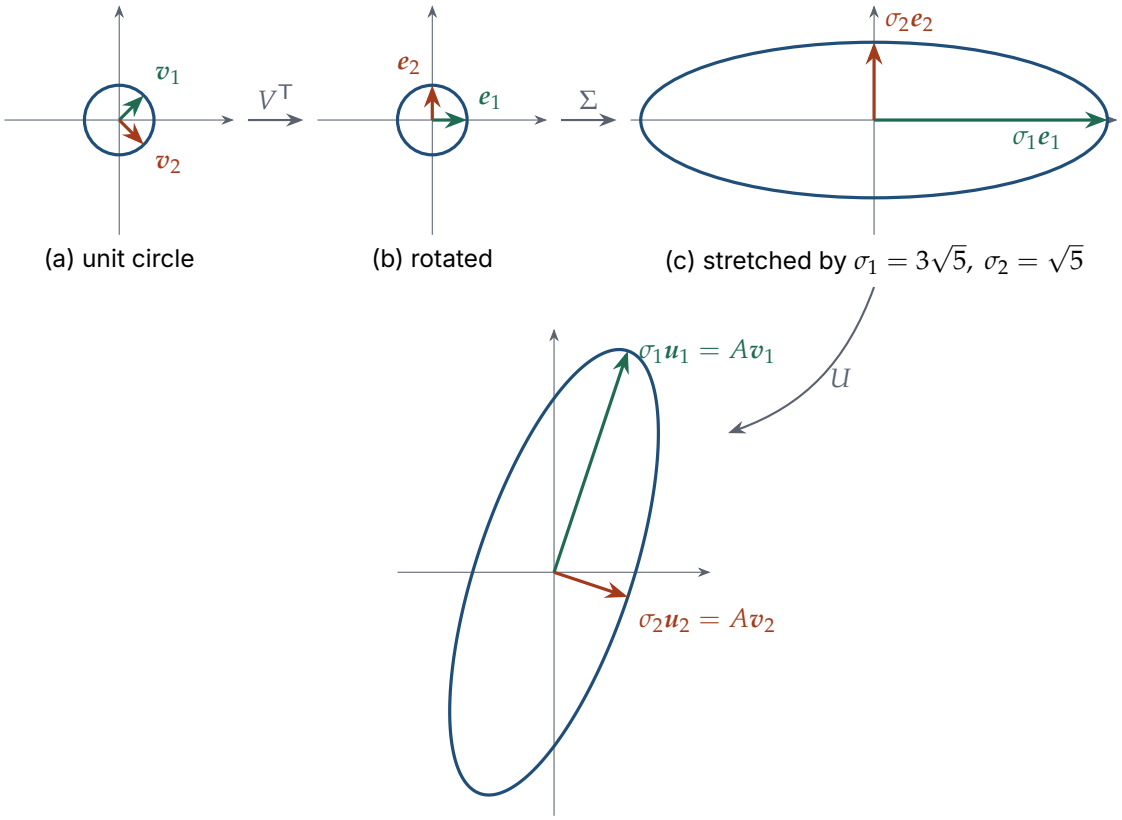
Always compute $u_i = Av_i/\sigma_i$ (or $v_i = A^{\mathsf{T}}u_i/\sigma_i$). Then the signs match automatically.

6.5 Geometry: circles become ellipses

Read $A\boldsymbol{x} = U\Sigma V^{\mathsf{T}}\boldsymbol{x}$ from right to left:

1. V^{T} is orthogonal: a rotation (or reflection) that turns $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots$ into $\boldsymbol{e}_1, \boldsymbol{e}_2, \ldots$;
2. Σ stretches along the axes: $\boldsymbol{e}_i \mapsto \sigma_i \boldsymbol{e}_i$ (and changes dimension from n to m);
3. U is orthogonal: it rotates $\boldsymbol{e}_1, \boldsymbol{e}_2, \ldots$ into $\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots$.

So *every* matrix is rotate–stretch–rotate. In particular A maps the unit sphere to an ellipsoid whose semi-axes are $\sigma_1\boldsymbol{u}_1, \sigma_2\boldsymbol{u}_2, \ldots$. Figure 6.2 shows this for $A = \left[\begin{smallmatrix} 3 & 0 \\ 4 & 5 \end{smallmatrix}\right]$.



(a) unit circle (b) rotated (c) stretched by $\sigma_1 = 3\sqrt{5}$, $\sigma_2 = \sqrt{5}$

(d) rotated by U: the image of the circle under A

Figure 6.2: $A = \left[\begin{smallmatrix} 3 & 0 \\ 4 & 5 \end{smallmatrix}\right] = U\Sigma V^{\mathsf{T}}$ acting on the unit circle, drawn to scale. The right singular vectors $\boldsymbol{v}_1, \boldsymbol{v}_2$ (a) are turned into the axes (b), stretched (c), and turned into the semi-axes $\sigma_1\boldsymbol{u}_1, \sigma_2\boldsymbol{u}_2$ of the image ellipse (d).

Theorem 6.2: Largest and smallest stretch. $\sigma_1 = \max_{\|\boldsymbol{x}\|=1} \|A\boldsymbol{x}\|$, attained at $\boldsymbol{x} = \boldsymbol{v}_1$. For a matrix with n columns and $\sigma_n > 0$, $\sigma_n = \min_{\|\boldsymbol{x}\|=1} \|A\boldsymbol{x}\|$, attained at $\boldsymbol{v}_n$.

Proof. $\|A\boldsymbol{x}\|^2 = \boldsymbol{x}^{\mathsf{T}}A^{\mathsf{T}}A\boldsymbol{x}$, a Rayleigh quotient of $A^{\mathsf{T}}A$ for unit $\boldsymbol{x}$. By Theorem 4.4 it lies between $\lambda_n = \sigma_n^2$ and $\lambda_1 = \sigma_1^2$, with the extremes at the eigenvectors $\boldsymbol{v}_n$ and $\boldsymbol{v}_1$. □

6.6 The SVD and the four fundamental subspaces

The singular vectors are orthonormal bases for all four subspaces of Chapter 2 at once:

Subspace	Orthonormal basis	Reason
row space $\mathcal{C}(A^\mathsf{T}) \subseteq \mathbb{R}^n$	$\boldsymbol{v}_1, \dots, \boldsymbol{v}_r$	$\boldsymbol{v}_i = A^\mathsf{T}\boldsymbol{u}_i/\sigma_i$
null space $\mathcal{N}(A) \subseteq \mathbb{R}^n$	$\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_n$	$A\boldsymbol{v}_i = \mathbf{0}$ for $i > r$
column space $\mathcal{C}(A) \subseteq \mathbb{R}^m$	$\boldsymbol{u}_1, \dots, \boldsymbol{u}_r$	$\boldsymbol{u}_i = A\boldsymbol{v}_i/\sigma_i$
left null space $\mathcal{N}(A^\mathsf{T}) \subseteq \mathbb{R}^m$	$\boldsymbol{u}_{r+1}, \dots, \boldsymbol{u}_m$	$A^\mathsf{T}\boldsymbol{u}_i = \mathbf{0}$ for $i > r$

The rank is the number of non-zero singular values. In Worked Example 6.2, $\boldsymbol{v}_3 = \frac{1}{3}(2, -2, -1)$ spans $\mathcal{N}(A)$; in Worked Example 6.3, $\boldsymbol{u}_2, \boldsymbol{u}_3$ span $\mathcal{N}(A^\mathsf{T})$.

Key idea. The SVD completes the "big picture" of Figure 2.1: A maps the orthonormal basis $\boldsymbol{v}_1, \dots, \boldsymbol{v}_r$ of the row space to the orthogonal basis $\sigma_1\boldsymbol{u}_1, \dots, \sigma_r\boldsymbol{u}_r$ of the column space, and sends $\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_n$ to zero. On the row space, A is invertible — and the pseudo-inverse of Chapter 9 is exactly that inverse.

6.7 SVD versus eigendecomposition

- If S is symmetric **positive semidefinite**, its spectral decomposition $Q\Lambda Q^\mathsf{T}$ *is* an SVD: $U = V = Q, \Sigma = \Lambda$.
- If S is symmetric but has negative eigenvalues, then $\sigma_i = |\lambda_i|$, $\boldsymbol{v}_i = \boldsymbol{q}_i$ and $\boldsymbol{u}_i = \operatorname{sign}(\lambda_i)\boldsymbol{q}_i$.
- For a general square matrix eigenvalues and singular values differ, although $|\det A| = \prod|\lambda_i| = \prod\sigma_i$ (Exercise 6.8).

Worked Example 6.5: A symmetric indefinite matrix.

$S = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ has $\lambda_1 = 3$ with $\boldsymbol{q}_1 = \frac{1}{\sqrt{2}}(1, 1)$ and $\lambda_2 = -1$ with $\boldsymbol{q}_2 = \frac{1}{\sqrt{2}}(1, -1)$.

Singular values. $\sigma_1 = 3, \sigma_2 = |-1| = 1$.

Vectors. $\boldsymbol{v}_1 = \boldsymbol{q}_1, \boldsymbol{u}_1 = S\boldsymbol{v}_1/3 = \boldsymbol{q}_1$; $\boldsymbol{v}_2 = \boldsymbol{q}_2, \boldsymbol{u}_2 = S\boldsymbol{v}_2/1 = -\boldsymbol{q}_2$.

Result. $S = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$. The minus sign moved from Λ into U.

Exercises

Exercise 6.1 $\star$ Find an SVD of $A = \begin{bmatrix} -3 & 0 \\ 0 & 2 \end{bmatrix}$.

Exercise 6.2 $\star$ Find an SVD of $A = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$. What are its rank and its four subspaces?

Exercise 6.3 $\star\star$ Compute the full SVD of $A = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix}$ and check your answer by multiplying out.

Exercise 6.4 $\star\star$ Compute the full SVD of the wide matrix $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$.

Exercise 6.5 $\star\star$ Find the SVD of the rank-one matrix $A = \begin{bmatrix} 2 & 4 \\ 1 & 2 \\ 2 & 4 \end{bmatrix}$ with as little work as possible. Give σ_1, $\boldsymbol{u}_1$, $\boldsymbol{v}_1$ and a basis for $\mathcal{N}(A)$ and $\mathcal{N}(A^{\mathsf{T}})$.

Exercise 6.6 $\star\star$ Show that $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ has $\sigma_1 = \sigma_2 = 1$ and that both $U = A$, $V = I$ and $U = I$, $V = A$ give valid SVDs. Explain the non-uniqueness.

Exercise 6.7 $\star\star\star$ (a) Show that all singular values of an orthogonal matrix are 1. (b) Show that $\|A\boldsymbol{x}\| \le \sigma_1 \|\boldsymbol{x}\|$ for every $\boldsymbol{x}$. (c) Show that if Q_1, Q_2 are orthogonal, then $Q_1 A Q_2$ has the same singular values as A.

Exercise 6.8 $\star\star\star$ (a) Show that A and A^{T} have the same singular values. (b) For square A, show $|\det A| = \sigma_1 \sigma_2 \cdots \sigma_n$. (c) Check (b) on $A = \left[\begin{smallmatrix} 3 & 0 \\ 4 & 5 \end{smallmatrix}\right]$.

CHAPTER 7

Norms, Low-Rank Approximation and Conditioning

The SVD is not only a factorisation; it is a measuring instrument. From the singular values alone you can read off the size of a matrix, how close it is to being singular, how sensitive linear systems with it are, and how well it can be compressed. This chapter collects these facts, culminating in the Eckart–Young theorem: truncating the SVD gives the best low-rank approximation.

7.1 Full, reduced and compact SVD

In the full SVD of an $m \times n$ matrix with $m > n$, the last $m - n$ columns of U multiply zero rows of Σ and contribute nothing. Dropping them gives a more economical factorisation. Three versions are in common use:

Version	U	Σ	V^{T}
full	$m \times m$	$m \times n$	$n \times n$
reduced (thin), $k = \min(m, n)$	$m \times k$	$k \times k$	$k \times n$
compact, $r = \operatorname{rank} A$	$m \times r$	$r \times r$	$r \times n$

The compact SVD $A = U_r \Sigma_r V_r^{\mathsf{T}}$ keeps only the non-zero singular values; U_r and V_r have orthonormal columns but are not square. In Python, `numpy.linalg.svd(A, full_matrices=False)` returns the reduced SVD.

Worked Example 7.1: Reduced SVD of a tall matrix.

From Worked Example 6.4, $A = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$ has the reduced (here also compact) SVD

$$A = \begin{bmatrix} \frac{2}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \sqrt{3} & 0 \\ 0 & 1 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

The column $\boldsymbol{u}_3$ is gone. The 3×2 matrix U_r satisfies $U_r^{\mathsf{T}} U_r = I_2$ but $U_r U_r^{\mathsf{T}} \neq I_3$: it is the projection onto $\mathcal{C}(A)$.

7.2 The outer-product expansion

Multiplying out the compact SVD column-times-row (Chapter 2, view 4) gives

$$\boxed{A = \sigma_1 \boldsymbol{u}_1 \boldsymbol{v}_1^{\mathsf{T}} + \sigma_2 \boldsymbol{u}_2 \boldsymbol{v}_2^{\mathsf{T}} + \cdots + \sigma_r \boldsymbol{u}_r \boldsymbol{v}_r^{\mathsf{T}}}$$

Each term is a rank-one matrix of "size" σ_i (each $\boldsymbol{u}_i \boldsymbol{v}_i^{\mathsf{T}}$ has unit Frobenius norm, see below), and the terms are ordered from most to least important. It is natural to stop the sum early.

7.3 Matrix norms

Definition 7.1: Spectral, Frobenius and nuclear norms. For an $m \times n$ matrix A:

- the **spectral norm** (operator 2-norm) $\|A\|_2 = \max_{\boldsymbol{x} \neq \boldsymbol{0}} \dfrac{\|A\boldsymbol{x}\|}{\|\boldsymbol{x}\|}$;
- the **Frobenius norm** $\|A\|_F = \left(\sum_{i,j} a_{ij}^2 \right)^{1/2}$;
- the **nuclear norm** $\|A\|_* = \sigma_1 + \cdots + \sigma_r$.

Theorem 7.2: Norms from singular values.

$$\|A\|_2 = \sigma_1, \qquad \|A\|_F = \sqrt{\sigma_1^2 + \cdots + \sigma_r^2}.$$

All three norms are unchanged by orthogonal factors: $\|Q_1 A Q_2\| = \|A\|$.

Proof. The first is Theorem 6.2. For the second, $\|A\|_F^2 = \sum_j \|\text{column } j\|^2 = \operatorname{tr}(A^{\mathsf{T}} A)$, and the trace is the sum of the eigenvalues σ_i^2 of $A^{\mathsf{T}} A$. Invariance holds because $Q_1 A Q_2$ has the same singular values (Exercise 6.7). □

Worked Example 7.2: Three norms of one matrix.

For $A = \left[\begin{smallmatrix} 3 & 0 \\ 4 & 5 \end{smallmatrix}\right]$ with $\sigma_1 = 3\sqrt{5}$, $\sigma_2 = \sqrt{5}$:

Spectral. $\|A\|_2 = 3\sqrt{5} \approx 6.708$.

Frobenius. $\sqrt{45 + 5} = \sqrt{50} \approx 7.071$. Direct check: $\sqrt{9 + 0 + 16 + 25} = \sqrt{50}$. ✓

Nuclear. $3\sqrt{5} + \sqrt{5} = 4\sqrt{5} \approx 8.944$.

In general $\|A\|_2 \leq \|A\|_F \leq \sqrt{r}\, \|A\|_2$. The Frobenius norm is easy to compute from the entries; the spectral norm is the one that measures the worst-case amplification $\|A\boldsymbol{x}\| \leq \|A\|_2 \, \|\boldsymbol{x}\|$.

7.4 The Eckart–Young–Mirsky theorem

Theorem 7.3: Best rank-k approximation. Let $A = \sum_{i=1}^{r} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^{\mathsf{T}}$ and for $k < r$ define the truncated SVD

$$A_k = \sigma_1 \boldsymbol{u}_1 \boldsymbol{v}_1^{\mathsf{T}} + \cdots + \sigma_k \boldsymbol{u}_k \boldsymbol{v}_k^{\mathsf{T}}.$$

Then A_k has rank k, and for every matrix B of rank at most k,

$$\|A - B\|_2 \geq \|A - A_k\|_2 = \sigma_{k+1}, \qquad \|A - B\|_F \geq \|A - A_k\|_F = \sqrt{\sigma_{k+1}^2 + \cdots + \sigma_r^2}.$$

Proof (spectral norm). $A - A_k = \sum_{i>k} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^{\mathsf{T}}$ is itself an SVD, with largest singular value σ_{k+1}; so $\|A - A_k\|_2 = \sigma_{k+1}$.

Now let rank $B \leq k$. Then $\dim \mathcal{N}(B) \geq n - k$. The space $W = \operatorname{span}(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{k+1})$ has dimension $k+1$, and $(n-k) + (k+1) > n$, so $\mathcal{N}(B)$ and W share a unit vector $\boldsymbol{x} = \sum_{i \leq k+1} c_i \boldsymbol{v}_i$ with $\sum c_i^2 = 1$. Then

$$\|(A - B)\boldsymbol{x}\|^2 = \|A\boldsymbol{x}\|^2 = \sum_{i \leq k+1} \sigma_i^2 c_i^2 \geq \sigma_{k+1}^2 \sum c_i^2 = \sigma_{k+1}^2.$$

Hence $\|A - B\|_2 \geq \sigma_{k+1}$. (The Frobenius case uses a similar but longer argument.) □

Key idea. The best rank-k approximation to a matrix is obtained by keeping the k largest singular values and zeroing the rest. The error is governed by the *discarded* singular values. If the singular values decay quickly, a small k gives an excellent approximation.

Worked Example 7.3: Best rank-one approximation.

$A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$, with $\sigma_1 = 5$, $\boldsymbol{u}_1 = \frac{1}{\sqrt{2}}(1,1)$, $\boldsymbol{v}_1 = \frac{1}{\sqrt{2}}(1,1,0)$ (Worked Example 6.2).

Approximation. $A_1 = 5\boldsymbol{u}_1\boldsymbol{v}_1^{\mathsf{T}} = \frac{5}{2}\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 2.5 & 2.5 & 0 \\ 2.5 & 2.5 & 0 \end{bmatrix}$.

Error matrix. $A - A_1 = \begin{bmatrix} 0.5 & -0.5 & 2 \\ -0.5 & 0.5 & -2 \end{bmatrix}$ $(= 3\boldsymbol{u}_2\boldsymbol{v}_2^{\mathsf{T}})$.

Error norms. $\|A - A_1\|_F = \sqrt{4 \cdot 0.25 + 2 \cdot 4} = \sqrt{9} = 3 = \sigma_2$, and $\|A - A_1\|_2 = \sigma_2 = 3$ (the error is rank one, so both norms agree).

Relative error. $\|A - A_1\|_F \,/\, \|A\|_F = 3/\sqrt{34} \approx 0.51$. With only two singular values that are not far apart, rank one is a poor approximation here; real data matrices usually have much faster decay.

Storage and energy. Storing A_k as $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k, \boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$ and $\sigma_1, \ldots, \sigma_k$ takes $k(m+n+1)$ numbers instead of mn. For a 1000×1000 image with $k = 50$: 100,050 numbers instead of 10^6, about

10%. The fraction of the "energy" $\|A\|_F^2$ captured by A_k is

$$\frac{\|A_k\|_F^2}{\|A\|_F^2} = \frac{\sigma_1^2 + \cdots + \sigma_k^2}{\sigma_1^2 + \cdots + \sigma_r^2}.$$

Figure 7.1 shows a matrix whose singular values decay extremely fast: the rank-one approximation already captures 97.8% of the energy.

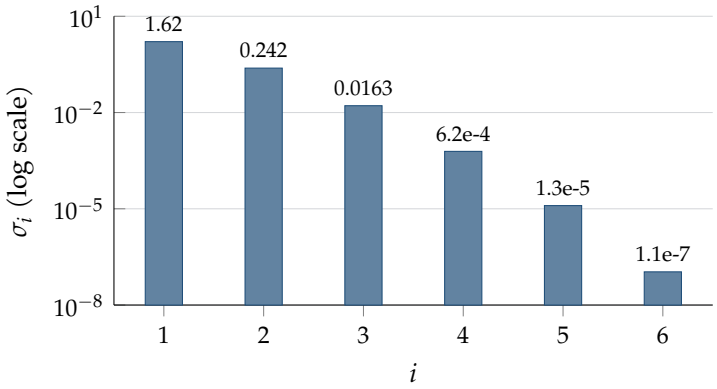


Figure 7.1: Singular values of the 6×6 Hilbert matrix $h_{ij} = 1/(i+j-1)$. Each is roughly 10–50 times smaller than the last. Although the matrix is invertible, it is very close to matrices of low rank.

7.5 Conditioning

Definition 7.4: Condition number. For a square invertible matrix, $\kappa(A) = \|A\|_2 \left\|A^{-1}\right\|_2 = \dfrac{\sigma_1}{\sigma_n}$. For a general matrix of rank r we use $\kappa(A) = \sigma_1/\sigma_r$.

($\left\|A^{-1}\right\|_2 = 1/\sigma_n$ because $A^{-1} = V\Sigma^{-1}U^{\mathsf{T}}$ has singular values $1/\sigma_i$, the largest being $1/\sigma_n$.) The condition number bounds how much relative errors in the data can be amplified in the solution:

Theorem 7.5: Sensitivity of $Ax = b$. If $A\boldsymbol{x} = \boldsymbol{b}$ and $A(\boldsymbol{x} + \delta\boldsymbol{x}) = \boldsymbol{b} + \delta\boldsymbol{b}$, then

$$\frac{\|\delta\boldsymbol{x}\|}{\|\boldsymbol{x}\|} \le \kappa(A)\,\frac{\|\delta\boldsymbol{b}\|}{\|\boldsymbol{b}\|}.$$

Proof. $\delta\boldsymbol{x} = A^{-1}\delta\boldsymbol{b}$, so $\|\delta\boldsymbol{x}\| \le \|\delta\boldsymbol{b}\|/\sigma_n$; and $\|\boldsymbol{b}\| = \|A\boldsymbol{x}\| \le \sigma_1\|\boldsymbol{x}\|$. Divide. □

Worked Example 7.4: An ill-conditioned system.

$A = \begin{bmatrix} 1 & 1 \\ 1 & 1.0001 \end{bmatrix}$. Its singular values are $\sigma_1 \approx 2.00005$ and $\sigma_2 \approx 0.00005$, so $\kappa(A) \approx 40{,}002$.

Original system. $\boldsymbol{b} = (2,\ 2.0001)$ gives $\boldsymbol{x} = (1, 1)$.

Perturbed system. $\boldsymbol{b} + \delta\boldsymbol{b} = (2,\ 2.0002)$: subtracting the equations, $0.0001x_2 = 0.0002$, so $x_2 = 2$ and $x_1 = 0$.

Amplification. The data changed by $\|\delta\boldsymbol{b}\| \,/\, \|\boldsymbol{b}\| \approx 0.0001/2.83 \approx 3.5 \times 10^{-5}$; the solution changed by $\|(-1,1)\| \,/\, \|(1,1)\| = 1$, i.e. 100%. The amplification factor $\approx 28{,}000$ is within the bound $\kappa \approx 40{,}000$.

Distance to singularity. By Eckart–Young with $k = n - 1$, the closest singular matrix to an invertible A is at distance exactly σ_n in the 2-norm. So σ_n measures *how far A is from being singular*, and $1/\kappa = \sigma_n/\sigma_1$ is that distance relative to the size of A. In floating-point arithmetic with machine precision $\varepsilon \approx 2.2 \times 10^{-16}$, a matrix with $\kappa \gtrsim 1/\varepsilon$ is *numerically singular*: its smallest singular values are indistinguishable from rounding noise. We then use the **numerical rank**, the number of singular values above a tolerance such as $\max(m,n)\,\varepsilon\,\sigma_1$.

Exercises

Exercise 7.1 $\star$ For $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$ (singular values 5 and 3) give $\|A\|_2$, $\|A\|_F$, $\|A\|_*$, the rank and $\kappa(A)$. Check $\|A\|_F$ from the entries.

Exercise 7.2 $\star$ Find the best rank-one approximation A_1 of $A = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$ and verify that $\|A - A_1\|_F = \sigma_2$.

Exercise 7.3 $\star\star$ An 800×600 greyscale image is stored as a rank-40 truncated SVD. How many numbers are stored, and what fraction of the original is that? For which k does the truncated SVD stop saving space?

Exercise 7.4 $\star\star$ A matrix has singular values $10, 6, 3, 1$. (a) What fraction of $\|A\|_F^2$ does A_2 capture? (b) Give $\|A - A_2\|_2$ and $\|A - A_2\|_F$, absolute and relative to $\|A\|_2$ and $\|A\|_F$.

Exercise 7.5 $\star\star$ (a) Show $\kappa(Q) = 1$ for every orthogonal Q. (b) Show that $\kappa(A^{\mathsf{T}}A) = \kappa(A)^2$ for a matrix with independent columns. (c) Why does (b) matter when solving least-squares problems?

Exercise 7.6 $\star\star\star$ Prove $\|A\|_F^2 = \sum_i \sigma_i^2$ using $\|A\|_F^2 = \operatorname{tr}(A^{\mathsf{T}}A)$ and the fact that $\operatorname{tr}(XY) = \operatorname{tr}(YX)$.

Exercise 7.7 $\star\star\star$ Show that the smallest singular value of an invertible $n \times n$ matrix equals $\min\{\|E\|_2 : A + E \text{ is singular}\}$, and find a smallest such E for $A = \left[\begin{smallmatrix} 3 & 0 \\ 4 & 5 \end{smallmatrix}\right]$.

CHAPTER 8

Least Squares

When $A\boldsymbol{x} = \boldsymbol{b}$ has no solution — more equations than unknowns, noisy data — we look for the $\boldsymbol{x}$ that makes $A\boldsymbol{x}$ as close as possible to $\boldsymbol{b}$. When it has infinitely many solutions, we look for the shortest. This chapter treats both problems with three tools: the normal equations, QR, and the SVD. The SVD answer will be the pseudo-inverse.

8.1 The least-squares problem

Definition 8.1: Least-squares solution. A vector $\hat{\boldsymbol{x}}$ is a **least-squares solution** of $A\boldsymbol{x} = \boldsymbol{b}$ if it minimises the squared residual

$$\|\boldsymbol{b} - A\boldsymbol{x}\|^2 = \sum_{i=1}^{m} \left(b_i - (A\boldsymbol{x})_i\right)^2.$$

The vector $\boldsymbol{e} = \boldsymbol{b} - A\hat{\boldsymbol{x}}$ is the **residual**.

As $\boldsymbol{x}$ ranges over $\mathbb{R}^n$, $A\boldsymbol{x}$ ranges over $\mathcal{C}(A)$. So we want the point of $\mathcal{C}(A)$ closest to $\boldsymbol{b}$ — the projection of Chapter 3. Hence:

Theorem 8.2: Normal equations. $\hat{\boldsymbol{x}}$ is a least-squares solution if and only if

$$\boxed{A^{\mathsf{T}}A\hat{\boldsymbol{x}} = A^{\mathsf{T}}\boldsymbol{b}}$$

Solutions always exist. The solution is unique if and only if the columns of A are independent, and then $\hat{\boldsymbol{x}} = (A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}\boldsymbol{b}$.

Proof. Geometric. $\|\boldsymbol{b} - A\boldsymbol{x}\|$ is smallest when $A\boldsymbol{x}$ is the orthogonal projection of $\boldsymbol{b}$ onto $\mathcal{C}(A)$, i.e. when $\boldsymbol{b} - A\boldsymbol{x} \perp \mathcal{C}(A)$, i.e. $A^{\mathsf{T}}(\boldsymbol{b} - A\boldsymbol{x}) = \boldsymbol{0}$.

Calculus. $f(\boldsymbol{x}) = \|\boldsymbol{b} - A\boldsymbol{x}\|^2 = \boldsymbol{x}^{\mathsf{T}}A^{\mathsf{T}}A\boldsymbol{x} - 2\boldsymbol{x}^{\mathsf{T}}A^{\mathsf{T}}\boldsymbol{b} + \boldsymbol{b}^{\mathsf{T}}\boldsymbol{b}$ has gradient $2A^{\mathsf{T}}A\boldsymbol{x} - 2A^{\mathsf{T}}\boldsymbol{b}$; setting it to zero gives the same equations, and f is convex because $A^{\mathsf{T}}A$ is PSD.

Existence was Exercise 5.5; uniqueness holds exactly when $A^{\mathsf{T}}A$ is invertible, i.e. (Theorem 5.3) when the columns are independent. □

8.2 Fitting a line

Given data $(t_1, y_1), \dots, (t_m, y_m)$, we look for the line $y = C + Dt$ minimising the sum of squared vertical errors. The unknowns are C, D and the equations $C + Dt_i = y_i$ form the system

$$\begin{bmatrix} 1 & t_1 \\ 1 & t_2 \\ \vdots & \vdots \\ 1 & t_m \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}.$$

Worked Example 8.1: Best line through three points.

Fit $y = C + Dt$ to the points $(0,6)$, $(1,0)$, $(2,0)$.

Set up. $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$, $\boldsymbol{b} = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}$ — exactly Worked Example 3.1.

Normal equations. $\begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \end{bmatrix}$, so $C = 5$, $D = -3$. The best line is $\boxed{y = 5 - 3t}$.

Residuals. Predictions $5, 2, -1$; residual $\boldsymbol{e} = (1, -2, 1)$; $\|\boldsymbol{e}\|^2 = 6$.

Check. $A^{\mathsf{T}}\boldsymbol{e} = (1-2+1,\ 0-2+2) = (0,0)$: the residual is orthogonal to both columns. Orthogonality to the column of ones means the residuals sum to zero — a quick check for any line fit with an intercept.

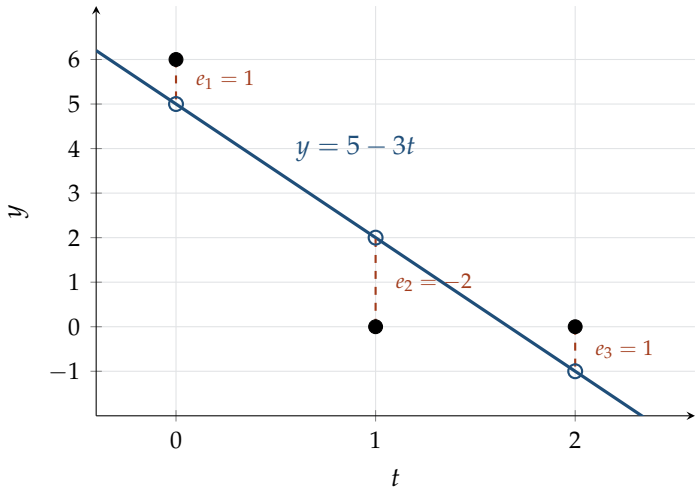


Figure 8.1: The least-squares line through $(0,6)$, $(1,0)$, $(2,0)$. Dots: data; circles: fitted values; dashed: residuals, whose squares sum to 6 — smaller than for any other line.

The same method fits any model that is *linear in the unknown coefficients*: a parabola $C + Dt + Et^2$ (columns 1, t_i, t_i^2), a sum of sines, and so on.

8.3 Least squares with QR

With the reduced QR factorisation $A = QR$ (Q: $m \times n$ with orthonormal columns, R: $n \times n$ invertible upper triangular) the normal equations simplify:

$$A^{\mathsf{T}}A\hat{\boldsymbol{x}} = A^{\mathsf{T}}\boldsymbol{b} \iff R^{\mathsf{T}}Q^{\mathsf{T}}QR\hat{\boldsymbol{x}} = R^{\mathsf{T}}Q^{\mathsf{T}}\boldsymbol{b} \iff \boxed{R\hat{\boldsymbol{x}} = Q^{\mathsf{T}}\boldsymbol{b}}$$

(we cancelled $Q^{\mathsf{T}}Q = I$ and the invertible R^{T}). Solve by back substitution.

Worked Example 8.2: The line fit via QR.

Same data as Worked Example 8.1.

Gram–Schmidt on the columns. $\boldsymbol{a}_1 = (1,1,1)$: $\boldsymbol{q}_1 = \frac{1}{\sqrt{3}}(1,1,1)$, $r_{11} = \sqrt{3}$. $\boldsymbol{a}_2 = (0,1,2)$: $r_{12} = \boldsymbol{q}_1 \cdot \boldsymbol{a}_2 = \frac{3}{\sqrt{3}} = \sqrt{3}$; $\boldsymbol{w}_2 = \boldsymbol{a}_2 - \sqrt{3}\boldsymbol{q}_1 = (-1,0,1)$; $r_{22} = \sqrt{2}$, $\boldsymbol{q}_2 = \frac{1}{\sqrt{2}}(-1,0,1)$.

Right-hand side. $Q^{\mathsf{T}}\boldsymbol{b} = \left(\frac{6}{\sqrt{3}}, \frac{-6}{\sqrt{2}}\right) = \left(2\sqrt{3}, -3\sqrt{2}\right)$.

Back substitution. $\begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{2} \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 2\sqrt{3} \\ -3\sqrt{2} \end{bmatrix}$: $D = -3$, then $\sqrt{3}C - 3\sqrt{3} = 2\sqrt{3}$, $C = 5$. ✓

8.4 Least squares with the SVD

Let $A = U\Sigma V^{\mathsf{T}}$ be the full SVD and write $\boldsymbol{c} = U^{\mathsf{T}}\boldsymbol{b}$ (coordinates of $\boldsymbol{b}$ in the $\boldsymbol{u}$-basis) and $\boldsymbol{y} = V^{\mathsf{T}}\boldsymbol{x}$ (coordinates of $\boldsymbol{x}$ in the $\boldsymbol{v}$-basis). Since U^{T} preserves lengths,

$$\|\boldsymbol{b} - A\boldsymbol{x}\|^2 = \left\|U^{\mathsf{T}}\boldsymbol{b} - \Sigma V^{\mathsf{T}}\boldsymbol{x}\right\|^2 = \|\boldsymbol{c} - \Sigma\boldsymbol{y}\|^2 = \sum_{i=1}^{r}(c_i - \sigma_i y_i)^2 + \sum_{i=r+1}^{m} c_i^2.$$

The problem has *decoupled*: each y_i appears in exactly one term.

- For $i \le r$, choose $y_i = c_i/\sigma_i$, making the i-th term zero.
- The terms c_i^2 for $i > r$ cannot be changed: they are the unavoidable residual, the part of $\boldsymbol{b}$ in $\mathcal{N}(A^{\mathsf{T}})$.
- For $i > r$ (if $r < n$) the y_i do not appear at all: they are free. These are the components of $\boldsymbol{x}$ in $\mathcal{N}(A)$.

Choosing the free $y_i = 0$ gives the shortest solution (since $\|\boldsymbol{x}\| = \|\boldsymbol{y}\|$):

Theorem 8.3: SVD solution of least squares. The minimum-norm least-squares solution of $A\boldsymbol{x} = \boldsymbol{b}$ is

$$\boxed{\boldsymbol{x}^{+} = \sum_{i=1}^{r} \frac{\boldsymbol{u}_i^{\mathsf{T}}\boldsymbol{b}}{\sigma_i}\boldsymbol{v}_i}$$

and the minimal residual is $\|\boldsymbol{b} - A\boldsymbol{x}^{+}\|^2 = \sum_{i>r}(\boldsymbol{u}_i^{\mathsf{T}}\boldsymbol{b})^2$. All least-squares solutions are $\boldsymbol{x}^{+} + \boldsymbol{0}$ with $\boldsymbol{0} \in \mathcal{N}(A)$.

Worked Example 8.3: SVD least squares for a tall matrix.

Solve $A\boldsymbol{x} \approx \boldsymbol{b}$ for $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$ and $\boldsymbol{b} = (1,2,3)$, using the SVD from Worked Example 6.4.

Coefficients. $\boldsymbol{u}_1^{\mathsf{T}}\boldsymbol{b} = \frac{1}{\sqrt{6}}(2+2+3) = \frac{7}{\sqrt{6}}$, $\boldsymbol{u}_2^{\mathsf{T}}\boldsymbol{b} = \frac{1}{\sqrt{2}}(0-2+3) = \frac{1}{\sqrt{2}}$.

Solution.

$$\hat{\boldsymbol{x}} = \frac{7/\sqrt{6}}{\sqrt{3}} \cdot \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix} + \frac{1/\sqrt{2}}{1} \cdot \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix} = \frac{7}{6}\begin{bmatrix}1\\1\end{bmatrix} + \frac{1}{2}\begin{bmatrix}1\\-1\end{bmatrix} = \begin{bmatrix}5/3\\2/3\end{bmatrix}.$$

Check with normal equations. $A^{\mathsf{T}}A = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$, $A^{\mathsf{T}}\boldsymbol{b} = (4,3)$: $\hat{\boldsymbol{x}} = \frac{1}{3}\left[\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right]\left[\begin{smallmatrix} 4 \\ 3 \end{smallmatrix}\right] = \frac{1}{3}(5,2)$. ✓

Residual. $\boldsymbol{e} = \boldsymbol{b} - A\hat{\boldsymbol{x}} = (1,2,3) - \left(\frac{7}{3}, \frac{2}{3}, \frac{5}{3}\right) = \frac{4}{3}(-1,1,1)$, a multiple of $\boldsymbol{u}_3 = \frac{1}{\sqrt{3}}(1,-1,-1)$, and $\|\boldsymbol{e}\| = \frac{4}{\sqrt{3}} = |\boldsymbol{u}_3^{\mathsf{T}}\boldsymbol{b}|$. ✓

8.5 Underdetermined systems and the minimum-norm solution

If A has more unknowns than equations (and full row rank), $A\boldsymbol{x} = \boldsymbol{b}$ has infinitely many exact solutions: one particular solution plus anything in $\mathcal{N}(A)$. By Section 3.5 the shortest is the one with no null-space component — the solution lying in the row space $\mathcal{C}(A^{\mathsf{T}})$. Writing $\boldsymbol{x} = A^{\mathsf{T}}\boldsymbol{w}$ and substituting, $AA^{\mathsf{T}}\boldsymbol{w} = \boldsymbol{b}$, so

$$\boldsymbol{x}^{+} = A^{\mathsf{T}}(AA^{\mathsf{T}})^{-1}\boldsymbol{b} \qquad \text{(full row rank)}.$$

Worked Example 8.4: Shortest solution of one equation.

Solve $x_1 + 2x_2 = 5$ with $\|\boldsymbol{x}\|$ as small as possible.

All solutions. $\boldsymbol{x} = (5,0) + t(-2,1)$, since $(-2,1)$ spans $\mathcal{N}(A)$ for $A = [1\ 2]$.

Minimise. $\|\boldsymbol{x}\|^2 = (5-2t)^2 + t^2$; derivative $-4(5-2t) + 2t = 10t - 20 = 0$ gives $t = 2$, so $\boldsymbol{x}^{+} = (1,2)$.

Row-space check. $(1,2)$ is a multiple of the row $(1,2)$. ✓ And $A^{\mathsf{T}}(AA^{\mathsf{T}})^{-1}\boldsymbol{b} = (1,2) \cdot \frac{1}{5} \cdot 5 = (1,2)$. ✓

When A is rank-deficient (neither independent columns nor independent rows), *both* difficulties occur: there is no exact solution and the least-squares solution is not unique. Theorem 8.3 handles this case too — and it is the reason the SVD is the method of last resort.

8.6 Which method?

	Normal equations	QR	SVD
Solves	$A^{\mathsf{T}}A\boldsymbol{x} = A^{\mathsf{T}}\boldsymbol{b}$	$R\boldsymbol{x} = Q^{\mathsf{T}}\boldsymbol{b}$	$\boldsymbol{x} = \sum \frac{\boldsymbol{u}_i^{\mathsf{T}}\boldsymbol{b}}{\sigma_i}\boldsymbol{v}_i$
Cost ($m \gg n$)	$\approx mn^2$ (cheapest)	$\approx 2mn^2$	several $\times mn^2$
Error amplified by	$\kappa(A)^2$	$\kappa(A)$	$\kappa(A)$
Rank-deficient A?	fails ($A^{\mathsf{T}}A$ singular)	needs pivoting	yes, min-norm solution
Reveals numerical rank?	no	partly	yes

Use QR as the default for well-posed problems, the SVD when the matrix may be rank-deficient or ill-conditioned, and the normal equations only when A is well conditioned and speed matters. The Lab Manual compares all three on an ill-conditioned polynomial fit.

Exercises

Exercise 8.1 $\star$ Fit a line $y = C + Dt$ to $(1,1)$, $(2,2)$, $(3,2)$ using the normal equations. Give the residual and check that it is orthogonal to both columns.

Exercise 8.2 $\star$ Find the minimum-norm solution of $x_1 + x_2 + x_3 = 3$.

Exercise 8.3 $\star\star$ Solve Exercise 8.1 again, this time with QR (Gram–Schmidt on the columns of A, then back substitution).

Exercise 8.4 $\star\star$ Fit a parabola $y = C + Dt + Et^2$ to the four points $(-1,4)$, $(0,6)$, $(1,0)$, $(2,6)$. Give the residual.

Exercise 8.5 $\star\star$ Show that the least-squares residual $\boldsymbol{e} = \boldsymbol{b} - A\hat{\boldsymbol{x}}$ lies in $\mathcal{N}(A^{\mathsf{T}})$, and that $\|\boldsymbol{b}\|^2 = \|A\hat{\boldsymbol{x}}\|^2 + \|\boldsymbol{e}\|^2$.

Exercise 8.6 $\star\star\star$ Prove that among all solutions of a consistent system $A\boldsymbol{x} = \boldsymbol{b}$, there is exactly one in the row space $\mathcal{C}(A^{\mathsf{T}})$, and that it is the one of minimal norm.

Exercise 8.7 $\star\star$ For $A = \begin{bmatrix} 1 & 1 \\ 1 & 1+\delta \end{bmatrix}$ with small $\delta > 0$, estimate $\kappa(A)$ and $\kappa(A^{\mathsf{T}}A)$. If you solve with the normal equations in arithmetic with 16 significant digits and $\delta = 10^{-6}$, roughly how many correct digits can you expect?

CHAPTER 9

The Moore–Penrose Pseudo-Inverse

An invertible matrix has an inverse that undoes it. A rectangular or singular matrix cannot be undone — it may destroy information (a non-trivial null space) and it may not reach every output (a proper column space). The pseudo-inverse is the best we can do: it undoes A where A is invertible, namely from the column space back to the row space, and sends everything else to zero. Applied to $\boldsymbol{b}$, it returns the minimum-norm least-squares solution of Chapter 8.

9.1 Definition via the SVD

Theorem 8.3 expressed the minimum-norm least-squares solution as $\boldsymbol{x}^+ = \sum_{i \le r} \frac{\boldsymbol{u}_i^{\mathsf{T}} \boldsymbol{b}}{\sigma_i} \boldsymbol{v}_i$. This is linear in $\boldsymbol{b}$, so it is a matrix times $\boldsymbol{b}$:

$$\boldsymbol{x}^+ = \Big(\sum_{i=1}^{r} \frac{1}{\sigma_i} \boldsymbol{v}_i \boldsymbol{u}_i^{\mathsf{T}} \Big) \boldsymbol{b}.$$

Definition 9.1: Moore–Penrose pseudo-inverse. Let $A = U \Sigma V^{\mathsf{T}}$ be an SVD of the $m \times n$ matrix A, with rank r. The **pseudo-inverse** of A is the $n \times m$ matrix

$$\boxed{A^+ = V \Sigma^+ U^{\mathsf{T}} = \sum_{i=1}^{r} \frac{1}{\sigma_i} \boldsymbol{v}_i \boldsymbol{u}_i^{\mathsf{T}}}$$

where Σ^+ is the $n \times m$ matrix with $1/\sigma_1, \ldots, 1/\sigma_r$ on its diagonal and zeros elsewhere.

To build Σ^+: *transpose* Σ, then invert the non-zero entries and leave the zeros. For example

$$\Sigma = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \end{bmatrix} \Longrightarrow \Sigma^+ = \begin{bmatrix} 1/5 & 0 \\ 0 & 1/3 \\ 0 & 0 \end{bmatrix}, \qquad \Sigma = \begin{bmatrix} \sqrt{6} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \Longrightarrow \Sigma^+ = \begin{bmatrix} 1/\sqrt{6} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

What A^+ does to the singular vectors is the reverse of what A does:

$$A \boldsymbol{v}_i = \sigma_i \boldsymbol{u}_i \text{ and } A^+ \boldsymbol{u}_i = \frac{1}{\sigma_i} \boldsymbol{v}_i \quad (i \le r); \qquad A \boldsymbol{v}_i = \boldsymbol{0} \ (i > r), \quad A^+ \boldsymbol{u}_i = \boldsymbol{0} \ (i > r).$$

Worked Example 9.1: Pseudo-inverse of an invertible matrix.
$A = \left[\begin{smallmatrix} 3 & 0 \\ 4 & 5 \end{smallmatrix}\right]$ with the SVD of Worked Example 6.1.

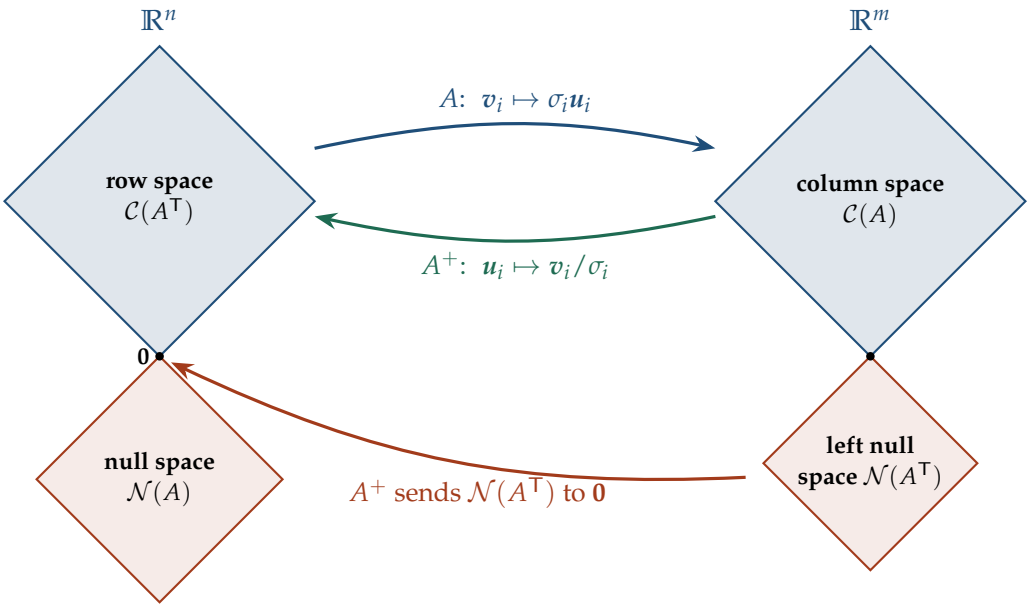


Figure 9.1: A maps the row space one-to-one onto the column space. A^+ inverts that map, and sends the left null space to zero. Every output of A^+ lies in the row space.

Formula. $A^+ = V\Sigma^{-1}U^\mathsf{T} = \dfrac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}\begin{bmatrix}\frac{1}{3\sqrt{5}} & 0\\ 0 & \frac{1}{\sqrt{5}}\end{bmatrix}\dfrac{1}{\sqrt{10}}\begin{bmatrix}1 & 3\\ 3 & -1\end{bmatrix}.$

Multiply. The scalar factors give $\frac{1}{\sqrt{2}\sqrt{10}\sqrt{5}} = \frac{1}{10}$, and

$$\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}\begin{bmatrix}\frac{1}{3} & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & 3\\ 3 & -1\end{bmatrix} = \begin{bmatrix}\frac{1}{3} & 1\\ \frac{1}{3} & -1\end{bmatrix}\begin{bmatrix}1 & 3\\ 3 & -1\end{bmatrix} = \begin{bmatrix}\frac{10}{3} & 0\\ -\frac{8}{3} & 2\end{bmatrix}.$$

So $A^+ = \dfrac{1}{10}\begin{bmatrix}\frac{10}{3} & 0\\ -\frac{8}{3} & 2\end{bmatrix} = \dfrac{1}{15}\begin{bmatrix}5 & 0\\ -4 & 3\end{bmatrix}.$

Compare. $A^{-1} = \dfrac{1}{15}\begin{bmatrix}5 & 0\\ -4 & 3\end{bmatrix}$ by the 2×2 formula ($\det A = 15$). Equal, as they must be.

9.2 The Penrose conditions

The SVD of A is not unique, so we should check that A^+ does not depend on which SVD we use. The cleanest way is an algebraic characterisation.

Theorem 9.2: Penrose conditions. For every matrix A there is exactly one matrix X satisfying

$$(1)\ AXA = A, \qquad (2)\ XAX = X, \qquad (3)\ (AX)^{\mathsf{T}} = AX, \qquad (4)\ (XA)^{\mathsf{T}} = XA,$$

and it is $X = A^{+}$.

Proof. *A^{+} satisfies them.* With $A = U\Sigma V^{\mathsf{T}}$ and $A^{+} = V\Sigma^{+}U^{\mathsf{T}}$: $AA^{+} = U\Sigma\Sigma^{+}U^{\mathsf{T}}$ and $A^{+}A = V\Sigma^{+}\Sigma V^{\mathsf{T}}$. Both $\Sigma\Sigma^{+}$ $(m \times m)$ and $\Sigma^{+}\Sigma$ $(n \times n)$ are diagonal with r ones followed by zeros, so (3) and (4) hold, and $\Sigma\Sigma^{+}\Sigma = \Sigma$, $\Sigma^{+}\Sigma\Sigma^{+} = \Sigma^{+}$ give (1) and (2).

Uniqueness. Suppose X and Y both satisfy (1)–(4). Using the conditions in turn,

$$X = XAX \overset{(3)}{=} X(AX)^{\mathsf{T}} = XX^{\mathsf{T}}A^{\mathsf{T}} \overset{(1)}{=} XX^{\mathsf{T}}(AYA)^{\mathsf{T}} = X(AX)^{\mathsf{T}}(AY)^{\mathsf{T}} \overset{(3)}{=} XAXAY \overset{(2)}{=} XAY,$$

$$Y = YAY \overset{(4)}{=} (YA)^{\mathsf{T}}Y = A^{\mathsf{T}}Y^{\mathsf{T}}Y \overset{(1)}{=} (AXA)^{\mathsf{T}}Y^{\mathsf{T}}Y = (XA)^{\mathsf{T}}(YA)^{\mathsf{T}}Y \overset{(4)}{=} XAYAY \overset{(2)}{=} XAY.$$

Hence $X = Y$. □

Key idea. To check that a candidate X is the pseudo-inverse, verify the four Penrose conditions. In practice, check (1) first — $AXA = A$ catches most mistakes — and then the symmetry of AX and XA.

9.3 Formulas for special cases

Theorem 9.3: Full-rank formulas.

1. If A is square and invertible, $A^{+} = A^{-1}$.
2. If A has **independent columns** (full column rank, $r = n$), then $A^{+} = (A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}$, and $A^{+}A = I_n$: A^{+} is a left inverse.
3. If A has **independent rows** (full row rank, $r = m$), then $A^{+} = A^{\mathsf{T}}(AA^{\mathsf{T}})^{-1}$, and $AA^{+} = I_m$: A^{+} is a right inverse.
4. If $A = \sigma \boldsymbol{u}\boldsymbol{v}^{\mathsf{T}}$ has rank one, $A^{+} = \dfrac{1}{\sigma}\boldsymbol{v}\boldsymbol{u}^{\mathsf{T}} = \dfrac{A^{\mathsf{T}}}{\|A\|_F^2}$.

Proof. (2) With the reduced SVD $A = U_n\Sigma_n V^{\mathsf{T}}$ (Σ_n invertible, V square): $(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}} = (V\Sigma_n^2 V^{\mathsf{T}})^{-1}V\Sigma_n U_n^{\mathsf{T}} = V\Sigma_n^{-2}V^{\mathsf{T}}V\Sigma_n U_n^{\mathsf{T}} = V\Sigma_n^{-1}U_n^{\mathsf{T}} = A^{+}$. (3) is (2) applied to A^{T} together with $(A^{\mathsf{T}})^{+} = (A^{+})^{\mathsf{T}}$ (Exercise 9.7). (1) is the case of both (2) and (3). (4) is the definition with one term; and $\|A\|_F^2 = \sigma^2$, $A^{\mathsf{T}} = \sigma\boldsymbol{v}\boldsymbol{u}^{\mathsf{T}}$. □

Further special cases: for a diagonal matrix, invert the non-zero diagonal entries and leave the zeros; the pseudo-inverse of a zero matrix is the zero matrix of transposed shape; for a non-zero column vector, $\boldsymbol{a}^{+} = \boldsymbol{a}^{\mathsf{T}} / \|\boldsymbol{a}\|^2$; for a scalar, $a^{+} = 1/a$ if $a \neq 0$ and $0^{+} = 0$.

Common pitfall. Do not use $(A^\mathsf{T}A)^{-1}A^\mathsf{T}$ when the columns of A are dependent — $A^\mathsf{T}A$ is singular and the formula is meaningless. And even when the columns are independent but nearly dependent, forming $A^\mathsf{T}A$ squares the condition number (Exercise 7.5). The SVD formula always works.

Worked Example 9.2: A wide matrix: three ways.

$A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$ with the SVD of Worked Example 6.2 ($\sigma_1 = 5, \sigma_2 = 3$).

Sum of outer products.

$$A^+ = \frac{1}{5}v_1u_1^\mathsf{T} + \frac{1}{3}v_2u_2^\mathsf{T} = \frac{1}{10}\begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix} + \frac{1}{18}\begin{bmatrix} 1 & -1 \\ -1 & 1 \\ 4 & -4 \end{bmatrix}$$
$$= \frac{1}{90}\begin{bmatrix} 14 & 4 \\ 4 & 14 \\ 20 & -20 \end{bmatrix} = \frac{1}{45}\begin{bmatrix} 7 & 2 \\ 2 & 7 \\ 10 & -10 \end{bmatrix}.$$

Right-inverse formula (the rows are independent). $AA^\mathsf{T} = \left[\begin{smallmatrix} 17 & 8 \\ 8 & 17 \end{smallmatrix}\right]$, $(AA^\mathsf{T})^{-1} = \frac{1}{225}\left[\begin{smallmatrix} 17 & -8 \\ -8 & 17 \end{smallmatrix}\right]$, and

$$A^\mathsf{T}(AA^\mathsf{T})^{-1} = \frac{1}{225}\begin{bmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{bmatrix}\begin{bmatrix} 17 & -8 \\ -8 & 17 \end{bmatrix} = \frac{1}{225}\begin{bmatrix} 35 & 10 \\ 10 & 35 \\ 50 & -50 \end{bmatrix} = \frac{1}{45}\begin{bmatrix} 7 & 2 \\ 2 & 7 \\ 10 & -10 \end{bmatrix}. \checkmark$$

Check $AA^+ = I_2$. Row 1 of A times A^+: $\frac{1}{45}(21 + 4 + 20,\ 6 + 14 - 20) = (1, 0)$; row 2 similarly gives $(0, 1)$. ✓

Worked Example 9.3: A tall matrix of full column rank.

$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$.

Left-inverse formula. $A^\mathsf{T}A = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$, $(A^\mathsf{T}A)^{-1} = \frac{1}{3}\left[\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right]$, so

$$A^+ = \frac{1}{3}\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \end{bmatrix}.$$

$A^+A = I_2$. $\frac{1}{3}\left[\begin{smallmatrix} 2+1 & -1+1 \\ -1+1 & 2+1 \end{smallmatrix}\right] = I_2$. ✓

AA^+ is a projection, not I_3. $AA^+ = \frac{1}{3}\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$: symmetric, trace $2 = \operatorname{rank} A$, and it

maps $(1,1,-1) \in \mathcal{N}(A^{\mathsf{T}})$ to $\mathbf{0}$.

Worked Example 9.4: A rank-deficient system.

Solve $A\boldsymbol{x} \approx \boldsymbol{b}$ with $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}$ and $\boldsymbol{b} = (1,2,4)$.

Diagnosis. $A = (1,2,3)^{\mathsf{T}}(1,2)$ has rank one: no independent columns, no independent rows. Neither full-rank formula applies.

Pseudo-inverse. Rank-one formula: $\|A\|_F^2 = (1+4+9)(1+4) = 70$, so $A^+ = \frac{1}{70}\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$.

Solution. $\boldsymbol{x}^+ = A^+\boldsymbol{b} = \frac{1}{70}(1+4+12,\ 2+8+24) = \frac{1}{70}(17,34) = \frac{17}{70}(1,2)$.

Residual. $A\boldsymbol{x}^+ = \frac{17}{70}\cdot 5\,(1,2,3) = \frac{17}{14}(1,2,3)$, so $\boldsymbol{e} = \boldsymbol{b} - A\boldsymbol{x}^+ = \left(-\frac{3}{14}, -\frac{3}{7}, \frac{5}{14}\right)$; check $\boldsymbol{e}\cdot(1,2,3) = \frac{-3-12+15}{14} = 0$. ✓

Other solutions. Every $\boldsymbol{x}^+ + t(-2,1)$ has the same residual, since $(-2,1) \in \mathcal{N}(A)$. Because $\boldsymbol{x}^+ \perp (-2,1)$, $\|\boldsymbol{x}^+ + t(-2,1)\|^2 = \|\boldsymbol{x}^+\|^2 + 5t^2$, smallest at $t = 0$.

Penrose check. $AA^+ = \frac{1}{14}\left[\begin{smallmatrix} 1&2&3\\2&4&6\\3&6&9 \end{smallmatrix}\right]$ and $A^+A = \frac{1}{5}\left[\begin{smallmatrix} 1&2\\2&4 \end{smallmatrix}\right]$ are symmetric; and $AA^+A = \frac{1}{14}(1,2,3)^{\mathsf{T}}(1,2,3)\cdot(1,2,3)^{\mathsf{T}}(1,2) = \frac{14}{14}A = A$. ✓

Worked Example 9.5: A singular square matrix.

$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ has no inverse. It is rank one with $\|A\|_F^2 = 4$, so $A^+ = A^{\mathsf{T}}/4 = \frac{1}{4}\left[\begin{smallmatrix} 1&1\\1&1 \end{smallmatrix}\right]$. For $\boldsymbol{b} = (1,3)$, $\boldsymbol{x}^+ = A^+\boldsymbol{b} = (1,1)$ and $A\boldsymbol{x}^+ = (2,2)$: the system $x_1 + x_2 = 1$, $x_1 + x_2 = 3$ is inconsistent, and the best compromise sets $x_1 + x_2$ to the average, 2, splitting it equally.

9.4 Projections and the general solution

Theorem 9.4: The four projections.

- $AA^+ = U_r U_r^{\mathsf{T}}$ is the orthogonal projection onto $\mathcal{C}(A)$;
- $I - AA^+$ is the orthogonal projection onto $\mathcal{N}(A^{\mathsf{T}})$;
- $A^+A = V_r V_r^{\mathsf{T}}$ is the orthogonal projection onto the row space $\mathcal{C}(A^{\mathsf{T}})$;
- $I - A^+A$ is the orthogonal projection onto $\mathcal{N}(A)$.

Consequently the set of *all* least-squares solutions is

$$\boldsymbol{x} = A^+\boldsymbol{b} + (I - A^+A)\mathbf{0}, \qquad \mathbf{0} \in \mathbb{R}^n \text{ arbitrary},$$

and $A\boldsymbol{x} = \boldsymbol{b}$ has an exact solution if and only if $AA^+\boldsymbol{b} = \boldsymbol{b}$.

Theorem 9.5: The pseudo-inverse solves least squares. For every $\boldsymbol{b}$, $\boldsymbol{x}^{+} = A^{+}\boldsymbol{b}$ is the unique vector that (a) minimises $\|\boldsymbol{b} - A\boldsymbol{x}\|$ and (b) among all such minimisers has the smallest norm.

Proof. This is Theorem 8.3 rewritten: $A^{+}\boldsymbol{b} = \sum_{i \le r} \sigma_i^{-1} \boldsymbol{v}_i (\boldsymbol{u}_i^{\mathsf{T}} \boldsymbol{b})$. Uniqueness: two minimum-norm minimisers would both lie in the row space (Exercise 8.6) and differ by a null-space vector, hence be equal. □

9.5 Algebraic properties — and some that fail

The pseudo-inverse behaves like an inverse in many ways:

$$(A^{+})^{+} = A, \qquad (A^{\mathsf{T}})^{+} = (A^{+})^{\mathsf{T}}, \qquad (cA)^{+} = c^{-1}A^{+} \ (c \neq 0),$$
$$\operatorname{rank} A^{+} = \operatorname{rank} A, \qquad (Q_1 A Q_2)^{+} = Q_2^{\mathsf{T}} A^{+} Q_1^{\mathsf{T}} \ \ (Q_1, Q_2 \text{ orthogonal}).$$

Also $A^{+} = (A^{\mathsf{T}}A)^{+}A^{\mathsf{T}} = A^{\mathsf{T}}(AA^{\mathsf{T}})^{+}$ always. But two familiar rules **fail**:

- $A^{+}A \neq I$ and $AA^{+} \neq I$ in general (they are projections);
- $(AB)^{+} \neq B^{+}A^{+}$ in general. Example: $A = \begin{bmatrix} 1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Then $AB = [1]$ and $(AB)^{+} = [1]$, but $A^{+} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $B^{+} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \end{bmatrix}$ and $B^{+}A^{+} = \begin{bmatrix} \frac{1}{2} \end{bmatrix}$.

9.6 The pseudo-inverse is not continuous

Consider $A_\varepsilon = \begin{bmatrix} 1 & 0 \\ 0 & \varepsilon \end{bmatrix}$. For $\varepsilon \neq 0$, $A_\varepsilon^{+} = \begin{bmatrix} 1 & 0 \\ 0 & 1/\varepsilon \end{bmatrix}$, which blows up as $\varepsilon \to 0$; but $A_0^{+} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$. A tiny change in A (from $\varepsilon = 0$ to $\varepsilon = 10^{-12}$) makes a huge change in A^{+}. Numerically, every exactly singular matrix becomes non-singular after rounding, so a naive pseudo-inverse would invert pure rounding noise. The cure, in Chapter 10, is to treat singular values below a tolerance as zero, or to regularise.

Exercises

Exercise 9.1 ⋆ Write down D^{+} for $D = \operatorname{diag}(2, 0, -5)$ and for the 2×3 matrix $E = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.

Exercise 9.2 ⋆ Find the pseudo-inverse of the column vector $\boldsymbol{a} = (1, 2, 2)$ and of the row vector $\boldsymbol{a}^{\mathsf{T}}$. Use it to find the best multiple t with $t\boldsymbol{a} \approx \boldsymbol{b} = (1, 1, 1)$.

Exercise 9.3 ⋆⋆ Compute A^{+} for $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}$ and use it to solve $A\boldsymbol{x} \approx \boldsymbol{b}$ with $\boldsymbol{b} = (1, 3, 5)$. What is the residual?

Exercise 9.4 ⋆⋆ Compute A^{+} for $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ (a) with the right-inverse formula and (b) from the SVD of Exercise 6.4.

Exercise 9.5 ⋆⋆ Verify all four Penrose conditions for $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}$ and $X = \frac{1}{4}\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$.

Exercise 9.6 ⋆⋆ Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$ (Worked Example 9.3) and $\boldsymbol{b} = (1, 1, 0)$. Find $\boldsymbol{x}^{+} = A^{+}\boldsymbol{b}$, the projection $AA^{+}\boldsymbol{b}$ and the residual. Is the system consistent?

Exercise 9.7 ⋆⋆⋆ Prove that $(A^{\mathsf{T}})^{+} = (A^{+})^{\mathsf{T}}$ and $(A^{+})^{+} = A$.

Exercise 9.8 ⋆⋆⋆ Show directly from the Penrose conditions that $P = AA^{+}$ satisfies $P^2 = P = P^{\mathsf{T}}$, and that $\mathcal{C}(P) = \mathcal{C}(A)$. Conclude that AA^{+} is the orthogonal projection onto $\mathcal{C}(A)$.

Exercise 9.9 ⋆⋆ Let $A = \begin{bmatrix} 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. Compute $(AB)^{+}$ and $B^{+}A^{+}$. Are they equal?

CHAPTER 10

Applications and Numerical Practice

This last chapter turns the theory into practice. Real data contain noise; real computers round. Both force us to think about *small* singular values: they make the pseudo-inverse unstable, and they are often not even reliably known. We look at numerical rank, at two ways to regularise (truncation and Tikhonov), at principal component analysis, and finally at how the SVD is actually computed.

10.1 Numerical rank and the tolerance

In floating-point arithmetic a matrix that should have rank r will usually have $\sigma_{r+1}, \sigma_{r+2}, \dots$ that are tiny but not zero — of the order of $\varepsilon\,\sigma_1$, where $\varepsilon \approx 2.2 \times 10^{-16}$ is the machine precision. Inverting them would produce numbers of size 10^{16} out of rounding noise. Every practical pseudo-inverse therefore uses a **tolerance** τ:

$$A_\tau^{+} = \sum_{\sigma_i > \tau} \frac{1}{\sigma_i} \boldsymbol{v}_i \boldsymbol{u}_i^{\mathsf{T}}.$$

The number of singular values above τ is the **numerical rank**. By default NumPy's `matrix_rank` uses $\tau = \max(m,n)\,\varepsilon\,\sigma_1$, and its `pinv` uses $\tau = 10^{-15}\sigma_1$. If your data are only accurate to, say, three digits, a tolerance of $10^{-3}\sigma_1$ is more honest.

> **Worked Example 10.1: Numerical rank.**
>
> A 3×3 matrix has computed singular values 3.2, 1.1 and 4×10^{-16}. With $\tau = 3 \cdot 2.2 \times 10^{-16} \cdot 3.2 \approx 2.1 \times 10^{-15}$ the numerical rank is 2: the third value is rounding noise, and A_τ^{+} uses only the first two terms. A naive inverse would contain a term of size $1/(4 \times 10^{-16}) = 2.5 \times 10^{15}$.

10.2 Noise amplification and truncated SVD

Suppose $\boldsymbol{b} = \boldsymbol{b}_{\text{true}} + \boldsymbol{\eta}$ where $\boldsymbol{\eta}$ is noise. The pseudo-inverse solution is

$$A^{+}\boldsymbol{b} = \sum_{i=1}^{r} \frac{\boldsymbol{u}_i^{\mathsf{T}}\boldsymbol{b}_{\text{true}}}{\sigma_i}\boldsymbol{v}_i + \sum_{i=1}^{r} \frac{\boldsymbol{u}_i^{\mathsf{T}}\boldsymbol{\eta}}{\sigma_i}\boldsymbol{v}_i.$$

The noise component along $\boldsymbol{v}_i$ is divided by σ_i. For small σ_i the noise term can swamp everything. The simplest remedy is the **truncated SVD** (TSVD) solution, which keeps only the k largest terms:

$$\boldsymbol{x}_k = \sum_{i=1}^{k} \frac{\boldsymbol{u}_i^{\mathsf{T}}\boldsymbol{b}}{\sigma_i}\boldsymbol{v}_i = A_k^{+}\boldsymbol{b}.$$

This introduces a *bias* (we ignore genuine information in the discarded directions) but removes the worst *variance*.

Worked Example 10.2: Truncation beats exact inversion.

$A = \operatorname{diag}(1,\ 10^{-3})$ and $\boldsymbol{x}_{\text{true}} = (1,1)$, so $\boldsymbol{b}_{\text{true}} = (1,\ 0.001)$. Measurement noise $\boldsymbol{\eta} = (0.01,\ 0.01)$ gives $\boldsymbol{b} = (1.01,\ 0.011)$.

Exact solution of the noisy system. $A^{+}\boldsymbol{b} = (1.01,\ 11)$. Error $\|(0.01,\ 10)\| \approx 10$.

TSVD with $k = 1$. $\boldsymbol{x}_1 = (1.01,\ 0)$. Error $\|(0.01,\ -1)\| \approx 1$.

Lesson. The second component of $\boldsymbol{b}$ is 90% noise; dividing it by $\sigma_2 = 10^{-3}$ is disastrous. Throwing that component away is ten times better — even though it means not even trying to recover x_2.

10.3 Tikhonov regularisation (ridge regression)

Instead of a hard cut-off we can penalise large solutions. **Tikhonov regularisation** with parameter $\lambda > 0$ solves

$$\min_{\boldsymbol{x}} \ \|A\boldsymbol{x} - \boldsymbol{b}\|^2 + \lambda\,\|\boldsymbol{x}\|^2 \quad \Longleftrightarrow \quad (A^{\mathsf{T}}A + \lambda I)\boldsymbol{x} = A^{\mathsf{T}}\boldsymbol{b}.$$

The matrix $A^{\mathsf{T}}A + \lambda I$ has eigenvalues $\sigma_i^2 + \lambda \geq \lambda > 0$, so it is always invertible, however singular A is. In statistics this is **ridge regression**.

Theorem 10.1: Tikhonov solution via the SVD.

$$\boldsymbol{x}_\lambda = \sum_{i=1}^{r} \frac{\sigma_i}{\sigma_i^2 + \lambda}(\boldsymbol{u}_i^{\mathsf{T}}\boldsymbol{b})\,\boldsymbol{v}_i = \sum_{i=1}^{r} f_i\,\frac{\boldsymbol{u}_i^{\mathsf{T}}\boldsymbol{b}}{\sigma_i}\boldsymbol{v}_i, \qquad f_i = \frac{\sigma_i^2}{\sigma_i^2 + \lambda}.$$

As $\lambda \to 0^{+}$, $\boldsymbol{x}_\lambda \to A^{+}\boldsymbol{b}$.

Proof. Substitute $A = U\Sigma V^{\mathsf{T}}$: $A^{\mathsf{T}}A + \lambda I = V(\Sigma^{\mathsf{T}}\Sigma + \lambda I)V^{\mathsf{T}}$ and $A^{\mathsf{T}}\boldsymbol{b} = V\Sigma^{\mathsf{T}}U^{\mathsf{T}}\boldsymbol{b}$. So $\boldsymbol{x}_\lambda = V(\Sigma^{\mathsf{T}}\Sigma + \lambda I)^{-1}\Sigma^{\mathsf{T}}U^{\mathsf{T}}\boldsymbol{b}$, whose i-th coordinate in the $\boldsymbol{v}$-basis is $\sigma_i(\boldsymbol{u}_i^{\mathsf{T}}\boldsymbol{b})/(\sigma_i^2 + \lambda)$ (and 0 for $i > r$). As $\lambda \to 0$, each $f_i \to 1$. □

The **filter factors** f_i are close to 1 for $\sigma_i^2 \gg \lambda$ (those components are kept) and close to 0 for $\sigma_i^2 \ll \lambda$ (those are suppressed); TSVD corresponds to a step function. Figure 10.1 compares them. Both methods have one tuning parameter (k or λ), usually chosen by cross-validation or by an estimate of the noise level.

Worked Example 10.3: Ridge on a single equation.

$A = [1\ \ 2]$, $b = 5$. Here $\sigma_1 = \sqrt{5}$, $u_1 = 1$, $\boldsymbol{v}_1 = \frac{1}{\sqrt{5}}(1,2)$, and $\boldsymbol{u}_1^{\mathsf{T}}\boldsymbol{b} = 5$.

Formula. $\boldsymbol{x}_\lambda = \dfrac{\sqrt{5}}{5+\lambda}\cdot 5 \cdot \dfrac{1}{\sqrt{5}}\begin{bmatrix}1\\2\end{bmatrix} = \dfrac{5}{5+\lambda}\begin{bmatrix}1\\2\end{bmatrix}$.

Limits. $\lambda \to 0$: $(1,2) = A^{+}b$, the minimum-norm solution of Worked Example 8.4. $\lambda = 5$: $(\tfrac{1}{2},1)$, shrunk halfway to zero. $\lambda \to \infty$: $\mathbf{0}$.

Check with the normal equations at $\lambda = 5$. $(A^{\mathsf{T}}A + 5I)\boldsymbol{x} = \left[\begin{smallmatrix}6&2\\2&9\end{smallmatrix}\right]\left[\begin{smallmatrix}1/2\\1\end{smallmatrix}\right] = \left[\begin{smallmatrix}5\\10\end{smallmatrix}\right] = A^{\mathsf{T}}b$. ✓

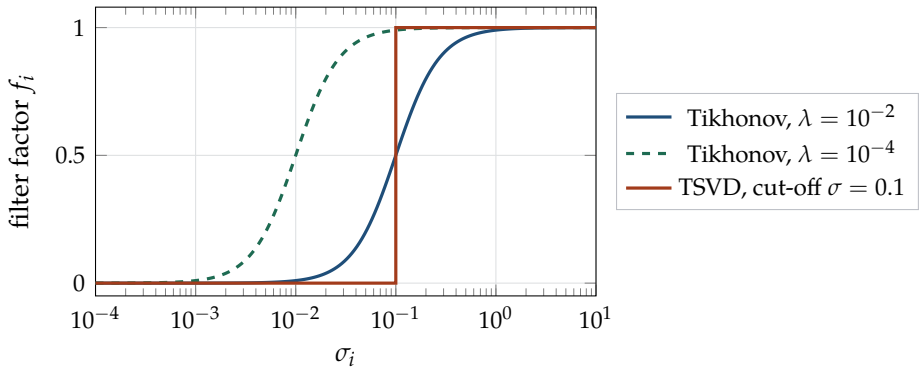


Figure 10.1: Filter factors. Tikhonov regularisation smoothly damps components with $\sigma_i \lesssim \sqrt{\lambda}$; truncation removes them abruptly.

10.4 Principal component analysis

Let the rows of X $(N \times d)$ be N data points in $\mathbb{R}^d$, and suppose the data are **centred**: each column has mean zero (subtract the column means first). The direction $\boldsymbol{w}$ (unit vector) along which the data vary most maximises the sum of squared projections

$$\sum_{k=1}^{N} (\boldsymbol{x}_k \cdot \boldsymbol{w})^2 = \|X\boldsymbol{w}\|^2,$$

which by Theorem 6.2 is $\boldsymbol{w} = \boldsymbol{v}_1$, the first right singular vector of X. More generally:

Theorem 10.2: PCA via the SVD. Let $X = U\Sigma V^{\mathsf{T}}$ be the SVD of the centred data matrix.

- The **principal directions** are $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots$ (the rows of V^{T}).
- The variance of the data along $\boldsymbol{v}_i$ is $\sigma_i^2/(N-1)$; the sample covariance matrix is $C = X^{\mathsf{T}}X/(N-1) = V\frac{\Sigma^{\mathsf{T}}\Sigma}{N-1}V^{\mathsf{T}}$.
- The fraction of variance explained by the first k components is $\sum_{i \le k} \sigma_i^2 \,/\, \sum_i \sigma_i^2$.
- The **scores** (coordinates of the data in the new axes) are $XV = U\Sigma$; keeping k columns gives the best k-dimensional representation (Eckart–Young).

Worked Example 10.4: PCA by hand.

Data points $(-2,-1)$, $(-1,-2)$, $(1,2)$, $(2,1)$, $(0,0)$.

Centre. The means of both coordinates are 0 already.

$X^{\mathsf{T}}X$. $\sum x^2 = 4+1+1+4 = 10$, $\sum y^2 = 10$, $\sum xy = 2+2+2+2 = 8$: $X^{\mathsf{T}}X = \left[\begin{smallmatrix} 10 & 8 \\ 8 & 10 \end{smallmatrix}\right]$.

Eigen-decomposition. $\lambda = 18, 2$ with $\boldsymbol{v}_1 = \frac{1}{\sqrt{2}}(1,1)$, $\boldsymbol{v}_2 = \frac{1}{\sqrt{2}}(1,-1)$. So $\sigma_1 = 3\sqrt{2}$, $\sigma_2 = \sqrt{2}$.

Variance explained. $18/(18+2) = 90\%$ by the first component. Sample variances $18/4 = 4.5$ and $2/4 = 0.5$.

Scores on PC1. $X\boldsymbol{v}_1 = \frac{1}{\sqrt{2}}(-3,-3,3,3,0) \approx (-2.12,-2.12,2.12,2.12,0)$. Projecting onto this single axis keeps 90% of the variance.

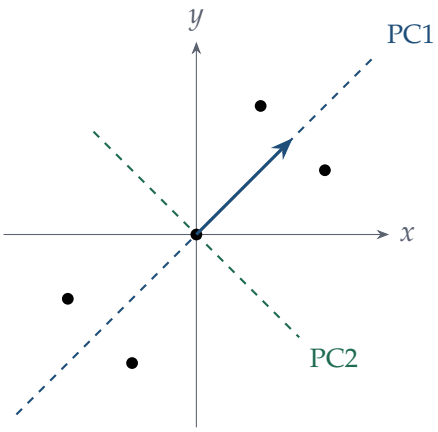


Figure 10.2: The five data points of Worked Example 10.4 with their principal axes. The spread along PC1 is three times the spread along PC2 (standard deviations $\sqrt{4.5}$ vs $\sqrt{0.5}$); the arrow has length one standard deviation along PC1.

10.5 Low-rank approximation in applications

- **Image compression.** A greyscale image is a matrix of pixel intensities. Its truncated SVD with $k \ll \min(m, n)$ often looks almost identical to the original at a fraction of the storage (Lab Manual, Part 7).
- **Recommender systems.** A users $\times$ items rating matrix is modelled as approximately low-rank: a few latent "taste" factors explain most preferences.
- **Latent semantic analysis.** A terms $\times$ documents count matrix is replaced by its rank-k approximation, merging synonyms into shared "concept" directions.
- **Denoising.** Signal often lies near a low-dimensional subspace while noise spreads over all singular values; truncation removes most of the noise.

10.6 How the SVD is really computed

Our hand method — eigen-decompose $A^{\mathsf{T}}A$ — is excellent on paper and dangerous in floating point, because forming $A^{\mathsf{T}}A$ squares the condition number and can destroy small singular values completely.

Worked Example 10.5: The Läuchli matrix.

$$A = \begin{bmatrix} 1 & 1 \\ \varepsilon & 0 \\ 0 & \varepsilon \end{bmatrix} \text{ with } \varepsilon = 10^{-8}.$$

Exact values. $A^{\mathsf{T}}A = \begin{bmatrix} 1+\varepsilon^2 & 1 \\ 1 & 1+\varepsilon^2 \end{bmatrix}$ has eigenvalues $2+\varepsilon^2$ and ε^2, so $\sigma_1 = \sqrt{2+\varepsilon^2} \approx 1.414$ and $\sigma_2 = \varepsilon = 10^{-8}$. The matrix has full rank, $\kappa \approx 1.4 \times 10^8$ — ill-conditioned, but harmless in double precision.

In floating point. $\varepsilon^2 = 10^{-16}$ is below half the machine precision, so the computer stores $1+\varepsilon^2$ as exactly 1. The computed $A^{\mathsf{T}}A$ is $\left[\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right]$, which is exactly singular: the computed

σ_2 is 0 and the rank appears to be 1.

Directly. An SVD algorithm working on A itself (as in NumPy) returns $\sigma_2 = 1.0000 \times 10^{-8}$, correct to all digits shown.

Production algorithms therefore work on A directly:

- **Golub–Kahan–Reinsch.** Orthogonal (Householder) transformations from left and right reduce A to bidiagonal form $B = Q_1^{\mathsf{T}} A Q_2$; a variant of the QR algorithm then drives B to diagonal form. This is LAPACK's `gesvd`; `gesdd` uses divide-and-conquer for the second phase. Cost: about $4mn^2 + 8n^3$ operations for $m \geq n$.
- **One-sided Jacobi.** Plane rotations are applied to pairs of columns of A until all columns are mutually orthogonal; their lengths are then the singular values. Slower but simple and very accurate, even for tiny singular values. You implement it in the Lab and Assignment 4.
- **Randomised SVD.** For huge matrices where only the top k singular triplets are wanted, multiply A by a random $n \times (k + p)$ matrix, orthonormalise the result, and compute a small SVD in that subspace.

10.7 Summary: what the SVD tells you

Question	Answer from $A = U\Sigma V^{\mathsf{T}}$
Rank	number of $\sigma_i > 0$ (numerically: $> \tau$)
Four fundamental subspaces	$\boldsymbol{v}_{1..r}$, $\boldsymbol{v}_{r+1..n}$, $\boldsymbol{u}_{1..r}$, $\boldsymbol{u}_{r+1..m}$
Size: $\|A\|_2$, $\|A\|_F$	σ_1, $\sqrt{\sum \sigma_i^2}$
Sensitivity	$\kappa = \sigma_1 / \sigma_r$
Distance to singularity / to rank k	σ_n / σ_{k+1}
Best rank-k approximation	$\sum_{i \leq k} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^{\mathsf{T}}$
Pseudo-inverse	$\sum_{i \leq r} \sigma_i^{-1} \boldsymbol{v}_i \boldsymbol{u}_i^{\mathsf{T}}$
Min-norm least squares	$\sum_{i \leq r} \sigma_i^{-1} (\boldsymbol{u}_i^{\mathsf{T}} \boldsymbol{b}) \boldsymbol{v}_i$
Regularised solution	$\sum f_i \sigma_i^{-1} (\boldsymbol{u}_i^{\mathsf{T}} \boldsymbol{b}) \boldsymbol{v}_i$, $f_i = \sigma_i^2 / (\sigma_i^2 + \lambda)$
Principal components	$\boldsymbol{v}_i$, variances $\sigma_i^2 / (N - 1)$

Exercises

Exercise 10.1 ⋆ Compute A_τ^+ for $A = \operatorname{diag}(1,\, 10^{-10})$ (a) with $\tau = 0$ and (b) with $\tau = 10^{-8}$. If $\boldsymbol{b} = (1,\, 10^{-9})$ where the second entry is pure measurement noise, compare the two solutions.

Exercise 10.2 ⋆⋆ For $A = \operatorname{diag}(1,\, 0.1)$, $\boldsymbol{b} = (1, 1)$ and $\lambda = 0.01$, compute the Tikhonov solution $\boldsymbol{x}_\lambda$ and the filter factors. Compare with $A^{-1}\boldsymbol{b}$.

Exercise 10.3 ⋆⋆ Perform PCA on the points $(2, 2)$, $(-2, -2)$, $(1, -1)$, $(-1, 1)$. Give the principal directions, the sample variances and the fraction of variance explained by the first component.

Exercise 10.4 ⋆⋆ A matrix has singular values $\sigma = (4,\, 2,\, 10^{-3})$ and $\boldsymbol{b}$ has coefficients $\boldsymbol{u}_i^{\mathsf{T}}\boldsymbol{b} = (8,\, 2,\, 10^{-3} + \eta)$, where $\eta = 10^{-2}$ is noise. Compute the coefficients y_i of $\boldsymbol{x}$ in the $\boldsymbol{v}$-basis for (a) the pseudo-inverse solution, (b) TSVD with $k = 2$, (c) Tikhonov with $\lambda = 10^{-4}$.

Exercise 10.5 ⋆⋆⋆ Show that the Tikhonov solution $\boldsymbol{x}_\lambda$ always lies in the row space of A, and that $\|\boldsymbol{x}_\lambda\|$ decreases as λ increases.

Exercise 10.6 ⋆⋆ For the Läuchli matrix of Worked Example 10.5, compute $\kappa(A)$ and $\kappa(A^{\mathsf{T}}A)$ for $\varepsilon = 10^{-8}$, and explain in terms of these numbers why the normal-equations approach fails in double precision while working with A succeeds.

APPENDIX A

Formula Sheet

Vectors	
Dot product, length	$\boldsymbol{x}\cdot\boldsymbol{y}=\boldsymbol{x}^{\mathsf{T}}\boldsymbol{y}=\sum x_i y_i, \quad \|\boldsymbol{x}\|=\sqrt{\boldsymbol{x}^{\mathsf{T}}\boldsymbol{x}}$
Angle	$\cos\theta=\dfrac{\boldsymbol{x}\cdot\boldsymbol{y}}{\|\boldsymbol{x}\|\,\|\boldsymbol{y}\|}$ (Cauchy–Schwarz: $\|\boldsymbol{x}\cdot\boldsymbol{y}\|\le\|\boldsymbol{x}\|\,\|\boldsymbol{y}\|$)
Norms	$\|\boldsymbol{x}\|_1=\sum\|x_i\|,\ \|\boldsymbol{x}\|_2=\sqrt{\sum x_i^2},\ \|\boldsymbol{x}\|_\infty=\max\|x_i\|$
Projection onto a line	$\boldsymbol{p}=\dfrac{\boldsymbol{a}^{\mathsf{T}}\boldsymbol{b}}{\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a}}\boldsymbol{a}, \quad P=\dfrac{\boldsymbol{a}\boldsymbol{a}^{\mathsf{T}}}{\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a}}$
Matrices and subspaces	
Product as outer products	$AB=\sum_k(\mathrm{col}_k A)(\mathrm{row}_k B)$
Transpose rules	$(AB)^{\mathsf{T}}=B^{\mathsf{T}}A^{\mathsf{T}}, \quad (A\boldsymbol{x})\cdot\boldsymbol{y}=\boldsymbol{x}\cdot(A^{\mathsf{T}}\boldsymbol{y})$
Dimensions (A: $m\times n$, rank r)	$\mathcal{C}(A^{\mathsf{T}})$: r, $\mathcal{N}(A)$: $n-r$, $\mathcal{C}(A)$: r, $\mathcal{N}(A^{\mathsf{T}})$: $m-r$
Orthogonality	$\mathcal{N}(A)\perp\mathcal{C}(A^{\mathsf{T}}), \quad \mathcal{N}(A^{\mathsf{T}})\perp\mathcal{C}(A)$
Orthogonality	
Orthonormal columns	$Q^{\mathsf{T}}Q=I$; square $\Rightarrow Q^{-1}=Q^{\mathsf{T}}$; $\|Q\boldsymbol{x}\|=\|\boldsymbol{x}\|$
Projection onto $\mathcal{C}(A)$	$P=A(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}=QQ^{\mathsf{T}};\ P^2=P=P^{\mathsf{T}}$
Gram–Schmidt	$\boldsymbol{w}_j=\boldsymbol{a}_j-\sum_{i<j}(\boldsymbol{q}_i^{\mathsf{T}}\boldsymbol{a}_j)\boldsymbol{q}_i,\ \boldsymbol{q}_j=\boldsymbol{w}_j/\|\boldsymbol{w}_j\|$
QR	$A=QR,\ r_{ij}=\boldsymbol{q}_i^{\mathsf{T}}\boldsymbol{a}_j,\ r_{jj}=\|\boldsymbol{w}_j\|$
Eigenvalues	
Characteristic equation	$\det(A-\lambda I)=0$; 2×2: $\lambda^2-(\mathrm{tr}\,A)\lambda+\det A=0$
Checks	$\sum\lambda_i=\mathrm{tr}\,A, \quad \prod\lambda_i=\det A$
Spectral theorem	$S=S^{\mathsf{T}}\Rightarrow S=Q\Lambda Q^{\mathsf{T}}=\sum\lambda_i\boldsymbol{q}_i\boldsymbol{q}_i^{\mathsf{T}}$
Rayleigh quotient	$\lambda_{\min}\le\dfrac{\boldsymbol{x}^{\mathsf{T}}S\boldsymbol{x}}{\boldsymbol{x}^{\mathsf{T}}\boldsymbol{x}}\le\lambda_{\max}$
PSD	$\boldsymbol{x}^{\mathsf{T}}S\boldsymbol{x}\ge0\iff\lambda_i\ge0$; $A^{\mathsf{T}}A$ is always PSD; $\mathcal{N}(A^{\mathsf{T}}A)=\mathcal{N}(A)$
Singular value decomposition	
SVD	$A=U\Sigma V^{\mathsf{T}}=\sum_{i=1}^{r}\sigma_i\boldsymbol{u}_i\boldsymbol{v}_i^{\mathsf{T}},\ \sigma_1\ge\cdots\ge\sigma_r>0$
Defining relations	$A\boldsymbol{v}_i=\sigma_i\boldsymbol{u}_i, \quad A^{\mathsf{T}}\boldsymbol{u}_i=\sigma_i\boldsymbol{v}_i$
Computing	σ_i^2, $\boldsymbol{v}_i$: eigen-pairs of $A^{\mathsf{T}}A$; $\boldsymbol{u}_i=A\boldsymbol{v}_i/\sigma_i$
Norms	$\|A\|_2=\sigma_1,\ \|A\|_F=\sqrt{\sum\sigma_i^2},\ \|A\|_*=\sum\sigma_i$
Condition number	$\kappa(A)=\sigma_1/\sigma_r$; $\kappa(A^{\mathsf{T}}A)=\kappa(A)^2$
Eckart–Young	$A_k=\sum_{i\le k}\sigma_i\boldsymbol{u}_i\boldsymbol{v}_i^{\mathsf{T}}$; $\|A-A_k\|_2=\sigma_{k+1},\ \|A-A_k\|_F=\sqrt{\sum_{i>k}\sigma_i^2}$

Storage of A_k	$k(m+n+1)$ numbers

Least squares and the pseudo-inverse

Normal equations	$A^{\mathsf{T}}A\hat{\boldsymbol{x}} = A^{\mathsf{T}}\boldsymbol{b}$; QR: $R\hat{\boldsymbol{x}} = Q^{\mathsf{T}}\boldsymbol{b}$
Pseudo-inverse	$A^{+} = V\Sigma^{+}U^{\mathsf{T}} = \sum_{i\le r}\sigma_i^{-1}\boldsymbol{v}_i\boldsymbol{u}_i^{\mathsf{T}}$
Penrose conditions	$AXA = A,\ XAX = X,\ (AX)^{\mathsf{T}} = AX,\ (XA)^{\mathsf{T}} = XA$
Full column rank	$A^{+} = (A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}},\ A^{+}A = I_n$
Full row rank	$A^{+} = A^{\mathsf{T}}(AA^{\mathsf{T}})^{-1},\ AA^{+} = I_m$
Rank one	$A^{+} = A^{\mathsf{T}}/\lVert A\rVert_F^2$
Projections	$AA^{+} \to \mathcal{C}(A),\ A^{+}A \to \mathcal{C}(A^{\mathsf{T}}),\ I - A^{+}A \to \mathcal{N}(A)$
Min-norm LS solution	$\boldsymbol{x}^{+} = A^{+}\boldsymbol{b} = \sum_{i\le r}\dfrac{\boldsymbol{u}_i^{\mathsf{T}}\boldsymbol{b}}{\sigma_i}\boldsymbol{v}_i$; all LS solutions: $\boldsymbol{x}^{+} + (I - A^{+}A)\mathbf{0}$
Tikhonov	$(A^{\mathsf{T}}A + \lambda I)\boldsymbol{x} = A^{\mathsf{T}}\boldsymbol{b},\ \boldsymbol{x}_\lambda = \sum\dfrac{\sigma_i}{\sigma_i^2+\lambda}(\boldsymbol{u}_i^{\mathsf{T}}\boldsymbol{b})\boldsymbol{v}_i$
PCA (centred X)	directions $\boldsymbol{v}_i$, variances $\sigma_i^2/(N-1)$, scores $XV = U\Sigma$

APPENDIX B

Solutions to the Exercises

Every exercise in the book is solved here, in the order it appears. For the hand-calculation exercises the solutions show the intermediate steps you should have on paper, not just the final answer — compare your work line by line. The page number after each solution points back to the exercise.

Chapter 1 Vectors, Inner Products and Norms

Solution 1.1 (p. 6). $\boldsymbol{u}\cdot\boldsymbol{v} = 2-2+0 = 0$, so the vectors are orthogonal and the angle is 90°. $\|\boldsymbol{u}\| = \sqrt{1+4+4} = 3$ and $\|\boldsymbol{v}\| = \sqrt{4+1} = \sqrt{5}$. Unit vectors: $\frac{1}{3}(1,-2,2)$ and $\frac{1}{\sqrt{5}}(2,1,0)$.

Solution 1.2 (p. 6). $\|\boldsymbol{x}\|_1 = 3+4+12 = 19$; $\|\boldsymbol{x}\|_2 = \sqrt{9+16+144} = \sqrt{169} = 13$; $\|\boldsymbol{x}\|_\infty = 12$. Indeed $12 \le 13 \le 19$.

Solution 1.3 (p. 6). $\boldsymbol{a}\cdot\boldsymbol{b} = 2+0+4 = 6$, $\boldsymbol{a}\cdot\boldsymbol{a} = 2$, so $\hat{t} = 3$ and $\boldsymbol{p} = (3,0,3)$. Then $\boldsymbol{e} = \boldsymbol{b}-\boldsymbol{p} = (-1,3,1)$ and $\boldsymbol{e}\cdot\boldsymbol{a} = -1+0+1 = 0$. ✓ Lengths: $\|\boldsymbol{b}\|^2 = 4+9+16 = 29$, $\|\boldsymbol{p}\|^2 = 18$, $\|\boldsymbol{e}\|^2 = 1+9+1 = 11$, and $18+11 = 29$. ✓

Solution 1.4 (p. 6). We need $\boldsymbol{x} = (x,y,z)$ with $x+y = 0$ and $y+z = 0$. Put $y = -1$: then $x = 1$ and $z = 1$, giving $(1,-1,1)$, whose length is $\sqrt{3}$. The unit vectors are $\pm\frac{1}{\sqrt{3}}(1,-1,1)$ — exactly two, because the vectors orthogonal to both form a line.

Solution 1.5 (p. 6). $\boldsymbol{u}\cdot\boldsymbol{v} = 4+10+18 = 32$; $\|\boldsymbol{u}\| = \sqrt{14}$, $\|\boldsymbol{v}\| = \sqrt{77}$, so $\|\boldsymbol{u}\|\,\|\boldsymbol{v}\| = \sqrt{1078} \approx 32.83 \ge 32$. ✓ $\cos\theta = 32/\sqrt{1078} \approx 0.9746$, so $\theta \approx 12.93^\circ$: the vectors point in nearly the same direction.

Solution 1.6 (p. 6). Dot products: $\boldsymbol{q}_1\cdot\boldsymbol{q}_2 = \frac{1}{9}(2+2-4) = 0$, $\boldsymbol{q}_1\cdot\boldsymbol{q}_3 = \frac{1}{9}(2-4+2) = 0$, $\boldsymbol{q}_2\cdot\boldsymbol{q}_3 = \frac{1}{9}(4-2-2) = 0$, and each $\boldsymbol{q}_i\cdot\boldsymbol{q}_i = \frac{1}{9}\cdot 9 = 1$. Three orthonormal vectors in $\mathbb{R}^3$ form a basis.
Coordinates: $c_1 = \boldsymbol{q}_1\cdot\boldsymbol{x} = \frac{1}{3}(3+6+6) = 5$, $c_2 = \frac{1}{3}(6+3-6) = 1$, $c_3 = \frac{1}{3}(6-6+3) = 1$.
Check: $5\boldsymbol{q}_1+\boldsymbol{q}_2+\boldsymbol{q}_3 = \frac{1}{3}\big((5,10,10)+(2,1,-2)+(2,-2,1)\big) = \frac{1}{3}(9,9,9) = (3,3,3)$. ✓
Also $c_1^2+c_2^2+c_3^2 = 27 = \|\boldsymbol{x}\|^2$.

Solution 1.7 (p. 6). (a) $\|\boldsymbol{u}+\boldsymbol{v}\|^2 = (\boldsymbol{u}+\boldsymbol{v})\cdot(\boldsymbol{u}+\boldsymbol{v}) = \boldsymbol{u}\cdot\boldsymbol{u}+2\,\boldsymbol{u}\cdot\boldsymbol{v}+\boldsymbol{v}\cdot\boldsymbol{v}$ and the middle term is 0.
(b) $\|\boldsymbol{u}-\boldsymbol{v}\|^2 = \|\boldsymbol{u}\|^2-2\boldsymbol{u}\cdot\boldsymbol{v}+\|\boldsymbol{v}\|^2 = 1-2\cos\theta+1$, since $\boldsymbol{u}\cdot\boldsymbol{v} = \|\boldsymbol{u}\|\,\|\boldsymbol{v}\|\cos\theta = \cos\theta$.
(c) $\|\boldsymbol{u}+\boldsymbol{v}\|^2 = \|\boldsymbol{u}\|^2+2\boldsymbol{u}\cdot\boldsymbol{v}+\|\boldsymbol{v}\|^2 \le \|\boldsymbol{u}\|^2+2\,\|\boldsymbol{u}\|\,\|\boldsymbol{v}\|+\|\boldsymbol{v}\|^2 = (\|\boldsymbol{u}\|+\|\boldsymbol{v}\|)^2$; take square roots (both sides are non-negative).

Chapter 2 Matrices, Rank and the Four Fundamental Subspaces

Solution 2.1 (p. 12). Columns: $1\cdot(1,3,5)-1\cdot(2,4,6)=(-1,-1,-1)$. Rows: $1-2=-1$, $3-4=-1$, $5-6=-1$. Both give $A\boldsymbol{x}=(-1,-1,-1)$.

Solution 2.2 (p. 12). $\begin{bmatrix}1\\3\end{bmatrix}\begin{bmatrix}5&6\end{bmatrix}+\begin{bmatrix}2\\4\end{bmatrix}\begin{bmatrix}7&8\end{bmatrix}=\begin{bmatrix}5&6\\15&18\end{bmatrix}+\begin{bmatrix}14&16\\28&32\end{bmatrix}=\begin{bmatrix}19&22\\43&50\end{bmatrix}$.
Entry check: $(AB)_{11}=1\cdot5+2\cdot7=19$, $(AB)_{12}=1\cdot6+2\cdot8=22$, $(AB)_{21}=3\cdot5+4\cdot7=43$, $(AB)_{22}=3\cdot6+4\cdot8=50$. ✓

Solution 2.3 (p. 12). **Elimination.** $R_2-2R_1=(0,0,1,2)$, $R_3-3R_1=(0,0,1,2)$, then $R_3-R_2=\mathbf{0}$: $R=\left[\begin{smallmatrix}1&2&0&1\\0&0&1&2\\0&0&0&0\end{smallmatrix}\right]$ (already reduced). Pivots in columns 1 and 3, so $r=2$.
Row space ($\dim 2$): $(1,2,0,1)$, $(0,0,1,2)$. **Column space** ($\dim 2$): columns 1 and 3 of A: $(1,2,3)$, $(0,1,1)$.
Null space ($\dim n-r=2$): free variables x_2,x_4. From R: $x_1=-2x_2-x_4$, $x_3=-2x_4$. Special solutions: $x_2=1,x_4=0$ gives $(-2,1,0,0)$; $x_2=0,x_4=1$ gives $(-1,0,-2,1)$.
Left null space ($\dim m-r=1$): row 3 = row 1 + row 2, so $\boldsymbol{y}=(1,1,-1)$.
Orthogonality: $(-2,1,0,0)\cdot(1,2,0,1)=0$, $(-2,1,0,0)\cdot(0,0,1,2)=0$, $(-1,0,-2,1)\cdot(1,2,0,1)=0$, $(-1,0,-2,1)\cdot(0,0,1,2)=-2+2=0$; and $(1,1,-1)\cdot(1,2,3)=0$, $(1,1,-1)\cdot(0,1,1)=0$. ✓

Solution 2.4 (p. 12). (a) $(A^{\mathsf{T}}A)^{\mathsf{T}}=A^{\mathsf{T}}(A^{\mathsf{T}})^{\mathsf{T}}=A^{\mathsf{T}}A$. (b) $\boldsymbol{x}^{\mathsf{T}}A^{\mathsf{T}}A\boldsymbol{x}=(A\boldsymbol{x})^{\mathsf{T}}(A\boldsymbol{x})=\|A\boldsymbol{x}\|^2$.
(c) $A^{\mathsf{T}}A=\begin{bmatrix}1&0\\2&1\\0&1\end{bmatrix}\begin{bmatrix}1&2&0\\0&1&1\end{bmatrix}=\begin{bmatrix}1&2&0\\2&5&1\\0&1&1\end{bmatrix}$, $AA^{\mathsf{T}}=\begin{bmatrix}1&2&0\\0&1&1\end{bmatrix}\begin{bmatrix}1&0\\2&1\\0&1\end{bmatrix}=\begin{bmatrix}5&2\\2&2\end{bmatrix}$. Both are symmetric.

Solution 2.5 (p. 13). Row 1 is twice row 2, so $M=\begin{bmatrix}2\\1\end{bmatrix}\begin{bmatrix}1&2&3\end{bmatrix}$, i.e. $\boldsymbol{u}=(2,1)$, $\boldsymbol{v}=(1,2,3)$.
Column space: the line through $(2,1)$. Row space: the line through $(1,2,3)$. Null space: the plane $x_1+2x_2+3x_3=0$ in $\mathbb{R}^3$, with basis $(-2,1,0)$, $(-3,0,1)$. Left null space: the line through $(1,-2)$ in $\mathbb{R}^2$ (perpendicular to $(2,1)$). Dimensions $1,1,2,1$ agree with $r=1$, $n=3$, $m=2$.

Solution 2.6 (p. 13). Every column of AB is A times a column of B, hence lies in $\mathcal{C}(A)$. So $\mathcal{C}(AB)\subseteq\mathcal{C}(A)$ and $\operatorname{rank}(AB)\le\operatorname{rank}A$.
Every row of AB is a row of A times B, i.e. a combination of rows of B; so the row space of AB lies inside the row space of B, and since row rank equals column rank, $\operatorname{rank}(AB)\le\operatorname{rank}B$.

Chapter 3 Orthogonality: Orthogonal Matrices, Projections and QR

Solution 3.1 (p. 18). The columns are $\frac{1}{3}(1,2,2)$, $\frac{1}{3}(2,1,-2)$, $\frac{1}{3}(2,-2,1)$ — the vectors of Exercise 1.6, shown there to be orthonormal. So $Q^{\mathsf{T}}Q = I$. Since Q is also symmetric, $Q^{-1} = Q^{\mathsf{T}} = Q$: this Q is a reflection (it equals $I - 2\boldsymbol{u}\boldsymbol{u}^{\mathsf{T}}$ with $\boldsymbol{u} = \frac{1}{\sqrt{3}}(1,-1,-1)$).

Solution 3.2 (p. 18). $P = \dfrac{\boldsymbol{a}\boldsymbol{a}^{\mathsf{T}}}{\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a}} = \dfrac{1}{9}\begin{bmatrix}1&2&2\\2&4&4\\2&4&4\end{bmatrix}$. Then $P^2 = \dfrac{\boldsymbol{a}(\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a})\boldsymbol{a}^{\mathsf{T}}}{(\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a})^2} = \dfrac{\boldsymbol{a}\boldsymbol{a}^{\mathsf{T}}}{\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a}} = P$. $\operatorname{tr} P = \frac{1}{9}(1+4+4) = 1$ (the dimension of the line). $P\boldsymbol{b} = \frac{1}{9}(5,10,10)$, as in Worked Example 1.4.

Solution 3.3 (p. 18). $\|\boldsymbol{a}_1\| = 2$, so $\boldsymbol{q}_1 = \frac{1}{2}(1,1,1,1)$.
$\boldsymbol{q}_1 \cdot \boldsymbol{a}_2 = 1$, so $\boldsymbol{w}_2 = (1,1,0,0) - \frac{1}{2}(1,1,1,1) = \frac{1}{2}(1,1,-1,-1)$, of length 1: $\boldsymbol{q}_2 = \frac{1}{2}(1,1,-1,-1)$.
$\boldsymbol{q}_1 \cdot \boldsymbol{a}_3 = 1$ and $\boldsymbol{q}_2 \cdot \boldsymbol{a}_3 = \frac{1}{2}(1+0-1-0) = 0$, so $\boldsymbol{w}_3 = (1,0,1,0) - \frac{1}{2}(1,1,1,1) = \frac{1}{2}(1,-1,1,-1)$, of length 1: $\boldsymbol{q}_3 = \frac{1}{2}(1,-1,1,-1)$.
Hence $A = QR$ with $Q = \frac{1}{2}\begin{bmatrix}1&1&1\\1&1&-1\\1&-1&1\\1&-1&-1\end{bmatrix}$ and $R = \begin{bmatrix}2&1&1\\0&1&0\\0&0&1\end{bmatrix}$.

Solution 3.4 (p. 18). $A^{\mathsf{T}}A = \begin{bmatrix}2&0\\0&1\end{bmatrix}$, $A^{\mathsf{T}}\boldsymbol{b} = (1+3,\,2) = (4,2)$, so $\hat{\boldsymbol{x}} = (2,2)$, $\boldsymbol{p} = A\hat{\boldsymbol{x}} = (2,2,2)$ and $\boldsymbol{e} = (-1,0,1)$. $A^{\mathsf{T}}\boldsymbol{e} = (-1+1,\,0) = (0,0)$. ✓

Solution 3.5 (p. 18). (a) $(I-P)^2 = I - 2P + P^2 = I - P$ and $(I-P)^{\mathsf{T}} = I - P^{\mathsf{T}} = I - P$.
(b) If $P\boldsymbol{x} = \lambda\boldsymbol{x}$ with $\boldsymbol{x} \neq \boldsymbol{0}$, then $\lambda\boldsymbol{x} = P\boldsymbol{x} = P^2\boldsymbol{x} = \lambda^2\boldsymbol{x}$, so $\lambda^2 = \lambda$, i.e. $\lambda \in \{0,1\}$.
(c) $(P\boldsymbol{b})^{\mathsf{T}}(I-P)\boldsymbol{b} = \boldsymbol{b}^{\mathsf{T}}P^{\mathsf{T}}(I-P)\boldsymbol{b} = \boldsymbol{b}^{\mathsf{T}}(P-P^2)\boldsymbol{b} = 0$.

Solution 3.6 (p. 18). (a) $1 = \det I = \det(Q^{\mathsf{T}}Q) = \det(Q^{\mathsf{T}})\det Q = (\det Q)^2$.
(b) If $Q\boldsymbol{x} = \lambda\boldsymbol{x}$ with $\boldsymbol{x} \neq \boldsymbol{0}$ and λ real, then $\|\boldsymbol{x}\| = \|Q\boldsymbol{x}\| = |\lambda|\,\|\boldsymbol{x}\|$, so $|\lambda| = 1$.
(c) $(Q_1Q_2)^{\mathsf{T}}(Q_1Q_2) = Q_2^{\mathsf{T}}Q_1^{\mathsf{T}}Q_1Q_2 = Q_2^{\mathsf{T}}Q_2 = I$.

Chapter 4 Eigenvalues, Eigenvectors and Symmetric Matrices

Solution 4.1 (p. 23). $\operatorname{tr} A = 7$, $\det A = 10$: $\lambda^2 - 7\lambda + 10 = (\lambda-5)(\lambda-2)$, so $\lambda = 5, 2$.
$A - 5I = \begin{bmatrix}-1&1\\2&-2\end{bmatrix}$ gives $(1,1)$; $A - 2I = \begin{bmatrix}2&1\\2&1\end{bmatrix}$ gives $(1,-2)$.
$(1,1)\cdot(1,-2) = -1 \neq 0$: not orthogonal. Not surprising, since A is not symmetric.

Solution 4.2 (p. 23). $\operatorname{tr} = 7$, $\det = 6$: $\lambda = 6, 1$. $S - 6I = \begin{bmatrix}-1&2\\2&-4\end{bmatrix}$ gives $(2,1)$; $S - I = \begin{bmatrix}4&2\\2&1\end{bmatrix}$ gives $(1,-2)$. So $\boldsymbol{q}_1 = \frac{1}{\sqrt{5}}(2,1)$, $\boldsymbol{q}_2 = \frac{1}{\sqrt{5}}(1,-2)$.
Check: $\dfrac{6}{5}\begin{bmatrix}4&2\\2&1\end{bmatrix} + \dfrac{1}{5}\begin{bmatrix}1&-2\\-2&4\end{bmatrix} = \dfrac{1}{5}\begin{bmatrix}25&10\\10&10\end{bmatrix} = \begin{bmatrix}5&2\\2&2\end{bmatrix}$. ✓

Solution 4.3 (p. 23). $S = J - I$, where J is the all-ones matrix with eigenvalues $3, 0, 0$. Hence S has eigenvalues $3-1 = 2$ and $0-1 = -1$ (twice); check $\operatorname{tr} S = 0 = 2-1-1$. Eigen-

vectors are those of J: $\boldsymbol{q}_1 = \frac{1}{\sqrt{3}}(1,1,1)$ for $\lambda = 2$, and for $\lambda = -1$ any orthonormal basis of the plane $x + y + z = 0$, e.g. $\boldsymbol{q}_2 = \frac{1}{\sqrt{2}}(1,-1,0)$, $\boldsymbol{q}_3 = \frac{1}{\sqrt{6}}(1,1,-2)$.

Solution 4.4 (p. 23). $\boldsymbol{x}_1 = S\boldsymbol{x}_0 = (5,2)$, $\boldsymbol{x}_2 = (29,14)$, $\boldsymbol{x}_3 = (173,86)$.
$R(\boldsymbol{x}_0) = 5$; $R(\boldsymbol{x}_1) = \dfrac{5\cdot 25 + 2\cdot 2\cdot 5\cdot 2 + 2\cdot 4}{29} = \dfrac{173}{29} \approx 5.9655$; $R(\boldsymbol{x}_2) = \dfrac{6221}{1037} \approx 5.99904$; $R(\boldsymbol{x}_3) = \dfrac{223949}{37325} \approx 5.999973$.
The true eigenvalue is 6. Errors 1, 0.0345, 0.00096, 0.000027 shrink by roughly $36 = (\lambda_1/\lambda_2)^2 = (6/1)^2$ per step. The vectors approach the direction $(2,1)$.

Solution 4.5 (p. 23). (a) Induction: $A^k\boldsymbol{x} = A(A^{k-1}\boldsymbol{x}) = A(\lambda^{k-1}\boldsymbol{x}) = \lambda^{k-1}A\boldsymbol{x} = \lambda^k\boldsymbol{x}$.
(b) If $\lambda = 0$ then $A\boldsymbol{x} = \boldsymbol{0}$ with $\boldsymbol{x} \neq \boldsymbol{0}$, so A is singular. Otherwise apply A^{-1} to $A\boldsymbol{x} = \lambda\boldsymbol{x}$: $\boldsymbol{x} = \lambda A^{-1}\boldsymbol{x}$.
(c) $(A + cI)\boldsymbol{x} = \lambda\boldsymbol{x} + c\boldsymbol{x}$. Shifting a matrix by cI shifts every eigenvalue by c and keeps the eigenvectors — which is why $J - I$ has eigenvalues those of J minus 1.

Solution 4.6 (p. 23). Let $S\boldsymbol{x} = \lambda\boldsymbol{x}$, $\boldsymbol{x} \neq \boldsymbol{0}$ (complex in general). Then $\bar{\boldsymbol{x}}^{\mathsf{T}}S\boldsymbol{x} = \lambda\,\bar{\boldsymbol{x}}^{\mathsf{T}}\boldsymbol{x} = \lambda\sum|x_i|^2$. Conjugating $S\boldsymbol{x} = \lambda\boldsymbol{x}$ (with S real) gives $S\bar{\boldsymbol{x}} = \bar{\lambda}\bar{\boldsymbol{x}}$; transposing and using $S^{\mathsf{T}} = S$: $\bar{\boldsymbol{x}}^{\mathsf{T}}S = \bar{\lambda}\bar{\boldsymbol{x}}^{\mathsf{T}}$. Hence also $\bar{\boldsymbol{x}}^{\mathsf{T}}S\boldsymbol{x} = \bar{\lambda}\,\bar{\boldsymbol{x}}^{\mathsf{T}}\boldsymbol{x}$. Since $\bar{\boldsymbol{x}}^{\mathsf{T}}\boldsymbol{x} = \sum|x_i|^2 > 0$, we get $\lambda = \bar{\lambda}$: λ is real.

Chapter 5 Positive Semidefinite Matrices and $A^{\mathsf{T}}A$

Solution 5.1 (p. 28). S_1: $\lambda = 3, 1$ — PD. S_2: tr $= 5$, det $= 0$, $\lambda = 5, 0$ — PSD (not PD). S_3: tr $= 2$, det $= -8$, $\lambda = 4, -2$ — indefinite. $S_4 = -\left[\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right]$ and $\left[\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right]$ is PD (Worked Example 5.1) — so S_4 is negative definite ($\lambda = -1, -3$).

Solution 5.2 (p. 28). $A^{\mathsf{T}}A = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$, eigenvalues $3, 1$. $AA^{\mathsf{T}} = \left[\begin{smallmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{smallmatrix}\right]$: trace 4; det $= 1(2-1) - 1(1-0) + 0 = 0$, so 0 is an eigenvalue, and the other two sum to 4. Checking $\lambda = 3$: $AA^{\mathsf{T}} - 3I = \left[\begin{smallmatrix} -2 & 1 & 0 \\ 1 & -1 & 1 \\ 0 & 1 & -2 \end{smallmatrix}\right]$ has $(1,2,1)$ in its null space; $\lambda = 1$: $(1,0,-1)$. So AA^{T} has eigenvalues $3, 1, 0$: the same non-zero ones plus one 0 (since $m - r = 1$).

Solution 5.3 (p. 28). Eigenvalues are $c + 2$ and $c - 2$ (shift of $\left[\begin{smallmatrix} 0 & 2 \\ 2 & 0 \end{smallmatrix}\right]$, whose eigenvalues are ± 2). (a) PD iff $c - 2 > 0$, i.e. $c > 2$. (b) PSD iff $c \geq 2$. (The minor test agrees: $c > 0$ and $c^2 - 4 > 0$.)

Solution 5.4 (p. 28). S^{-1} has eigenvalues $1/\lambda_i > 0$ (Exercise 4.5) and is symmetric, so it is PD. For the sum: $\boldsymbol{x}^{\mathsf{T}}(S+T)\boldsymbol{x} = \boldsymbol{x}^{\mathsf{T}}S\boldsymbol{x} + \boldsymbol{x}^{\mathsf{T}}T\boldsymbol{x} > 0$ for $\boldsymbol{x} \neq \boldsymbol{0}$.
ST need not even be symmetric: with $S = \left[\begin{smallmatrix} 1 & 0 \\ 0 & 2 \end{smallmatrix}\right]$, $T = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right]$ (both PD), $ST = \left[\begin{smallmatrix} 2 & 1 \\ 2 & 2 \end{smallmatrix}\right]$, which is not symmetric, so it is not PD in our sense. (Its eigenvalues are nonetheless positive.)

Solution 5.5 (p. 28). The first part is the proof of Theorem 5.3, part 2. Taking orthogonal complements of the equal null spaces gives equal row spaces: $\mathcal{C}\big((A^{\mathsf{T}}A)^{\mathsf{T}}\big) = \mathcal{C}(A^{\mathsf{T}})$, and $(A^{\mathsf{T}}A)^{\mathsf{T}} = A^{\mathsf{T}}A$, so $\mathcal{C}(A^{\mathsf{T}}A) = \mathcal{C}(A^{\mathsf{T}})$. Since $A^{\mathsf{T}}\boldsymbol{b} \in \mathcal{C}(A^{\mathsf{T}}) = \mathcal{C}(A^{\mathsf{T}}A)$, there is an $\boldsymbol{x}$ with $A^{\mathsf{T}}A\boldsymbol{x} = A^{\mathsf{T}}\boldsymbol{b}$. So the normal equations are always solvable, even when A has dependent

columns (the solution is then not unique).

Solution 5.6 (p. 28). By Theorem 5.4, each $\boldsymbol{u}_i$ is a unit eigenvector of AA^T with eigenvalue λ_i, and $\boldsymbol{u}_i^\mathsf{T}\boldsymbol{u}_j = \boldsymbol{v}_i^\mathsf{T}A^\mathsf{T}A\boldsymbol{v}_j/\sqrt{\lambda_i\lambda_j} = \lambda_j\boldsymbol{v}_i^\mathsf{T}\boldsymbol{v}_j/\sqrt{\lambda_i\lambda_j} = \delta_{ij}$.
Finally $A^\mathsf{T}\boldsymbol{u}_i = A^\mathsf{T}A\boldsymbol{v}_i/\sqrt{\lambda_i} = \lambda_i\boldsymbol{v}_i/\sqrt{\lambda_i} = \sqrt{\lambda_i}\,\boldsymbol{v}_i$. Together with $A\boldsymbol{v}_i = \sqrt{\lambda_i}\,\boldsymbol{u}_i$ this says A maps $\boldsymbol{v}_i$ to $\sigma_i\boldsymbol{u}_i$ and A^T maps $\boldsymbol{u}_i$ back to $\sigma_i\boldsymbol{v}_i$, with $\sigma_i = \sqrt{\lambda_i}$.

Chapter 6 The Singular Value Decomposition

Solution 6.1 (p. 36). $A^\mathsf{T}A = \operatorname{diag}(9,4)$, so $\sigma_1 = 3$ with $\boldsymbol{v}_1 = \boldsymbol{e}_1$ and $\sigma_2 = 2$ with $\boldsymbol{v}_2 = \boldsymbol{e}_2$. Then $\boldsymbol{u}_1 = A\boldsymbol{e}_1/3 = (-1,0)$ and $\boldsymbol{u}_2 = A\boldsymbol{e}_2/2 = (0,1)$:

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Singular values are never negative; the sign goes into U.

Solution 6.2 (p. 36). $A^\mathsf{T}A = \left[\begin{smallmatrix} 0 & 0 \\ 0 & 4 \end{smallmatrix}\right]$: $\sigma_1 = 2$ with $\boldsymbol{v}_1 = \boldsymbol{e}_2$, and $\sigma_2 = 0$ with $\boldsymbol{v}_2 = \boldsymbol{e}_1$. $\boldsymbol{u}_1 = A\boldsymbol{e}_2/2 = \boldsymbol{e}_1$; complete with $\boldsymbol{u}_2 = \boldsymbol{e}_2$. So $A = I \cdot \operatorname{diag}(2,0) \cdot \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$, i.e. $A = 2\,\boldsymbol{e}_1\boldsymbol{e}_2^\mathsf{T}$. Rank 1. Row space: span of $\boldsymbol{e}_2$; null space: span of $\boldsymbol{e}_1$; column space: span of $\boldsymbol{e}_1$; left null space: span of $\boldsymbol{e}_2$. Note A has only the eigenvalue 0, yet $\sigma_1 = 2$.

Solution 6.3 (p. 36). $A^\mathsf{T}A = \left[\begin{smallmatrix} 5 & 3 \\ 3 & 5 \end{smallmatrix}\right]$: $\lambda = 8, 2$, so $\sigma_1 = 2\sqrt{2}$, $\sigma_2 = \sqrt{2}$, with $\boldsymbol{v}_1 = \frac{1}{\sqrt{2}}(1,1)$, $\boldsymbol{v}_2 = \frac{1}{\sqrt{2}}(1,-1)$.
$\boldsymbol{u}_1 = A\boldsymbol{v}_1/\sigma_1 = \frac{1}{\sqrt{2}\cdot 2\sqrt{2}}(4,0) = (1,0)$; $\boldsymbol{u}_2 = A\boldsymbol{v}_2/\sigma_2 = \frac{1}{\sqrt{2}\cdot\sqrt{2}}(0,-2) = (0,-1)$.
$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{bmatrix}\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$. Check: $U\Sigma = \left[\begin{smallmatrix} 2\sqrt{2} & 0 \\ 0 & -\sqrt{2} \end{smallmatrix}\right]$, and times V^T: $\left[\begin{smallmatrix} 2 & 2 \\ -1 & 1 \end{smallmatrix}\right]$. ✓

Solution 6.4 (p. 36). Use $AA^\mathsf{T} = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$: $\lambda = 3, 1$, so $\sigma_1 = \sqrt{3}$, $\sigma_2 = 1$, $\boldsymbol{u}_1 = \frac{1}{\sqrt{2}}(1,1)$, $\boldsymbol{u}_2 = \frac{1}{\sqrt{2}}(1,-1)$.
$\boldsymbol{v}_1 = A^\mathsf{T}\boldsymbol{u}_1/\sqrt{3} = \frac{1}{\sqrt{6}}(1,1,2)$; $\boldsymbol{v}_2 = A^\mathsf{T}\boldsymbol{u}_2/1 = \frac{1}{\sqrt{2}}(1,-1,0)$.
$\boldsymbol{v}_3$ spans $\mathcal{N}(A)$: $x_1 + x_3 = 0$, $x_2 + x_3 = 0$ gives $\boldsymbol{v}_3 = \frac{1}{\sqrt{3}}(1,1,-1)$. (Check: orthogonal to $\boldsymbol{v}_1$: $1+1-2=0$; to $\boldsymbol{v}_2$: $1-1=0$.)
$U = \frac{1}{\sqrt{2}}\left[\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right]$, $\Sigma = \left[\begin{smallmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{smallmatrix}\right]$, $V = [\boldsymbol{v}_1\ \boldsymbol{v}_2\ \boldsymbol{v}_3]$. Check: $\sqrt{3}\boldsymbol{u}_1\boldsymbol{v}_1^\mathsf{T} + \boldsymbol{u}_2\boldsymbol{v}_2^\mathsf{T} = \frac{1}{2}\left[\begin{smallmatrix} 1 & 1 & 2 \\ 1 & 1 & 2 \end{smallmatrix}\right] + \frac{1}{2}\left[\begin{smallmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{smallmatrix}\right]$. ✓

Solution 6.5 (p. 36). $A = \boldsymbol{x}\boldsymbol{y}^\mathsf{T}$ with $\boldsymbol{x} = (2,1,2)$, $\boldsymbol{y} = (1,2)$. So $\sigma_1 = \|\boldsymbol{x}\|\,\|\boldsymbol{y}\| = 3\sqrt{5}$, $\boldsymbol{u}_1 = \frac{1}{3}(2,1,2)$, $\boldsymbol{v}_1 = \frac{1}{\sqrt{5}}(1,2)$, and $A = \sigma_1\boldsymbol{u}_1\boldsymbol{v}_1^\mathsf{T}$. Check: $3\sqrt{5}\cdot\frac{1}{3}\cdot\frac{1}{\sqrt{5}} = 1$. ✓
$\mathcal{N}(A)$: spanned by $\boldsymbol{v}_2 = \frac{1}{\sqrt{5}}(2,-1)$. $\mathcal{N}(A^\mathsf{T}) = \{\boldsymbol{y} : 2y_1 + y_2 + 2y_3 = 0\}$, e.g. $\boldsymbol{u}_2 = \frac{1}{\sqrt{2}}(1,0,-1)$ and $\boldsymbol{u}_3 = \boldsymbol{u}_1 \times \boldsymbol{u}_2 = \frac{1}{3\sqrt{2}}(-1,4,-1)$.

Solution 6.6 (p. 36). $A^\mathsf{T}A = I$, so both eigenvalues are 1 and $\sigma_1 = \sigma_2 = 1$. $A \cdot I \cdot I^\mathsf{T} = A$ and $I \cdot I \cdot A^\mathsf{T} = A^\mathsf{T} = A$, and A, I are orthogonal — so both are SVDs. Because the singular value 1 is repeated, *every* orthonormal basis of $\mathbb{R}^2$ is a valid choice of $\boldsymbol{v}_1, \boldsymbol{v}_2$ (then $\boldsymbol{u}_i = A\boldsymbol{v}_i$). This is the situation of Worked Example 4.3: repeated values leave only the subspace determined.

Solution 6.7 (p. 36). (a) $Q^\mathsf{T}Q = I$ has all eigenvalues 1, so all $\sigma_i = 1$.
(b) For $\boldsymbol{x} \neq \boldsymbol{0}$, $\|A\boldsymbol{x}\| = \|\boldsymbol{x}\| \, \|A(\boldsymbol{x}/\|\boldsymbol{x}\|)\| \leq \|\boldsymbol{x}\| \, \sigma_1$ by Theorem 6.2.
(c) $Q_1 A Q_2 = (Q_1 U)\Sigma(Q_2^\mathsf{T} V)^\mathsf{T}$, and $Q_1 U$, $Q_2^\mathsf{T} V$ are orthogonal (products of orthogonal matrices). By uniqueness of singular values this is an SVD with the same Σ.

Solution 6.8 (p. 36). (a) $A^\mathsf{T} = (U\Sigma V^\mathsf{T})^\mathsf{T} = V\Sigma^\mathsf{T}U^\mathsf{T}$ is an SVD of A^T with Σ^T, which has the same diagonal.
(b) $\det A = \det U \det \Sigma \det V^\mathsf{T} = (\pm 1)(\sigma_1 \cdots \sigma_n)(\pm 1)$, using $\det Q = \pm 1$ (Exercise 3.6).
(c) $\det A = 15$ and $\sigma_1\sigma_2 = 3\sqrt{5} \cdot \sqrt{5} = 15$. ✓

Chapter 7 Norms, Low-Rank Approximation and Conditioning

Solution 7.1 (p. 41). $\|A\|_2 = 5$; $\|A\|_F = \sqrt{25+9} = \sqrt{34} \approx 5.831$ (entries: $9+4+4+4+9+4 = 34$ ✓); $\|A\|_* = 8$; rank 2; $\kappa = 5/3$.

Solution 7.2 (p. 41). $A_1 = \sigma_1 \boldsymbol{u}_1 \boldsymbol{v}_1^\mathsf{T} = 3\sqrt{5} \cdot \frac{1}{\sqrt{10}} \begin{bmatrix} 1 \\ 3 \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \end{bmatrix} = \frac{3}{2} \begin{bmatrix} 1 & 1 \\ 3 & 3 \end{bmatrix}$.
$A - A_1 = \begin{bmatrix} 1.5 & -1.5 \\ -0.5 & 0.5 \end{bmatrix}$, with $\|\cdot\|_F^2 = 2.25 + 2.25 + 0.25 + 0.25 = 5$. So $\|A - A_1\|_F = \sqrt{5} = \sigma_2$. ✓

Solution 7.3 (p. 41). $40(800 + 600 + 1) = 56{,}040$ numbers versus 480,000: about 11.7%. It stops saving space when $k(m + n + 1) \geq mn$, i.e. $k \geq 480000/1401 \approx 342.6$, so from $k = 343$ on.

Solution 7.4 (p. 41). $\|A\|_F^2 = 100 + 36 + 9 + 1 = 146$. (a) $(100 + 36)/146 \approx 93.2\%$. (b) $\|A - A_2\|_2 = \sigma_3 = 3$, relative $3/10 = 30\%$; $\|A - A_2\|_F = \sqrt{9+1} = \sqrt{10} \approx 3.162$, relative $\sqrt{10/146} \approx 26.2\%$.

Solution 7.5 (p. 41). (a) All singular values are 1, so $\sigma_1/\sigma_n = 1$: orthogonal matrices are perfectly conditioned.
(b) $A^\mathsf{T}A = V\Sigma^\mathsf{T}\Sigma V^\mathsf{T}$ has singular values σ_i^2, so $\kappa(A^\mathsf{T}A) = \sigma_1^2/\sigma_n^2 = \kappa(A)^2$.
(c) The normal equations $A^\mathsf{T}A\boldsymbol{x} = A^\mathsf{T}\boldsymbol{b}$ use $A^\mathsf{T}A$. If $\kappa(A) = 10^8$, then $\kappa(A^\mathsf{T}A) = 10^{16} \approx 1/\varepsilon$: all accuracy can be lost. QR and SVD methods work with A directly and only suffer $\kappa(A)$.

Solution 7.6 (p. 41). $A^\mathsf{T}A = V\Sigma^\mathsf{T}\Sigma V^\mathsf{T}$. Then $\operatorname{tr}(A^\mathsf{T}A) = \operatorname{tr}\left(V(\Sigma^\mathsf{T}\Sigma V^\mathsf{T})\right) = \operatorname{tr}\left(\Sigma^\mathsf{T}\Sigma V^\mathsf{T}V\right) = \operatorname{tr}(\Sigma^\mathsf{T}\Sigma) = \sum \sigma_i^2$. And $\operatorname{tr}(A^\mathsf{T}A) = \sum_j (A^\mathsf{T}A)_{jj} = \sum_j \sum_i a_{ij}^2 = \|A\|_F^2$.

Solution 7.7 (p. 41). Singular $A+E$ means $B=A+E$ has rank $\le n-1$, so by Eckart–Young $\|E\|_2 = \|A-B\|_2 \ge \sigma_n$; equality is attained by $B=A_{n-1}$, i.e. $E=-\sigma_n \boldsymbol{u}_n \boldsymbol{v}_n^{\mathsf{T}}$.
For the example, $E=-\sqrt{5}\,\boldsymbol{u}_2\boldsymbol{v}_2^{\mathsf{T}} = -\frac{1}{2}\begin{bmatrix}3\\-1\end{bmatrix}\begin{bmatrix}1 & -1\end{bmatrix} = \begin{bmatrix}-1.5 & 1.5\\ 0.5 & -0.5\end{bmatrix}$, and $A+E = \begin{bmatrix}1.5 & 1.5\\ 4.5 & 4.5\end{bmatrix}$ has determinant 0. ✓ $\|E\|_2=\sqrt{5}$.

Chapter 8 Least Squares

Solution 8.1 (p. 46). $A=\begin{bmatrix}1&1\\1&2\\1&3\end{bmatrix}$, $\boldsymbol{b}=(1,2,2)$. $A^{\mathsf{T}}A=\begin{bmatrix}3&6\\6&14\end{bmatrix}$, $A^{\mathsf{T}}\boldsymbol{b}=(5,11)$. $\det=6$, so $\begin{bmatrix}C\\D\end{bmatrix}=\frac{1}{6}\begin{bmatrix}14&-6\\-6&3\end{bmatrix}\begin{bmatrix}5\\11\end{bmatrix}=\frac{1}{6}\begin{bmatrix}4\\3\end{bmatrix}=\begin{bmatrix}2/3\\1/2\end{bmatrix}$.
Line: $y=\frac{2}{3}+\frac{1}{2}t$. Predictions $\frac{7}{6},\frac{5}{3},\frac{13}{6}$; residual $\boldsymbol{e}=(-\frac{1}{6},\frac{1}{3},-\frac{1}{6})$. Orthogonality: $-\frac{1}{6}+\frac{1}{3}-\frac{1}{6}=0$ and $-\frac{1}{6}+\frac{2}{3}-\frac{1}{2}=0$. ✓

Solution 8.2 (p. 46). $A=[1\ 1\ 1]$, $AA^{\mathsf{T}}=3$, so $\boldsymbol{x}^{+}=A^{\mathsf{T}}\cdot\frac{1}{3}\cdot 3=(1,1,1)$. It is in the row space (a multiple of $(1,1,1)$), and every other solution is $(1,1,1)$ plus a vector orthogonal to it, hence longer.

Solution 8.3 (p. 46). $\boldsymbol{q}_1=\frac{1}{\sqrt{3}}(1,1,1)$, $r_{11}=\sqrt{3}$. $r_{12}=\boldsymbol{q}_1\cdot(1,2,3)=\frac{6}{\sqrt{3}}=2\sqrt{3}$; $\boldsymbol{w}_2=(1,2,3)-(2,2,2)=(-1,0,1)$, $r_{22}=\sqrt{2}$, $\boldsymbol{q}_2=\frac{1}{\sqrt{2}}(-1,0,1)$.
$Q^{\mathsf{T}}\boldsymbol{b}=(\frac{5}{\sqrt{3}},\frac{1}{\sqrt{2}})$. Back substitution: $\sqrt{2}D=\frac{1}{\sqrt{2}}\Rightarrow D=\frac{1}{2}$; $\sqrt{3}C+2\sqrt{3}\cdot\frac{1}{2}=\frac{5}{\sqrt{3}}\Rightarrow C=\frac{5}{3}-1=\frac{2}{3}$. ✓

Solution 8.4 (p. 46). $A=\begin{bmatrix}1&-1&1\\1&0&0\\1&1&1\\1&2&4\end{bmatrix}$, $\boldsymbol{b}=(4,6,0,6)$. $A^{\mathsf{T}}A=\begin{bmatrix}4&2&6\\2&6&8\\6&8&18\end{bmatrix}$, $A^{\mathsf{T}}\boldsymbol{b}=(16,8,28)$.
Solve (e.g. by elimination): $(C,D,E)=(3,-1,1)$. Check: $4\cdot3+2(-1)+6\cdot1=16$, $2\cdot3+6(-1)+8\cdot1=8$, $6\cdot3+8(-1)+18\cdot1=28$. ✓
Parabola $y=3-t+t^2$; predictions $5,3,3,5$; residual $\boldsymbol{e}=(-1,3,-3,1)$, $\|\boldsymbol{e}\|^2=20$.

Solution 8.5 (p. 46). The normal equations say $A^{\mathsf{T}}(\boldsymbol{b}-A\hat{\boldsymbol{x}})=\boldsymbol{0}$, i.e. $\boldsymbol{e}\in\mathcal{N}(A^{\mathsf{T}})$. Since $A\hat{\boldsymbol{x}}\in\mathcal{C}(A)\perp\mathcal{N}(A^{\mathsf{T}})$, Pythagoras gives $\|\boldsymbol{b}\|^2=\|A\hat{\boldsymbol{x}}+\boldsymbol{e}\|^2=\|A\hat{\boldsymbol{x}}\|^2+\|\boldsymbol{e}\|^2$.

Solution 8.6 (p. 46). Let $\boldsymbol{x}$ be any solution and split $\boldsymbol{x}=\boldsymbol{x}_r+\boldsymbol{x}_n$ with $\boldsymbol{x}_r\in\mathcal{C}(A^{\mathsf{T}})$, $\boldsymbol{x}_n\in\mathcal{N}(A)$. Then $A\boldsymbol{x}_r=A\boldsymbol{x}=\boldsymbol{b}$, so a row-space solution exists. If $\boldsymbol{x}_r,\boldsymbol{x}_r'$ are two row-space solutions, $\boldsymbol{x}_r-\boldsymbol{x}_r'$ lies in $\mathcal{C}(A^{\mathsf{T}})\cap\mathcal{N}(A)=\{\boldsymbol{0}\}$ (the two spaces are orthogonal), so they are equal. For any solution $\boldsymbol{x}=\boldsymbol{x}_r+\boldsymbol{x}_n$, Pythagoras gives $\|\boldsymbol{x}\|^2=\|\boldsymbol{x}_r\|^2+\|\boldsymbol{x}_n\|^2\ge\|\boldsymbol{x}_r\|^2$, with equality only if $\boldsymbol{x}_n=\boldsymbol{0}$.

Solution 8.7 (p. 46). A is symmetric with eigenvalues $\lambda=\frac{(2+\delta)\pm\sqrt{4+\delta^2}}{2}\approx 2$ and $\approx\delta/2$, so $\kappa(A)\approx 4/\delta$ and $\kappa(A^{\mathsf{T}}A)\approx 16/\delta^2$. With $\delta=10^{-6}$: $\kappa(A)\approx 4\times10^{6}$, $\kappa(A^{\mathsf{T}}A)\approx 1.6\times10^{13}$. A rule of thumb is that you lose about $\log_{10}\kappa$ digits: about $16-13=3$ correct digits with the normal equations, versus about $16-7=9$ with QR or SVD.

Chapter 9 The Moore–Penrose Pseudo-Inverse

Solution 9.1 (p. 52). $D^+ = \operatorname{diag}(\frac{1}{2}, 0, -\frac{1}{5})$. (Negative entries are fine: $D = U\Sigma V^\mathsf{T}$ just has a sign in U, and the formula reduces to inverting non-zero entries.) E^+ is 3×2: $\begin{bmatrix} 1/4 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$.

Solution 9.2 (p. 52). $\boldsymbol{a}^+ = \boldsymbol{a}^\mathsf{T} / \|\boldsymbol{a}\|^2 = \frac{1}{9}[1\ 2\ 2]$, and $(\boldsymbol{a}^\mathsf{T})^+ = \frac{1}{9}\boldsymbol{a}$. The best multiple is $t = \boldsymbol{a}^+\boldsymbol{b} = \frac{5}{9}$ — the projection coefficient of Worked Example 1.4.

Solution 9.3 (p. 52). Rank one with $\|A\|_F^2 = 4$: $A^+ = \frac{1}{4}\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$. Then $\boldsymbol{x}^+ = \frac{1}{4}(4,4) = (1,1)$, $A\boldsymbol{x}^+ = (2,2,0)$ and $\boldsymbol{e} = (-1,1,5)$, with $\|\boldsymbol{e}\|^2 = 27$. The third equation $0 = 5$ cannot be helped at all; the first two are averaged.

Solution 9.4 (p. 52). (a) $AA^\mathsf{T} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, inverse $\frac{1}{3}\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$; $A^\mathsf{T}(AA^\mathsf{T})^{-1} = \frac{1}{3}\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 2 & -1 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}$.
(b) $\frac{1}{\sqrt{3}}\boldsymbol{v}_1\boldsymbol{u}_1^\mathsf{T} + \frac{1}{1}\boldsymbol{v}_2\boldsymbol{u}_2^\mathsf{T} = \frac{1}{\sqrt{3}}\cdot\frac{1}{\sqrt{6}\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 2 & 2 \end{bmatrix} + \frac{1}{2}\begin{bmatrix} 1 & -1 \\ -1 & 1 \\ 0 & 0 \end{bmatrix} = \frac{1}{6}\begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 2 & 2 \end{bmatrix} + \frac{1}{6}\begin{bmatrix} 3 & -3 \\ -3 & 3 \\ 0 & 0 \end{bmatrix} = \frac{1}{6}\begin{bmatrix} 4 & -2 \\ -2 & 4 \\ 2 & 2 \end{bmatrix}$, the same. ✓

Solution 9.5 (p. 53). $AX = \frac{1}{4}\begin{bmatrix} 2 & 2 & 0 \\ 2 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ — symmetric, (3) ✓. $XA = \frac{1}{4}\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ — symmetric, (4) ✓.
(1) $AXA = A(XA) = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}\frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix} = A$. ✓ (2) $XAX = (XA)X = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}\frac{1}{4}\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} = \frac{1}{8}\begin{bmatrix} 2 & 2 & 0 \\ 2 & 2 & 0 \end{bmatrix} = X$. ✓

Solution 9.6 (p. 53). $\boldsymbol{x}^+ = \frac{1}{3}(2-1+0,\ -1+2+0) = (\frac{1}{3}, \frac{1}{3})$. $A\boldsymbol{x}^+ = (\frac{1}{3}, \frac{1}{3}, \frac{2}{3})$, residual $\boldsymbol{e} = (\frac{2}{3}, \frac{2}{3}, -\frac{2}{3})$, a multiple of $(1,1,-1)$, which spans $\mathcal{N}(A^\mathsf{T})$. Since $\boldsymbol{e} \neq \boldsymbol{0}$, the system $(x_1 = 1, x_2 = 1, x_1 + x_2 = 0)$ is inconsistent.

Solution 9.7 (p. 53). From $A = U\Sigma V^\mathsf{T}$: $A^\mathsf{T} = V\Sigma^\mathsf{T}U^\mathsf{T}$ is an SVD of A^T, so $(A^\mathsf{T})^+ = U(\Sigma^\mathsf{T})^+V^\mathsf{T}$. Since $(\Sigma^\mathsf{T})^+ = (\Sigma^+)^\mathsf{T}$ (both are $m \times n$ with $1/\sigma_i$ on the diagonal), $(A^\mathsf{T})^+ = (V\Sigma^+U^\mathsf{T})^\mathsf{T} = (A^+)^\mathsf{T}$.
$A^+ = V\Sigma^+U^\mathsf{T}$ is an SVD of A^+ (after reordering the diagonal to be decreasing, which permutes columns of U and V consistently). Its pseudo-inverse is $U(\Sigma^+)^+V^\mathsf{T} = U\Sigma V^\mathsf{T} = A$. Alternatively: the Penrose conditions for the pair (A, A^+) are symmetric under swapping A and X — conditions (1)↔(2) and (3)↔(4) — so A satisfies the conditions to be the pseudo-inverse of A^+, and by uniqueness it is.

Solution 9.8 (p. 53). $P^\mathsf{T} = P$ is condition (3). $P^2 = AA^+AA^+ = (AA^+A)A^+ = AA^+ = P$ by (1). Every column of $P = A(A^+)$ is A times something, so $\mathcal{C}(P) \subseteq \mathcal{C}(A)$; conversely $A = PA$ by (1), so every column of A is P times something and $\mathcal{C}(A) \subseteq \mathcal{C}(P)$. A symmetric idempotent matrix is the orthogonal projection onto its column space (Exercise 3.5 and Theorem 3.3).

Solution 9.9 (p. 53). $AB = [1]$, so $(AB)^+ = [1]$. $A^+ = \frac{1}{2}\begin{bmatrix}1\\1\end{bmatrix}$ and $B^+ = [1\ \ 0]$, so $B^+A^+ = [1\ \ 0]\cdot\frac{1}{2}\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}\frac{1}{2}\end{bmatrix}$. Not equal: the reverse-order law fails.

Chapter 10 Applications and Numerical Practice

Solution 10.1 (p. 59). (a) $\operatorname{diag}(1, 10^{10})$, giving $\boldsymbol{x} = (1,\ 10)$. (b) $\operatorname{diag}(1, 0)$, giving $\boldsymbol{x} = (1,\ 0)$. With $\tau = 0$ the noise 10^{-9} is amplified into a component of size 10; with the tolerance it is ignored.

Solution 10.2 (p. 59). $f_1 = \frac{1}{1.01} \approx 0.990$, $f_2 = \frac{0.01}{0.01+0.01} = 0.5$. $\boldsymbol{x}_\lambda = \left(\frac{1}{1.01}\cdot 1,\ \frac{0.1}{0.02}\cdot 1\right) \approx (0.990,\ 5)$, while $A^{-1}\boldsymbol{b} = (1, 10)$. The well-determined component is almost unchanged; the poorly determined one is halved.

Solution 10.3 (p. 59). Mean $(0, 0)$. $\sum x^2 = 4+4+1+1 = 10$, $\sum y^2 = 10$, $\sum xy = 4+4-1-1 = 6$: $X^{\mathsf{T}}X = \left[\begin{smallmatrix}10 & 6\\ 6 & 10\end{smallmatrix}\right]$, eigenvalues 16 and 4, $\boldsymbol{v}_1 = \frac{1}{\sqrt{2}}(1,1)$, $\boldsymbol{v}_2 = \frac{1}{\sqrt{2}}(1,-1)$. Sample variances $16/3 \approx 5.33$ and $4/3 \approx 1.33$. Explained by PC1: $16/20 = 80\%$.

Solution 10.4 (p. 59). (a) $y = (8/4,\ 2/2,\ 0.011/0.001) = (2,\ 1,\ 11)$.
(b) $y = (2,\ 1,\ 0)$.
(c) $y_i = \sigma_i(\boldsymbol{u}_i^{\mathsf{T}}\boldsymbol{b})/(\sigma_i^2+\lambda)$: $y_1 = 4\cdot 8/16.0001 \approx 2.0000$, $y_2 = 2\cdot 2/4.0001 \approx 0.99998$, $y_3 = 10^{-3}\cdot 0.011/(10^{-6}+10^{-4}) = 1.1\times 10^{-5}/1.01\times 10^{-4} \approx 0.109$.
The noise-free answer is $y = (2, 1, 1)$. TSVD and Tikhonov both suppress the noisy third component (at the cost of losing its true value 1); the pseudo-inverse gets it wildly wrong (11).

Solution 10.5 (p. 59). By Theorem 10.1, $\boldsymbol{x}_\lambda$ is a combination of $\boldsymbol{v}_1, \dots, \boldsymbol{v}_r$, which span $\mathcal{C}(A^{\mathsf{T}})$. Its squared norm is $\sum_{i\le r}\left(\sigma_i\boldsymbol{u}_i^{\mathsf{T}}\boldsymbol{b}/(\sigma_i^2+\lambda)\right)^2$; each term is a non-increasing function of $\lambda > 0$ (strictly decreasing when $\boldsymbol{u}_i^{\mathsf{T}}\boldsymbol{b} \neq 0$), so the sum decreases.

Solution 10.6 (p. 59). $\kappa(A) = \sqrt{2+\varepsilon^2}/\varepsilon \approx 1.41\times 10^8$ and $\kappa(A^{\mathsf{T}}A) = \kappa(A)^2 \approx 2\times 10^{16}$. Double precision has $\varepsilon_{\text{mach}} \approx 2.2\times 10^{-16}$, i.e. $1/\varepsilon_{\text{mach}} \approx 4.5\times 10^{15} < \kappa(A^{\mathsf{T}}A)$: $A^{\mathsf{T}}A$ is numerically singular and the information in σ_2 is lost when it is formed. $\kappa(A) \approx 10^8 \ll 1/\varepsilon_{\text{mach}}$, so an algorithm that works on A itself still has about 8 correct digits to spare.

APPENDIX C

Further Reading

These books go further than this course, in the directions you are most likely to want next.

- **Gilbert Strang**, *Introduction to Linear Algebra*. The "four fundamental subspaces" viewpoint used throughout this book. Its chapters on the SVD and on least squares are the natural next step for the theory.
- **Gilbert Strang**, *Linear Algebra and Learning from Data*. The SVD, low-rank approximation, PCA and the pseudo-inverse in the context of data science and machine learning.
- **Lloyd N. Trefethen and David Bau III**, *Numerical Linear Algebra*. Short, beautifully written lectures on how the QR factorisation, least squares and the SVD are computed in floating point, and why conditioning and stability matter.
- **Gene H. Golub and Charles F. Van Loan**, *Matrix Computations*. The standard reference for algorithms, including bidiagonalisation, the Golub–Kahan SVD and Jacobi methods.
- **Adi Ben-Israel and Thomas N. E. Greville**, *Generalized Inverses: Theory and Applications*. Everything about the pseudo-inverse and its relatives.
- **Per Christian Hansen**, *Discrete Inverse Problems: Insight and Algorithms*. Regularisation with the truncated SVD and Tikhonov's method, including how to choose the parameter.